



Exploring the World of Science

Division C Rules Manual

Division C (Gr. 9-12)

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WELCOME TO THE 2027 SCIENCE OLYMPIAD!

This Rules Manual will help you prepare to compete in Invitational, Regional, State and National Tournaments held across the United States annually. Each Science Olympiad event has a corresponding page on the Science Olympiad national website complete with free resources, training handouts and useful links. All users of this manual are subject to the Terms of Use Agreement. To compete, users must first join the Science Olympiad program in their home state and become registered members.

See our website for info on Membership, Policies and Terms of Use at www.soinc.org

Division C (Grades 9-12) Membership Rules

A team may have up to fifteen (15) members. A maximum of seven (7) 12th grade students is permitted on a Division C team.

Division B (Grades 6-9) Membership Rules

A team may have up to fifteen (15) members. A maximum of five (5) 9th grade students is permitted on a Division B team. Because middle schools that do not have grades 7, 8 or 9 are at a slight disadvantage, they may invite any combination of up to five (5) of their last year's 6th, 7th or 8th grade students to be part of the team. Possible examples can be found on the Science Olympiad website.

Students Below Grade Level Designations

Science Olympiad encourages students to participate in the Division that matches current Science Olympiad grade level designations. However, to support the inclusion of students who wish to participate in Science Olympiad, schools with grade levels lower than those stated in a Division are permitted to invite members below the grade level designations. Participation is limited to age-appropriate events (as determined by a coach, principal or tournament director) and prohibited where safety is a concern (such as the use of chemicals). See Team Qualifications for more information.

Science Olympiad Team Membership

Science Olympiad requires that all teams (up to 15 members) competing in any Science Olympiad Tournament (Invitational, Regional, State or National) must be a member of Science Olympiad and pay the national fee (currently \$75, paid as part of the state membership). There is no exception to this requirement, regardless of what teams from the same school are called (Varsity, JV, Alternate Team, Extra Team, Team Two, Team B). No school, region or state Science Olympiad organization is allowed to alter or amend these national membership requirements. Please see the Science Olympiad Copyrights and Use Statement outlining use of Science Olympiad Rules and procedures at sanctioned tournaments.

Find more Science Olympiad team information under the Policies section of the national website: Code of Ethics & Rules, Scoring Guidelines, Home & Virtual Schools, Small Schools, All Stars, Copyrights and Use, Lasers, Building Policy, Eye Protection, Significant Figures and Wristband Procedures.

SCIENCE OLYMPIAD KITS AND RESOURCES AVAILABLE NOW!

Please visit store.soinc.org to purchase 2027 coaching manuals, video downloads, test packets and other event resources for Elementary, Division B, and Division C Science Olympiad. Order the only officially licensed Science Olympiad Kits, supplies and parts for a variety of 2027 events with your Fall Early Bird Savings: Save 12% on your Ward's Science Olympiad Kit order at wardsci.com/scienceolympiad with promo code SOVIP2027. Don't wait! This limited-time offer ends 12/31/26.



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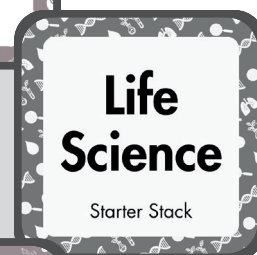
Exploring the World of Science

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Mark Your Calendar

- **7/13/27 – 7/15/27 – Virtual Bootcamp:** Get early access to the 2028 Rules, learn directly from Science Olympiad experts, and choose live sessions based on your interests. Registration includes instructional videos, live sessions, and a hard copy draft of the 2028 Draft Rules.

Try one of the Clinics below – or do both!

- **7/19/27 – 7/20/27 – Content Clinic:** Take a deeper dive into the 2028 Core Knowledge and Lab events. Coaches will work with content experts and Event Supervisors to explore event content, resources, activities, and strategies they can bring back to their teams.
- **7/21/27 – 7/23/27 – Summer Build Clinic:** Get hands-on experience with the 2028 Build events! Coaches will work alongside expert Event Supervisors to build, test, and learn strategies for the upcoming season while collaborating with fellow coaches.

All Summer Workshop opportunities are designed for coaches and educators, not students.



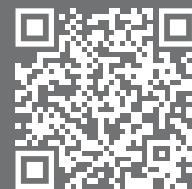
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SCIENCE OLYMPIAD

DIVISION C RULES MANUAL

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- Please read the General Rules on the next page as they apply to all events. Note: all changes are in bold.
- Please visit the official Science Olympiad web site: www.soinc.org for Membership Information, Team Size Requirements, Rules Corrections, Rules Clarifications, New Store Items, news, tips, resources, and other valuable information.

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GENERAL RULES, CODE OF ETHICS, AND SPIRIT OF THE PROBLEM

The goal of competition is to give one's best effort while displaying honesty, integrity, and good sportsmanship. Everyone is expected to display courtesy and respect as outlined in the Science Olympiad Pledges. Teams are expected to make an honest effort to follow the rules and the spirit of the problem (not interpret the rules so they have an unfair advantage). Failure by a participant, coach, or guest to abide by these codes, accepted safety procedures, or rules below, may result in an assessment of penalty points or, in rare cases, disqualification by the Tournament Director from the event, the tournament, or future tournaments.

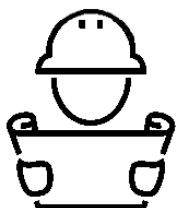
1. Actions and items (e.g., tools, notes, resources, supplies, electronics, etc.) are permitted, unless they are explicitly excluded in the rules, are unsafe, or violate the spirit of the problem.
2. While competing in an event, participants may not leave without the event supervisor's approval and must not receive any external assistance. All electronic devices capable of external communication as well as calculator applications on multipurpose devices (e.g., laptop, phone, tablet) are not permitted unless expressly permitted in the event rule or by an event supervisor. Cell phones, if not permitted, must be turned off. At the discretion of the event supervisor, participants may be required to place their cell phones in a designated location.
3. Participants, coaches and other adults are responsible for ensuring that any applicable school or Science Olympiad policy, law, or regulation is not broken. All Science Olympiad content such as policies, requirements, rule corrections and rule clarifications on www.soinc.org must be treated as if it were included in the printed rules.
4. All pre-built devices presented for judging must be constructed, impounded, and operated by one or more of the 15 current team members unless stated otherwise in the rules. If a device has been removed from the event area, appeals related to that device will not be considered.
5. During the tournament, participants are only permitted to practice with any built or designed device at a Tournament event venue prior to competing if the Tournament Director makes the facilities open to all teams to practice.
6. Officials are encouraged to apply the least restrictive penalty for rules infractions - see examples in the Scoring Guidelines. Event supervisors must provide prompt notification of any penalty, disqualification or tier ranking.
7. State and Regional Tournament Directors must notify teams of any site-dependent rule or other rule modification with as much notice as possible, ideally at least 30 days prior to the tournament.

While a Science Olympiad tournament typically consists of 23 different events, those 23 events can be classified into one of four event types. This information is being provided so that Science Olympiad participants more easily can identify events that they may enjoy competing in regardless of the event content, coaches can approach coaching from the perspective of event type as opposed to event content, and teams can be aware of how the format of the tournament they are intending to compete may affect available events. The symbol to the left of each description has been added to the upper right-hand corner of each Event Rule to identify the event by event type.



Core Knowledge Event: An event where participants are given a set of topics that they are expected to research and master the factual content. Mastery is demonstrated at a tournament by taking a paper-pencil, station, or computer test.

Core Knowledge Events can be run regardless of the tournament format that has been chosen by the Tournament Director.



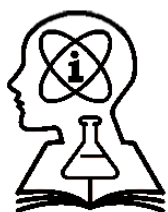
Build Event: An event where participants are given some specifications about a device or object they are expected to design, create, and test in advance of the tournament. The devices or objects are often modified on site to account for an unknown parameter prior to testing or evaluation.

In some cases, Build Events may or may not be run depending upon the format of Science Olympiad tournament being conducted. The Tournament Director will make these decisions to ensure safety and fairness for all teams. If a Build Event is not to be run at a tournament, the Tournament Director will notify all teams in advance of the given tournament.



Laboratory/Hands-On Event: An event where participants are given a general topic in which they will be expected to deepen their content knowledge of the topic and associated research techniques prior to the tournament. At the tournament they will be assessed by the completion of a hands-on task, which may or may not require a written report, within a defined timeframe.

Depending upon the format of Science Olympiad Tournament being held, there may be some alterations to or cancelation of Lab Events. To the greatest extent possible, Tournament Directors will work to ensure Lab Events are conducted; though, that may mean in some cases participants will be working with previously collected data and hands-on activities will be omitted. The Tournament Director will make these decisions to ensure safety and fairness for all teams. If a Lab Event is altered or not to be run at a tournament, the Tournament Director will notify all teams in advance of the given tournament.



Hybrid Event: An event which contains elements from two, or more, of the above event types in combination. The most common combination mixes elements of a Core Knowledge event with elements of a Building or Lab event.

As with the previous events, Hybrid Events may be altered to fit the format of the Science Olympiad Tournament being held. This may mean that Lab or Build elements of the event are modified or not conducted. The Tournament Director will make these decisions to ensure safety and fairness for all teams.

If a Hybrid Event is not to be run at a tournament, the Tournament Director will notify all teams in advance of the given tournament.



1. **DESCRIPTION:** Participants will be assessed on their understanding of the anatomy and physiology for the **respiratory, digestive and immune systems of the human body.**

A TEAM OF UP TO: 2

APPROXIMATE TIME: 50 minutes

CALCULATOR: Class II

2. **EVENT PARAMETERS:**

- a. Each team may bring writing utensils.
 - b. Each team may bring two Class II calculators.
 - c. Each team may bring one (1) 8.5" x 11" sheet of paper that may contain information on both sides in any form and from any source. The sheet of paper may be laminated or placed in a sheet protector to increase durability. Affixed labels, as well as multiple sheets of paper (whether in a single sheet protector or not) are prohibited.
3. **THE COMPETITION:** This Event may be administered as a written test or as a series of lab-practical stations which can include but are not limited to experiments, scientific apparatus, models, illustrations, specimens, data collection and analysis, and problems for students to solve. **Content topics will include the following:**

- a. **RESPIRATORY SYSTEM:**

- i. Understand the anatomy and function of the respiratory system at macroscopic and cellular levels
- ii. Mechanisms of breathing, including inspiratory and expiratory mechanics, pressure gradients, lung compliance, elastic recoil, and airway resistance
- iii. Compare and contrast the walls of the pharynx, larynx, trachea, bronchi, bronchioles, and alveoli
- iv. Quantitatively understand and be able to interpret measures and patterns of breathing, including lung volumes and capacities, spirometry, flow-volume loops, and differentiation of obstructive versus restrictive patterns
- v. Explain mechanisms of gas exchange and oxygen transport, including diffusion principles and oxygen-hemoglobin dissociation curves
- vi. Explain the effects of exercise and high altitude on the respiratory system
- vii. Understand the following respiratory disorders: asthma, chronic obstructive pulmonary disease (COPD), emphysema, cystic fibrosis, pneumonia, and sleep apnea

State & National Level only

- viii. *Regulation and control of respiration, including respiratory centers, chemoreceptors, blood chemistry, and respiratory rhythm*
- ix. *Additional diseases/disorders: tuberculosis, pulmonary edema, pleurisy, pulmonary fibrosis, pneumoconiosis (limited to silicosis, coal worker's lung, asbestosis)*

National Level Only

- x. *Additional diseases/disorders: acute respiratory distress syndrome, infant respiratory distress syndrome, pulmonary hypoplasia, small cell lung cancer, and non-small cell lung cancer*
- xi. *Prevention, diagnosis, and treatment for all diseases/disorders listed above*

- b. **DIGESTIVE SYSTEM:**

- i. Understand the anatomy and functions of the **alimentary tract and the spatial relationships between organs**
 - (1) **Layers of the abdominal wall**
 - (2) **Distinguish between intra-peritoneal organs and retro-peritoneal organs**
- ii. Understand the basic anatomy of the component parts of the alimentary canal and accessory organs of digestion (e.g., salivary glands, liver, gallbladder, pancreas)
- iii. Anatomy of the four layers of the wall of the alimentary canal
- iv. Compare and contrast the lining of the esophagus, stomach, small intestine, and large intestine
- v. Mechanisms of gastrointestinal motility, regulation of digestion by hormones and the enteric nervous system
- vi. Compare and contrast mechanical and chemical digestion
- vii. Understand the physiology of chemical digestion of proteins, fats, and carbohydrates
- viii. Understand physiology of nutrient, vitamin, mineral, and water absorption, including the role of fiber
- ix. Explain the effects of exercise and obesity on the digestive system



- x. Understanding of digestive system diseases/disorders: gastroesophageal reflux disease (GERD), peptic ulcer disease, lactose intolerance, ulcerative colitis, Crohn's disease, celiac disease, appendicitis, and diverticular disease (limited to diverticulosis, complicated and uncomplicated diverticulitis, symptomatic uncomplicated diverticular disease, segmental colitis associated with diverticulosis, smoldering diverticulitis)

State & National Level only

- xi. **Cellular-level organization and function of the liver (e.g., Kupffer cells) and pancreas, including roles in energy metabolism (e.g., glucose homeostasis) and cholesterol balance (e.g., HDL, LDL)**
- xii. **Effects of metabolic syndrome and toxins (limited to acetaminophen, anabolic steroids, alcohol, heavy metals, bisphenols, and polychlorinated biphenyls) on the liver and pancreas**
- xiii. **Role of intestinal microbes in digestion and defense**
- xiv. **Additional diseases/disorders: pancreatitis, metabolic dysfunction-associated steatotic liver disease, cirrhosis, hepatitis, cancers (limited to colorectal, gastric, and hepatic)**

National Level Only

- xv. **Additional diseases/disorders: motility disorders of the digestive system (limited to achalasia, Hirschsprung disease), cholecystitis and biliary system disorders (limited to choledocholithiasis, ascending cholangitis, Mirizzi syndrome, primary biliary cholangitis, primary sclerosing cholangitis, Gilbert syndrome, Crigler-Najjar syndrome), cancers (limited to esophageal, pancreatic and neuroendocrine)**
- xvi. **Prevention, diagnosis, and treatment for all diseases/disorders listed above**

c. IMMUNE SYSTEM:

- i. Anatomy and functions of the immune system (bone marrow, thymus, spleen, lymph nodes, skin)
- ii. Anatomy and physiology of innate immune system (e.g., anatomical barriers, antimicrobial proteins, complement system, Toll-like receptors, inflammation, fever, and innate immune cells)
- iii. Anatomy and physiology of adaptive immune system (e.g., B and T cells, antigen recognition, maturation, selection, tolerance, antigen presentation, activation, and memory)
- iv. Physiology of the immune response, and integration of the innate and adaptive immune systems
- v. Compare and contrast active and passive immunity, understand the role of immunization
- vi. Compare and contrast the general structure as well as the general roles of the 5 human immunoglobulin classes (IgG, IgM, IgA, IgD, and IgE).
- vii. Understanding of immune system diseases/disorders: acquired and genetic immunodeficiencies (limited to HIV/AIDS, common variable immunodeficiency, severe combined immunodeficiency), autoimmune diseases (limited to multiple sclerosis, rheumatoid arthritis, Hashimoto thyroiditis, Graves' Disease, Sjogren's syndrome), and type I-IV hypersensitivities (limited to asthma, anaphylaxis, urticaria, myasthenia gravis, Arthus reaction, serum sickness, contact dermatitis, celiac disease)

State & National Level Only

- viii. **Immune signaling molecules and pathways (e.g., cytokines, interferons, JAK-STAT pathway, cGAS-STING pathway), understand the molecular structural variation in immunoglobulin constant regions, understand the detailed structure and function of immunoglobulins (including the IgG and IgA subclasses), and the roles of the immune molecules and pathways in all physiologic and pathologic processes listed in this section**
- ix. **Types of organ transplantation (e.g., allograft and autograft), mechanisms of rejection (i.e., hyperacute, acute, chronic), prevention of rejection, and graft-versus-host disease**
- x. **Additional diseases/disorders: Type I diabetes, systemic lupus erythematosus, dermatomyositis, scleroderma, psoriasis, and psoriatic arthritis, erythroblastosis fetalis, immunodeficiencies (limited to DiGeorge's syndrome, Bruton's agammaglobulinemia, Job's syndrome, Bloom's syndrome, Wiskott-Aldrich Syndrome, MHC class I and class II deficiency)**



National Level Only

- xi. Tumor immunology, including immune surveillance, tumor evasion, and cancer immunotherapies (therapeutic cancer vaccines, cytokine therapies, antibody therapies, CAR T-cell therapy)
- xii. Compare and contrast vaccine routes and the role of vaccine adjuvants in common use
- xiii. Suboptimal Antibodies: Understand immunological imprinting and antibody-dependent enhancement
- xiv. Additional diseases/disorders: Leukemia (Acute lymphoblastic, acute myeloid, chronic lymphocytic, and chronic myeloid), severe congenital neutropenia, cyclic neutropenia, diseases caused by superantigens (limited to staph enterotoxin food poisoning and toxic shock syndrome)
- xv. Prevention, diagnosis, and treatment for all diseases/disorders listed above

4. **SCORING:**

- a. High Score wins.
- b. Preselected questions will be used to break ties.





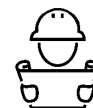
1. **DESCRIPTION:** Teams will demonstrate an understanding of stellar evolution in normal & starburst galaxies.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 50 minutes

CALCULATOR: Class IV

2. **EVENT PARAMETERS:** Each team may bring one of the following options containing information in any form from any source: a computer/tablet and a three-ring binder, two computers/tablets of any kind, or two three-ring binders.
 - a. If three ring binders are used, they may be of any size and the information contained should be attached using the available rings. The information or pages may be removed during the event. Sheet protectors and laminated sheets are allowed.
 - b. Each team may bring two stand-alone Class IV calculators. If the participants are using a computer or tablet, they may use the calculator app or other program on their device in place of a stand-alone calculator.
 - c. Participants using computers/tablets as a resource should have all information stored so that it is available to them offline. Unauthorized AI tools (e.g., ChatGPT, or other trained models) are not allowed to be used to answer questions under any circumstances during the event.
3. **THE COMPETITION:** Using information which may include Hertzsprung-Russell diagrams, spectra, light curves, proper motions, cosmological distance equations and relationships, stellar magnitudes and classification, multi-wavelength (gamma-ray, X-ray, UV, optical, IR, radio) and multi-messenger (gravitational waves, neutrinos, cosmic rays) observational data, charts, and graphs, teams will compete in activities and answer questions related to:
 - a. Stellar evolution including stellar classification, spectral features and chemical composition, luminosity, blackbody radiation, spectroscopy, H-R diagrams, Cepheid and RR Lyrae variables, Type Ia supernovas, neutron stars, ultra-luminous X-ray sources (ULXs), gravitational waves; normal & starburst galaxies including morphological classification, galactic structure (e.g., stellar populations, globular clusters) and interactions (e.g., tidal disruptions, collisions, mergers).
 - b. Use orbital mechanics, Kepler's laws, rotation and circular motion to analyze the orbital motions of binary and galactic systems; use parallax, spectroscopic parallax, the distance modulus, period-luminosity relations (Cepheids and RR Lyrae), the Tully-Fisher relationship, Type Ia supernovas, and Hubble's law to calculate cosmological distances; and use the Stefan-Boltzmann law and Wien's displacement law to answer questions relating to the content areas outlined above.
 - c. Identify and answer questions relating to the content areas outlined above for the following objects: Andromeda Galaxy (M31), Sombrero Galaxy, M51, NGC 4536, MCG+07-33-027, NGC 1569, Antennae Galaxies, Arp 143, Arp 147, Cartwheel Galaxy, M82 (X-1, X-2, SN 2014J), GW170817, Terzan 5.
4. **SCORING:**
 - a. High Score wins.
 - b. Preselected questions will be used to break ties.



1. **DESCRIPTION:** Teams will build a cantilevered beam or truss structure that extends from a vertical Testing Wall and supports a load at a specified height and distance from the Testing Wall. The structure must meet the requirements specified in these rules to achieve the highest score, which is a combination of Structural Efficiency and Load Score Bonus.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 10 minutes

EYE PROTECTION: B

IMPOUND: No

2. EVENT PARAMETERS:

- a. Each team may enter only one structure in the competition.
- b. Eye Protection B protective eyewear must be worn during competition (required ANSI marking: Z87+).
- c. Design Knowledge: Participants must be able to answer questions on construction & operation.
- d. At in person tournaments, Test Apparatus will be supplied by the Event Supervisor. Participants may not measure or adjust Test Apparatus or bring any apparatus, equipment, tools or instruments.

3. CONSTRUCTION PARAMETERS:

- a. Single Structure: no separate, loose or detachable parts or pieces.
- b. The structure is constructed of wood, bonded by adhesive (7.a. – 7.d.). No other materials permitted.
- c. Structure attaches to the testing wall by resting on the Mounting Bolt Assembly (6.b.v.).
- d. Structure supports only the Loading Block of the Loading Assembly (6.c.i.) which supports the balance of the Loading Assembly.
- e. Structure supports the Loading Assembly (6.c.) with the centerline of the chain being:
 - i. At least 40 cm but no more than 45 cm when measured horizontally from the Testing Wall (6.b.).
 - ii. No more than 2.5 cm horizontally from either side of a perpendicular line extending from the centerline of the Testing Wall.

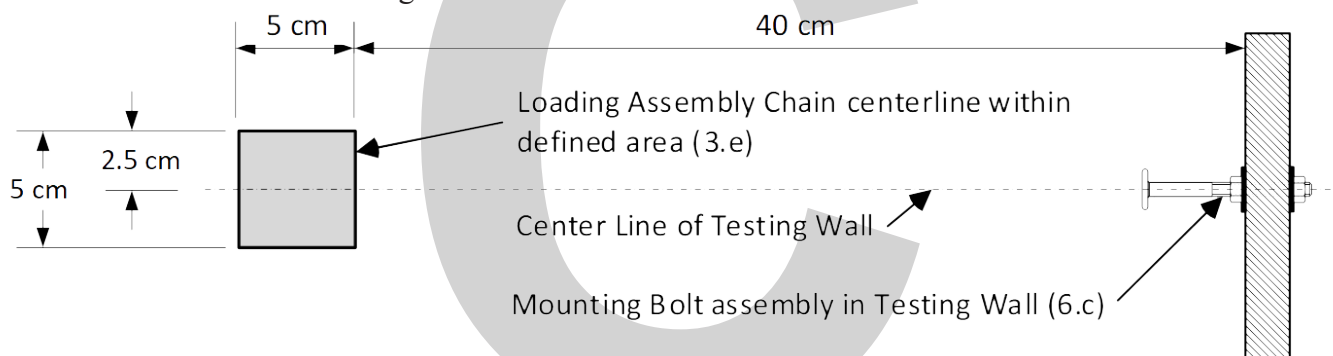
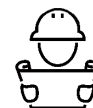


Figure 1: Top view of Testing Wall (not to scale)

- f. During the Competition testing, the Boomilever may ONLY touch the Testing Wall surface OUTSIDE the Vertical Contact Width Lines (6.b.iv.). Contact with the sides, top or bottom of the Testing Wall is not allowed.
- g. **Prior to loading sand, the bottom of the Loading Block must be supported ABOVE the horizontal plane extending from the centerline of the Mounting Bolt. During loading, the bottom of the loading block may go below the horizontal plane extending from the centerline of the Mounting Bolt.**
- h. Base Option:
 - i. During the Competition testing, the Boomilever may ONLY touch the Testing Wall ABOVE the 15 cm Horizontal Contact Depth Lines (6.b.iii.). Structures touching below the 15 cm Horizontal Contact Depth Lines shall be placed in Tier 2.
- i. Load Scored BONUS option scored if the following requirements are met in place of 3.h.:
 - i. During the Competition testing, the Boomilever may ONLY touch the Testing Wall ABOVE the 10 cm Horizontal Contact Depth Line (6.b.iii.). Structures touching below the 10 cm Horizontal Contact Depth Lines shall be scored based on 3.h. and
 - ii. The structure must hold 15 kg to earn the Load Scored Bonus. If the structure does not hold the 15 kg, the Boomilever will be scored using the Load Supported without the Load Scored Bonus. No construction violation will be recorded for not holding the 15 kg. Structures in Tier 2 can receive the bonus if tiered for rules other than 3.h.i.

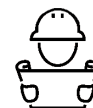


4. COMPETITION:

- a. Prior to competition, the Event Supervisor will:
 - i. Verify all Test Apparatus are available and properly set-up per section 6.
 - ii. Verify the mass of the Loading Assembly meets requirements of 6.c.v.
 - iii. Verify that the combined mass of the Loading Assembly (6.c.) and sand (6.d.) is at least 15,100 g. but no more than 15,200 g.
- b. Check-in:
 - i. Once participants begin check-in they may NOT leave or gain any outside assistance, materials or communications (including cell phone communication) until Testing is completed.
 - ii. No alterations to the structure may be made once check-in begins.
 - iii. Event Supervisor will verify:
 - (1) Participants have proper Eye Protection
 - (2) Structure meets Construction parameters
 - (3) Teams have submitted their Estimated Load Supported to be used as a tie breaker
 - iv. Participants will place the structure on the Structure Scale so the Event Supervisor can determine the structure mass to the nearest 0.01 gram or best precision available.
- c. Testing:
 - i. Teams will have 6 minutes to set up and test their Structure on the Testing Wall with the Loading Assembly. If necessary, participants may disassemble the Loading Assembly but must reassemble in the same order as presented by the Event Supervisor (6.c.). The bucket must be mounted to allow enough clearance above the floor for the bucket to tilt or the Structure to deflect.
 - ii. The Event Supervisor will check the following before loading of sand begins:
 - (1) Conformity to the requirements 3.c., 3.d., 3.e., 3.f., 3.g., 3.h.
 - (2) Conformity to requirements 3.i.i. to determine if structure meets construction requirements for Load Score Bonus.
 - (3) The bucket is suspended to have clearance above the floor for testing. (**e.g. 2 inches**)
 - iii. Structure may not be adjusted once loading of sand has begun.
 - iv. Participants will load sand and may stabilize the Bucket from movement. Bucket may only be stabilized from movement by the tips of Bucket Stabilizing Sticks (6.e.). The Bucket may not be lifted or pushed.
 - v. Loading stops immediately when the Structure fails, the load is supported by anything other than the Structure, the Structure receives any vertical support other than the Mounting Bolt, or time expires. At the Supervisors discretion, sand may be removed from the bucket if pouring continued after the Structure fails or time expires.
 - vi. Load Supported is determined and recorded to the nearest gram. Load Supported is the combined mass of the Loading Assembly (6.c.) and Sand (6.d.) in the Bucket, with any parts of the Structure in the Bucket removed prior to measurement.
 - vii. The Minimum Load Supported is the mass of the Loading Assembly (6.c.). If a Structure cannot hold the Loading Assembly (6.c.), the team will be scored as participation only.
 - viii. The Maximum Load Supported is 15,000 g.
 - ix. Test Data Review. The Event Supervisor will review with the team the data recorded on their score sheet. Once data is acknowledged by the team, the Testing is complete.
 - x. Teams who wish to file an appeal must leave their structure with the Event Supervisor.

5. SCORING:

- a. High Score Wins. $\text{Score} = \text{Load Scored (5.c.)} / \text{Structure Mass (4.b.iv.)}$.
- b. Load Scored Bonus: meeting the Load Scored BONUS construction parameters (3.i.) Bonus = 7,500 g.
- c. Load Scored = Load Supported (4.c.vi.) + Load Scored Bonus (5.b.).
- d. Structures will be placed in one of 3 categories.
 - i. Tier 1: meeting all construction and competition requirements.
 - ii. Tier 2: NOT meeting any one or more of the construction and/or competition requirements.
 - iii. Unable to be loaded, not supporting the minimum load (4.c.vii.), no eye protection, structure does not fit on test stand or for safety concerns. Scored for participation only.
- e. Tiebreakers
 - i. Lowest Structure mass.
 - ii. Estimated Load Supported that is the closest to actual load supported.



6. TEST APPARATUS:

- a. In-Person tournaments: The Event Supervisor will provide all Test Apparatus. At the Event Supervisor's discretion, more than one Test Apparatus may be used. Satellite Tournaments: the team coach or designee will supply the Test Apparatus.
- b. The Testing Wall
 - i. Solid and rigid surface at least 40.0 cm wide x 30.0 cm high. Constructed of $\frac{3}{4}$ " grade plywood or other suitable material, with a smooth, hard, low friction surface that does not bend when loaded. The Testing Wall must be adjusted to be plumb vertical by the Event Supervisor utilizing a 9" torpedo level or similar method.
 - ii. One Mounting Bolt hole no larger than 0.266-inch diameter must be drilled through the wall located approximately 5.0 cm below the top of the Testing Wall and halfway between the sides of the wall. The horizontal and vertical centerlines of the Mounting Bolt hole must be marked on the face of the Testing Wall.
 - iii. Horizontal Contact Depth Lines must be clearly visible on the Testing Wall. They must be drawn at 10 cm, 15 cm, and 20 cm and below the center of the Mounting Bolt hole.
 - iv. Two vertical Contact Width Lines must be clearly visible on the Testing Wall. They will be drawn 4.0 cm to the right and left side of the center of the Mounting Bolt hole.

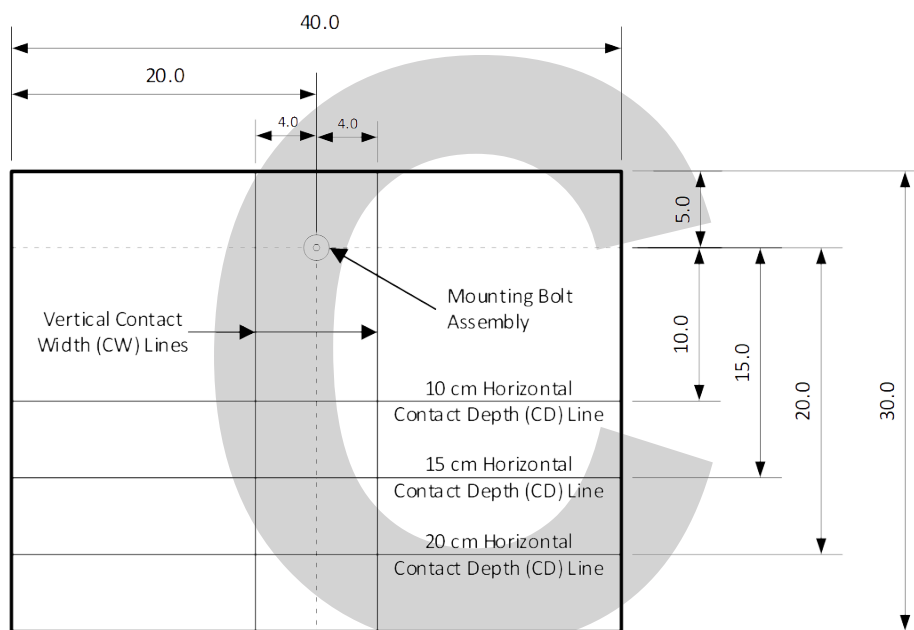


Figure 2 – Testing Wall (not to scale, dimensions in cm)



v. The Mounting Bolt Assembly

(1) The Mounting Bolt Assembly will consist of:

- (a) Connector Bolt, $\frac{1}{4}$ -20 x 2.36 inches (60 mm) long, with a 0.66-inch (17 mm) diameter head and 0.78 inch (20 mm) unthreaded section, 1 required. (Figure 3)

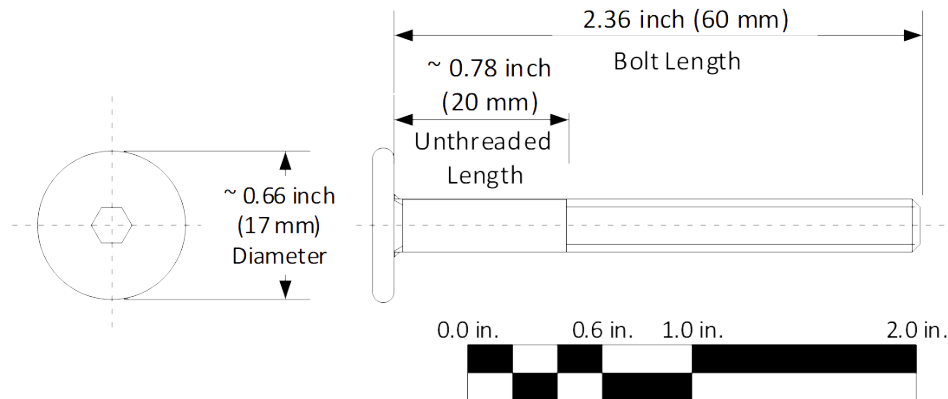


Figure 3 - Connector Bolt, $\frac{1}{4}$ -20 x 2.36 inches (60 mm) long

- (b) Hex Nut, American National Standard, $\frac{1}{4}$ - 20, 0.23-inch-thick maximum, 2 required.
 (c) Flat Washer, USS $\frac{1}{4}$ inch, approx. $\frac{3}{4}$ inch outside diameter x 0.09-inch-thick maximum, 2 required.
- (2) The Mounting Bolt Assembly will be secured in place on the Testing Wall by the Event Supervisor as follows:
- (a) The Connector Bolt will have a hex nut and flat washer on the front side of the Testing Wall and a hex nut and flat washer on the back side of the Testing Wall.
 (b) Connector Bolt is installed to allow 3.0 cm $\pm$ 0.1 cm (1.18 inch $\pm$ 0.03 inch) clearance between the closest face of the Connector Bolt to the Testing Wall and the Testing Wall face (Figure 4).
 (c) The nuts must be tightened firmly to hold the bolt in place during the competition.
 (d) Teams may not adjust or disassemble the Mounting Bolt Assembly.

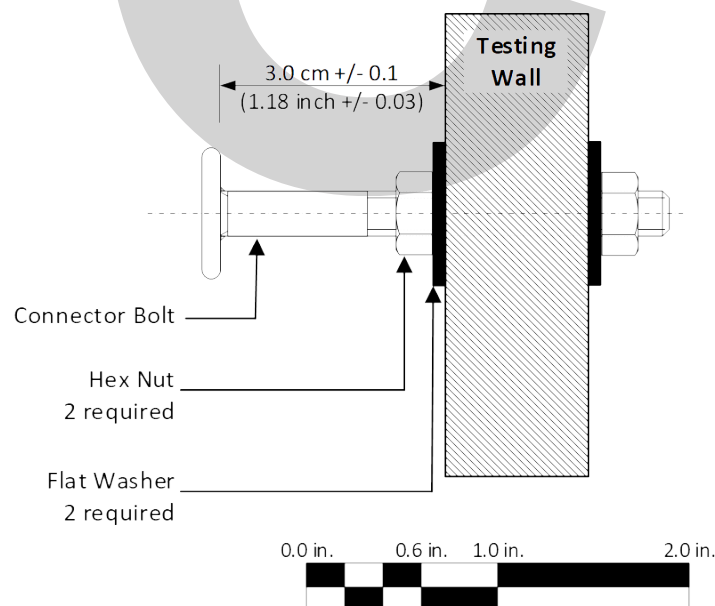


Figure 4 – Mounting Bolt Assembly in Testing Wall



- c. The Loading Assembly will consist of:
- A square Loading Block measuring 5 cm x 5 cm x approximately 2 cm high with a hole no larger than 0.266-inch diameter drilled in the center of the 5 cm x 5 cm faces for a $\frac{1}{4}$ " threaded eyebolt.
 - A $\frac{1}{4}$ inch threaded eyebolt (1-inch nominal eye outside diameter) with an overall length no longer than 3 inches, and a $\frac{1}{4}$ inch wing nut. The Loading Block must be mounted on the eye bolt and be trapped between the "eye" of the eye bolt and the wing nut.

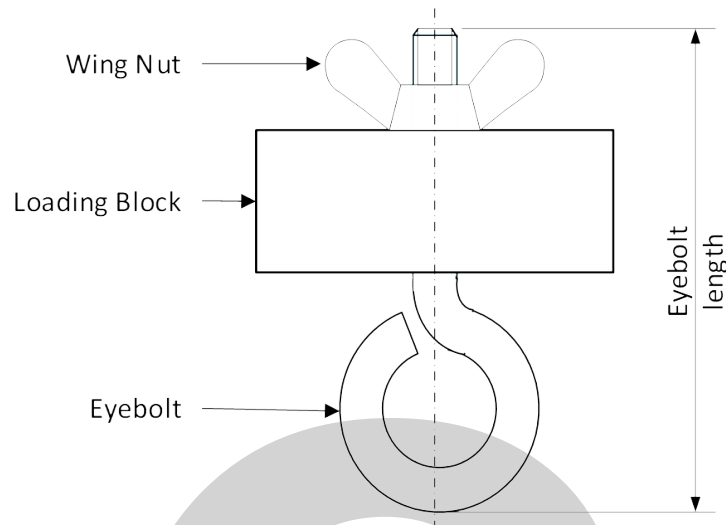


Figure 5 – Loading Block with eyebolt and wing nut (not to scale)

- A chain and S-hook that are suspended from the eyebolt on the Loading Block.
- A five-gallon plastic bucket with handle and hook to be suspended from the chain.
- The total combined mass of the Loading Assembly may not exceed 1,500 g.

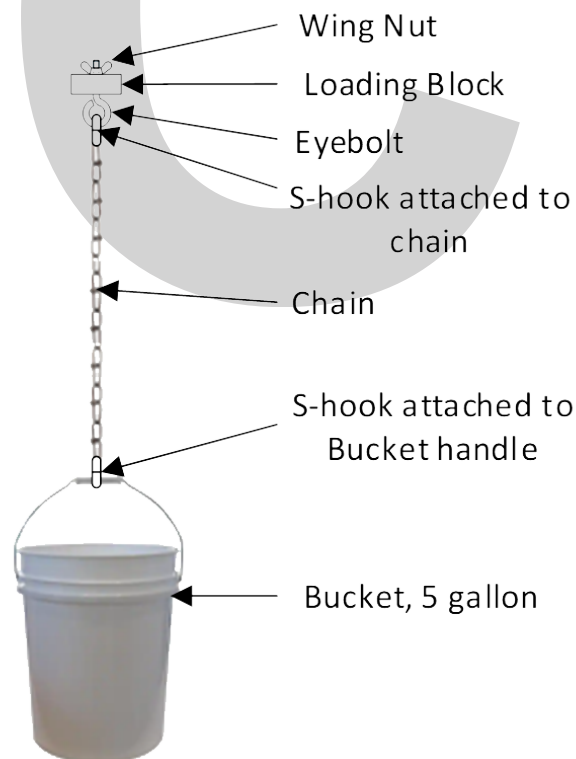
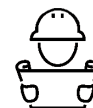


Figure 6 – Loading Assembly (not to scale)



- d. Sand: Load will be applied using Sand.
- e. Two (2) Bucket Stabilizing Sticks each made from a piece of $\frac{1}{2}$ " dowel approximately 18 inches long with a spring-type doorstop screwed into one end.

$\frac{1}{2}$ " dowel approximately 18 inches long

spring-type doorstop screwed into one end.

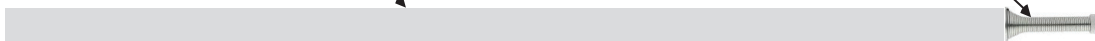


Figure 7 – Stabilizing Stick (not to scale)

- f. Structure scale: A digital scale is recommended. The scale is recommended to have a minimum range of 0 to 100 grams with a recommended resolution of 0.01 gram and minimum resolution of 0.1 grams.
 - g. Sand scale and load verification: A digital scale is recommended. The scale is recommended to have a range of 0 to 25,000 grams and a minimum range of 0 to 16,000 grams. The scale is recommended to have a resolution of 1 gram and a minimum resolution of 10 grams.
 - h. A digital countdown timer is recommended. Must be capable of being set to 6 minutes. Minimum resolution: 1 second
7. **DEFINITIONS:**
- a. *Wood* is defined as the hard, fibrous substance making up the greater part of the stems, branches, trunks, and roots of trees beneath the bark. Wood does NOT include bark, particleboard, wood composites, bamboo or grasses, paper, commercially laminated wood (i.e., plywood), or pieces formed of sawdust, wood shavings, and adhesive. Wood may never be painted, soaked, or coated in glue, chemically modified, color enhanced, or have tape/preprinted/paper labels affixed. Ink barcodes or markings from the construction process may be left on the wood.
 - b. *Wood Size*: There are no limits on the cross-sectional sizes of individual pieces of wood.
 - c. *Lamination*: multiple layers of wood may be bonded together using adhesives by the team without restriction. Commercially laminated wood is not allowed.
 - d. *Adhesive* is a substance used to join two or more materials together and may be used only for this purpose. Any commercially available adhesive may be used (e.g., glue, cement, cyanoacrylate, epoxy, hot melt, polyurethane, and super glue). Adhesive tapes are not allowed.
 - e. *Connector Bolt* can be found at several stores including Ace Hardware, Fleet Farm, Home Depot, Lowes, Menards, and Amazon. The bolt is sold under the following brand names: Hillman, Everbilt, and Midwest Fastener. The Connector Bolt is typically in the specialty fasteners section in a drawer labeled "Furniture Parts". The bolt comes in at least 4 finishes: Black, Brass, Bronze and Nickel. The black finish being preferred, but any of the listed finishes may be used.
 - f. *Sand* is defined as a clean, dry, free-flowing material of a similar density and flowability characteristics to play sand or steel shot.

This event is supported by the Cleveland-Cliffs Foundation



1. **DESCRIPTION:** Participants will demonstrate their knowledge of plant life and general botany principles.

ATEAM OF UPTO: 2

APPROXIMATE TIME: 50 minutes

CALCULATOR: Class II

2. **EVENT PARAMETERS:**

- a. Each team may bring writing utensils.
- b. Each team may bring two Class II calculators.
- c. Each team may bring one (1) 8.5" x 11" sheet of paper that may contain information on both sides in any form and from any source. The sheet of paper may be laminated or placed in a sheet protector to increase durability. Affixed labels, as well as multiple sheets of paper (whether in a single sheet protector or not) are prohibited.
- d. Event Supervisors will provide live/preserved specimens, pictures, tables, graphs of data, microscopes, slides, and any other required equipment for the event. If used, toxic/irritating plants or specimens in liquid (e.g., Algae, protists) must be in closed, non-breakable containers.

3. **THE COMPETITION:**

- a. This event may be run as a paper pencil test or as a station test.
- b. Participants will be expected to understand the structure of plant cells, roots, stems, leaves, spore forming bodies and flowers, aspects of plant growth and differentiation, and the transport and storage of gases, water, and nutrition throughout the plant body.
- c. Participants should also have a broad knowledge of the major divisions between groups of plants (i.e., algae vs. multicellular plants, monocot vs. dicot, embryophytes vs. cryptogams, woody vs. herbaceous plants).
- d. In addition to the above listed topics, participants should know:
 - i. The history of botany
 - ii. Basic plant genetics and reproduction
 - iii. Photosynthesis
 - iv. Differences between the major taxonomic groups of plants
 - v. Paleo-botany and plant evolution
 - vi. The role of plants in global energy and nutrient cycles
 - vii. Use of plant materials by animals and humans
 - viii. Competition in the plant community
 - ix. Genetically Modified Organisms (GMOs)
 - x. Production of foodstuffs and plant products
 - xi. Plant diseases; including nutrient deficiencies and infections
 - xii. Principles of horticulture and aquaculture
 - xiii. Plant biochemistry
 - xiv. The roles of plants in medicine and environmental management
 - xv. Importance of plant diversity

4. **SCORING:**

- a. High Score wins.
- b. Preselected questions will be used to break ties.



1. **DESCRIPTION:** Teams will complete one or more tasks and answer a series of questions involving the science processes of chemistry focused in the areas of kinetics and **gases**.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 50 minutes

EYE PROTECTION: C

CALCULATOR: Class III

2. **EVENT PARAMETERS:**

- a. Each participant must bring a writing utensil.
- b. Each participant may bring a Class III calculator.
- c. Each participant may bring one 8.5" x 11" sheet of paper, which may be in a sheet protector sealed by tape or laminated, with information on both sides in any form and from any source.
- d. Teams should bring any or all of the items listed on the Division C Chemistry Events Lab Equipment List, posted on soinc.org. Teams not bringing these items will be at a disadvantage, as they are not provided.
- e. Participants must bring safety equipment and must wear goggles, an apron or a lab coat and have skin covered from the neck down to the wrist and toes. Gloves are optional, but if the host requires a specific type, they will notify teams. Pants should be loose fitting; if the host has more specific guidelines, they will notify teams in advance of the tournament. Shoulder length or longer hair must be tied back. Participants removing safety clothing/ goggles or unsafely handling materials, or equipment will be penalized, up to and including possible disqualification.
- f. Supervisors will provide any required reagents, additional glassware, and/or references that are needed for the tasks (e.g., Periodic Table, table of standard reduction potentials, any constants needed).

3. **THE COMPETITION:**

- a. The competition will consist of a series of tasks focused in the areas of **gases** and kinetics. These tasks could include hands-on activities, questions on listed topics, interpretation of data (e.g., graphs, diagrams, tables), or observation of an established and running experiment. At least two activities, one on each topic are required.
- b. Teams may be asked to collect data using a probe ware set-up demonstrated by the Supervisor(s). Following a demonstration of the sensors/probes, participants may be given data sets to interpret.
- c. Nomenclature, formula writing, & stoichiometry (mole conversions & percentage yield) are essential tools of chemistry & may be included in the event.
- d. Participants are expected to know the symbols and charges of the following ions at the different competition levels:
 - i. Regional: monatomic cations and anions of main group elements, acetate, ammonium, bicarbonate, carbonate, cyanide, hydroxide, nitrate, phosphate, sulfate.
 - ii. **State and National: chromate, dichromate, permanganate, peroxide, thiocyanate, common transition metal ions ($\text{Cu}^+/\text{Cu}^{2+}$, $\text{Fe}^{2+}/\text{Fe}^{3+}$, Ag^+ , Zn^{2+} , etc.), polyatomic ions involving -ite and -ate suffixes and hypo- and per- prefixes. With a periodic table, participants should be able to obtain charges for monatomic ions (e.g., Na^+ , S^{2-}).**
- e. **Gases:** Participants will complete experimental tasks and answer questions related to the physical properties of gases, effect of greenhouse gases and ozone depletion on our climate, behavior of gases described by the following: Avogadro's Law, Boyle's Law, Charles' Law, Dalton's Law, Gay-Lussac's Law, Graham's Law, and the Ideal Gas Law. Students may be expected to complete labs/activities such as: Determine the: density of a gas, partial pressure of a gas, molar mass of a gas, relative rates of diffusion. Examine the relationship between: Pressure and volume, Pressure and temperature, Temperature and volume
- f. Participants will demonstrate an understanding of the principles of kinetics by:
 - i. Measuring reaction rates and identifying how and why reaction conditions (temperature, concentration, particle size, and catalysts) affect reaction rates.
 - ii. **At the State and National Levels:** teams will be asked to determine **differential and integrated** rate laws from actual experimentation or data provided, and teams should also be able to determine rate constants with correct units. Teams may also be asked to use experimentally-determined rate laws to validate proposed reaction mechanisms **and determine the activation energy of a reaction.**



4. **SCORING:**

- High Score wins.
- Points will be divided evenly between kinetics and **gases**.
- Time may be limited at each task but will not be used as a tiebreaker or for scoring.
- Preselected questions will be used to break ties.
- A penalty of up to 10% may be given if the area is not cleaned up as instructed.

C



1. **DESCRIPTION:** Participants will complete **hands-on** tasks and answer questions about electricity and electronics.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 50 minutes

CALCULATOR: Class III

2. **EVENT PARAMETERS:**

- a. Each team may bring writing utensils.
- b. Each team may bring two Class III calculators for use during any part of the event.
- c. Each team may bring a collection of notes and resources, written/printed on paper, of any size containing information in any form and from any source. Binders, **books**, notebooks, folders, sheet protectors, lamination, tabs, and labels are permitted. Participants are responsible for organizing and containing their notes efficiently. They may separate or remove the pages from containers for use during any part of the event.
- d. Event Supervisors must provide all supplies and measurement devices required for the hands-on tasks.
- e. Participants may bring their own basic multimeters for use in place of provided ones at the discretion of the Event Supervisor.

3. **THE COMPETITION:**

Part I: Written Test

- a. The written test will assess the team's knowledge of electricity and **electronics**.
- b. Unless otherwise requested, answers must be in metric units with appropriate significant figures.
- c. The test will consist of at least one question from each of the following areas only:
 - i. Historical perspective of the electricity discoveries made by Ampere, Coulomb, Kirchhoff, Volta, Ohm, **Hertz, and Boole**
 - ii. Properties of electric charges/fields, sources/hazards of static electricity, and Coulomb's Law
 - iii. Characteristics, sources, uses, and hazards of direct current (DC) and alternating current (AC)
 - iv. Concepts, **calculations**, and units of current, voltage, resistance, power, and energy
 - v. Calculations, constructions, and configurations of **DC circuits** and individual components, **including series and parallel circuits, open and short circuits, Ohm's Law**, and circuit schematics
 - vi. Fundamental electrical characteristics of silicon PN junctions and **uses of single-junction devices** (e.g., diodes, LEDs)
 - vii. Circuit analysis using Kirchhoff's Voltage & Current Laws
 - viii. **Digital logic gates, truth tables, universal gate sets, boolean expressions, De Morgan's Law, and combinational logic circuits** (e.g., multiplexers, adders)
 - ix. **States/Nationals only: Clock signals, timing diagrams, propagation delay, rise and fall times, race conditions, operation and applications of flip-flops**
 - x. **States/National only: DC biasing and gain of PNP/NPN bipolar junction transistors and the implementation of up to 2-input logic gates with transistor-level circuits**
 - xi. **States/Nationals only: Ideal operating principles and applications of operational amplifiers**
- d. Topics not included in the competition are: semiconductors (beyond those listed above), AC circuit theory, frequency analysis, **capacitance**, inductance, non-linear devices (**beyond those listed above**), 3 phase power, **Norton/Thevenin equivalents**, and oscilloscopes.
- e. **Questions will be limited to the following mathematical content:**
 - i. **Functions, algebraic manipulations, and systems of equations**
 - ii. **Geometry and trigonometry required for resolving vectors into component parts**
 - iii. **Representation of whole numbers in binary and the addition of binary numbers, not including negative numbers, floating point representations, or any arithmetic operations other than addition**
 - iv. **Logical operations as boolean expressions, including De Morgan's Law and truth tables**
 - v. **No questions may require the use of complex numbers, matrix algebra, or calculus**

Part II: Hands-On Tasks

- a. The hands-on portion will consist of at least one task for the teams to complete.
- b. Participants must be familiar with the operation of breadboards and multimeters and how to use them. Participants may ask event supervisors for details of the internal wiring of any breadboards used for the tasks.



- c. The hands-on tasks, or stations, may include but are not limited to:
 - i. Determine the value of a mystery resistor in a circuit using only voltage measurements.
 - ii. Calculate the power supplied to a circuit.
 - iii. Given some wires, batteries, resistors, and 2 LEDs, hook them up so the LEDs are equally bright.
 - iv. Given a USB charger, read the label and provide details of the various output power levels it can provide and calculate how long it would take to charge a specific battery.
 - v. **Construct a logic circuit to mimic a given truth table.**
 - vi. **Troubleshoot a faulty circuit.**
 - vii. **Wire a 3-way switch using SPDT switches.**
 - viii. **Create a specified equivalent resistance from given individual resistors.**
 - d. **Participants must not be exposed to hazardous conditions including soldering, high power circuits, and AC circuits.**
4. **SCORING:**
- a. High Score wins.
 - b. Points will be awarded for correct answers, measurements, calculations, and data analysis.
 - c. The written portion of the competition will account for 50-75% of each team's score. The hands-on portion of the competition will account for the remaining 25-50% of each team's score.
 - d. Preselected questions will be used as tiebreakers.

This event is supported by the Kinder Morgan Foundation





1. **DESCRIPTION:** Participants will cryptanalyze and decode encrypted messages using cryptanalysis techniques for historical and modern advanced ciphers.

A TEAM OF UP TO: 3

APPROXIMATE TIME: 50 minutes

2. **EVENT PARAMETERS:**

- a. Teams must bring only writing utensils.
- b. No resource materials or other tools, except those provided by the Event Supervisor, may be used.
- c. The exam packet must be printed single-sided to facilitate separation and writing answers by individual team members. Using a separate answer sheet is not recommended.
- d. The exam packet will include a copy for each team member of a resource sheet with the Morse Code Table, English/Spanish letter frequencies, Porta Table, and Baconian mappings as needed for the questions on the exam. The resource sheets may be printed double-sided.

3. **THE COMPETITION:**

- a. This event consists of participants decrypting ciphertext on a written or computer-based exam.
- b. Teams will begin the event simultaneously at the indication of the Event Supervisor.
- c. Teams must not open the exam packet nor write anything prior to the “start” signal, nor may they write anything after the “stop” signal.
- d. Participants are allowed to separate the pages of the test to be free to answer the questions in any order, working individually or in groups, attempting whichever of the questions seem right for them.
- e. The code types that may be used at Division C Regional Tournaments are as follows:
 - i. Monoalphabetic substitution using K1, K2, K3, or random alphabets as defined by the American Cryptogram Association (ACA) with or without a hint
 - (1) Aristocrats - ciphertext with spaces included.
 - (2) Patristocrats - ciphertext with spaces removed and letters grouped in sets of 5 encoded using a K1 or K2 alphabet.
 - ii. For aristocrats encoded using a K1, K2 or K3 alphabet, the answer requested can be the keyword or key phrase used to construct the alphabet instead of the deciphered text.
 - iii. The Baconian Cipher - decrypting ciphertext encoded with the a and b values represented as one or more letters, glyphs, symbols, or character rendering variations (e.g., bold, underline, italic). Word Baconian Ciphers will include a “crib” of at least 4 letters.
 - iv. Xenocrypt (maximum of 2) - messages in Spanish encoded with a K1 or K2 English keyword alphabet.
 - v. Cryptanalysis of the Fractionated Morse Cipher - decrypting Morse code ciphertext encoded as letters and spaces with a “crib” of at least 4 plaintext characters.
 - vi. Cryptarithms - determining mapping values to letters in base 10 (decimal) mathematical equations and decoding a word or phrase using that mapping.
 - vii. The Porta Cipher - Decrypting ciphertext given a key.
 - viii. Cryptanalysis of the Complete Columnar Transposition Cipher - Decrypting ciphertext encoded in 9 columns or less given a “crib” which is no shorter than one less than the number of columns used.
 - ix. The Nihilist Cipher - Decrypting ciphertext given the keys.
 - x. The 5x5 Checkerboard Cipher – Decrypting ciphertext given the Polybius key. **For the Checkerboard cipher, all letters used for the row and column keywords must be present in the ciphertext.**
 - xi. **Cryptanalysis of the Homophonic Cipher - Decrypting ciphertext given a “crib” which indicates at least 2 letters of the keyword. For the Homophonic cipher, the keyword must have four unique letters.**
- f. The code types that may be used on the exam at State and National competitions are as follows:
 - i. All Regional code types.
 - ii. Xenocrypt - at the State and National levels, at least two cryptograms and no more than three will be in Spanish, encoded using a K1 or K2 English keyword alphabet.
 - iii. Cryptanalysis of the Porta Cipher with a “crib” of at least 3 plaintext characters.
 - iv. Cryptanalysis of the Nihilist Cipher with a “crib” that is no shorter than double than the length of the keyword used.
 - v. Cryptanalysis of the Complete Columnar Transposition Cipher - Decrypting ciphertext encoded in 11 columns or less given a “crib” which is no shorter than three less than the number of columns used.



- vi. Cryptanalysis of the 5x5 Checkerboard Cipher encoded with two 5-letter keywords and a Polybius key given a “crib” of at least 5 characters.
 - vii. **The Homophonic Cipher - Decrypting ciphertext given at least 2 letters of the keyword.**
 - g. For Aristocrats, Patristocrats, and Xenocrypts, no letter can ever decrypt to itself.
 - h. The first question of the exam will be timed.
 - i. The first question will be the decoding of an Aristocrat as defined by 3.e.i.(1).
 - ii. A team member should signal when his or her team has broken the cryptogram.
 - iii. Before the exam begins, the Event Supervisor will announce the nature of the signal that must be used (e.g., shouting “time”, or quietly raising hand).
 - iv. The time in seconds, to the precision of the device used, to solve the cryptogram will be recorded by the Event Supervisor or designee.
 - v. If a team gets the timed question wrong, they may attempt to answer the question repeatedly without penalty. The timing bonus will be calculated from the start of the event until the question is successfully answered by the team with two or fewer errors, or until 10 minutes have elapsed. After 10 minutes, the timed question can still be answered but the timing bonus is zero.
 - i. Up to three questions which are not Aristocrats, Patristocrats, or Xenocrypts will be marked on the exam as special bonus questions. **At least one special bonus question will use the Homophonic Cipher (3.e.xi. or 3.f.vii.).**
 - j. For Cryptanalysis problems providing a “crib” (3.e.iii., 3.e.v., 3.e.xi., 3.f.iii., 3.f.iv., 3.f.vi.) with the exception of the Complete Columnar Cipher (3.e.viii., 3.f.v.), the placement of the “crib” on the ciphertext will be clearly identified.
4. **SCORING:**
- a. The high score wins.
 - b. Final Score = Exam Score + Timing Bonus + Special Bonus.
 - c. The scores for each question will be added together to determine the Exam Score.
 - d. Unless otherwise specified, the final points will be determined based on the number of errors found in the decoded plaintext as is appropriate to the question.
 - i. Two or fewer errors will be scored as correct and result in full credit.
 - ii. Each additional error results in a penalty of 100 points but the penalty will not exceed the value of the question. For example, a 400-point question with 5 errors earns 100 points [400 - 3(100)] whereas the same 400-point question with 7 errors would earn 0 points, not -100 points.
 - e. For answers involving the keyword or key phrase for a K1, K2 or K3 alphabet (3.e.ii.) the final points will be determined based on the number of errors found in the keyword or key phrase. For the Cryptarithm (3.e.vi.), the final points will be determined based on the number of errors found in the decoded word or phrase.
 - i. Zero errors are required for full credit.
 - ii. Each error results in a penalty of 100 points but the penalty will not exceed the value of the question. For example, a 500-point question with eight (8) errors would earn 0 points, not -300 points.
 - f. A Timing Bonus can be earned based on the number of seconds it takes a team to correctly decode the first question. The timing bonus is equal to $2 \times (600 - \text{number of seconds})$. For example, 6 minutes = $2 \times (600 - 360) = 480$ points.
 - g. A Special Bonus can be earned by solving any of the questions marked as special bonus questions with no penalty points. The bonus will be awarded as follows: One solved = 150 points, Two solved = 400 points, All three solved = 750 points.
 - h. Scoring example: Team A earns 3600 points on the exam and solved the timed question in 435 seconds and solved one Special Bonus question

Exam Score	=	3600 points
+ Timing Bonus $2(600-435)$	=	330 points
+ Special Bonus (One=150)	=	150 points
Final Score		4080 points
 - i. Tiebreakers: For teams that are tied, select questions predetermined by the Event Supervisor will be used to break the tie using the following criteria in this order: score, degree of correctness and number attempted.



1. **DESCRIPTION:** Teams will answer questions, solve problems, and analyze data pertaining to classic, evolutionary, and molecular genetics.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 50 minutes

CALCULATOR: Class II

2. **EVENT PARAMETERS:**

- a. Each team may bring writing utensils.
- b. Each team may bring two Class II calculators.
- c. Each team may bring one (1) 8.5" x 11" sheet of paper that may contain information on both sides in any form and from any source. The sheet of paper may be laminated or placed in a sheet protector to increase durability. Affixed labels, as well as multiple sheets of paper (whether in a single sheet protector or not) are prohibited.

3. **THE COMPETITION:**

- a. This Event will be administered as a written test which may be arranged in stations. **Questions will emphasize process skills such as quantitative reasoning, making calculations, analyzing and interpreting experimental results, and drawing evidence-based conclusions.** The Event will cover topics listed below without overemphasis on any one particular topic. The list of topics and subtopics should be considered exhaustive.
- b. For each of the following topics, participants will be expected to use quantitative reasoning and computational skills, analyze and interpret experimental results, and draw evidence-based conclusions.
 - i. Mendelian Genetics
 - (1) Describe Mendel's Laws of Inheritance and their implications in heredity. Describe inheritance patterns which violate these laws (i.e. linkage, incomplete and codominance, complementation).
 - (2) Use provided information to construct Punnett Squares of mono-, di- and trihybrid crosses.
 - (3) Predict genotypes and phenotypes of offspring and compute their likelihood based on Punnett Squares and experimental data using probability rules.
 - (4) Evaluate pedigrees to predict modes of inheritance (i.e. dominant vs recessive traits, autosomal vs sex-linked traits) and construct pedigrees based on data from a case study.
 - (5) Understand epistasis and its implications in predicting genetic outcomes.
 - (6) State and Nationals only: Map two (State) or three (National) genes on a chromosome based on recombination frequency data.
 - ii. Mitosis and Meiosis
 - (1) Understand, compare, and contrast the major stages and key structures in mitosis and meiosis.
 - (2) Describe the effects of nondisjunction
 - (3) Use data from a Karyotypes to detect chromosomal abnormalities.
 - (4) State and Nationals Only: Understand somatic recombination in the context of immune cell V(D)J recombination and immunoglobulin class switching.
 - iii. Population & Evolutionary Genetics
 - (1) Understand the assumptions and calculations that underlie the Hardy-Weinberg equilibrium and the consequences of violating these assumptions. Be able to make these calculations given data.
 - (2) Understand and explain the relationships between changing allele frequencies and genetic drift, founder effects, bottlenecks, migration, and relative fitness.
 - (3) Explain how additive alleles generate continuous variation and estimate the number of genes contributing to a trait.
 - (4) Understand the role gene duplication plays in the evolution of the genome and new genes. Be able to identify homologs, orthologs, and paralogs.
 - (5) Understand how a phylogenetic tree shows evolutionary histories and interpret phylogenies of nuclear, organellar, and extrachromosomal elements. Describe the basic steps of constructing a phylogenetic tree beginning with sequence data input.
 - (6) State and Nationals only: Understand how heritability is used to explore phenotypic variation in a population. Calculate broad-sense heritability, narrow-sense heritability, and realized heritability.
 - iv. Molecular Biology of DNA
 - (1) Be able to identify and explain the components of DNA (i.e. structure of the nucleotides and backbone), directionality of DNA (5' and 3' ends), rules in the context of DNA structure and



- base pairing
- (2) Understand the stages of DNA replication, from assembly of the pre-replication complex to termination, understand in detail the processes that happen at the replication fork **for bacteria, archaea, and eukarya.**
 - (3) Understand how DNA polymerases achieve fidelity of replication, and the mechanisms behind DNA proofreading.
 - (4) Identify and understand the organization of DNA structure, such as plasmids, chromatin complexes, euchromatin and heterochromatin, and chromosomes.
 - (5) Understand how UVA and UVB, ionizing radiation, and intercalating agents cause DNA damage; explain the mechanisms of nucleotide excision repair, non-homologous end-joining, and homology-directed repair in bacteria and eukarya.
 - (6) Identify, explain, and classify DNA mutations on the DNA level (i.e. chromosomal rearrangements, insertions, deletions, and substitutions), and the potential impact of the resulting protein sequence (i.e. frameshift, silent, missense, and nonsense mutations).
 - (7) **State and Nationals only: Describe the end replication problem and the mechanism of telomere synthesis**
- v. **Eukaryotic** gene expression and regulation
- (1) Explain the flow of genetic information according to the Central Dogma of Molecular Biology
 - (2) Transcription: Understand and explain initiation, elongation, and termination stages of transcription, the mechanism of RNA polymerase.
 - (3) Understand the role and general locations of cis- and trans-regulatory elements (i.e. promoters, enhancers, silencers, riboswitches, transcription factors, epigenetics). Explain how transcription factors are used in yeast two-hybrid screening.
 - (4) **RNA maturation: Understand and explain the steps of post-transcriptional RNA processing (i.e. capping, polyadenylation, and splicing)**
 - (5) Translation: Understand and explain the initiation, elongation and termination stages of translation, the mechanism of the ribosome, the role mRNA and tRNA play in translation, and how translation is regulated.
 - (6) **State and Nationals only: Compare and contrast mobile genetic elements (i.e. retroviruses, retrotransposons, and DNA transposons). Describe their mechanism of movement, including enzymes involved.**
- vi. Technology and Techniques
- (1) Polymerase chain reaction (PCR): Describe what occurs and which molecules are involved in the denaturation, annealing, and extension steps of a PCR reaction; Explain the role of temperature in PCR; Identify the components needed for a PCR reaction; Identify applications of PCR (i.e. Sanger sequencing, error-prone PCR, reverse transcriptase-qPCR) and experimental questions which could be addressed by these methods.
 - (2) Next-gen (Illumina) vs. third-gen (Nanopore) sequencing: Describe the major platforms used for Next-Gen (Illumina) sequencing. Compare and contrast these methods in terms of how they are performed, data produced, and questions the data addresses.
 - (3) **Molecular cloning: Describe the major steps in cloning a gene of interest to be expressed in a specific Organism. Understand the utility of plasmids, restriction enzymes, PCR, Gibson assembly, Golden Gate cloning, stable cell lines, transient transfection of mammalian cells in the context of molecular cloning.**
 - (4) **Knockout/knockdowns: Understand how knockout, knockdown, and complementation experiments are used to determine gene function for sufficiency and necessity. Compare and contrast knockdowns and knockouts and explain why one may be used over the other. State and Nationals only: Describe techniques used to generate knockdown/knockouts (i.e. RNAi, homologous recombination including Cre-Lox, CRISPR).**
 - (5) State and Nationals only: Describe the basic procedure, applications, and limitations of the following techniques: ChIP-seq, Hi-C, RNA-Seq (single-cell vs bulk).

4. **SCORING:**

- a. High Score wins.
- b. Preselected questions will be used as tiebreakers.
- c. Points will be awarded for quality and accuracy of answers, quality of supporting reasoning, and the use of proper scientific methods.



1. **DESCRIPTION:** Participants will use their investigative skills in the scientific study of disease, injury, health, and disability in populations or groups of people.

ATEAM OF UPTO: 2

APPROXIMATE TIME: 50 minutes

CALCULATOR: Class II

2. **EVENT PARAMETERS:**

- a. Each team may bring writing utensils.
- b. Each team may bring two Class II calculators.
- c. Each team may bring one (1) 8.5" x 11" sheet of paper that may contain information on both sides in any form and from any source. The sheet of paper may be laminated or placed in a sheet protector to increase durability. Affixed labels, as well as multiple sheets of paper (whether in a single sheet protector or not) are prohibited.

3. **THE COMPETITION:**

- a. This event addresses three topics related to disease, injury, health, and disability in populations or groups of people. Each part should count approximately equally towards a team's final score. Questions should be process-oriented and involve skills in evaluation and interpretation. Matching pathogens with specific diseases (i.e. – What causes X disease?) or knowledge of signs, symptoms or epidemiologic characteristics such as incubation or latency periods or infectious dose is not part of this event. However, it is appropriate to provide this information as background information and expect competitors to be able to use it.

- b. The topics for this event are as follows:

- i. Background & Surveillance

- (1) Understand the Clinical Approach (health of individuals) vs Public Health Approach (health of populations).
- (2) Understand the history and development of epidemiology.
- (3) Understand the roles of epidemiology in public health and the steps in solving health problems.
- (4) Understand the Natural History and Spectrum of Disease. Understand in broad terms the impact of infectious (bacterial, viral, fungal, protist and prion diseases) and noninfectious causes of disease (such as accidents, exposures, and toxicities).
- (5) Understand the basic epidemiological and public health terms found in the glossary of CDC's Principles of Epidemiology in Public Health Practice (e.g., outbreak, epidemic, pandemic, surveillance, risk, vector, etc.).
- (6) Understand the role of Surveillance in identifying health problems, the 5-Step Process for Surveillance, the types of surveillance and the attributes of a surveillance system.

- ii. Outbreak Investigation

- (1) Analyze actual or hypothetical outbreaks given in case scenarios.
- (2) Understand Experimental and Observational studies and the Types of Epidemiological Studies – (e.g., case control, cohort, ecological, cross-sectional). Know the advantages and disadvantages of each. Recognize various fundamental study designs and identify which is appropriate to use in analysis of presented outbreak scenarios.
- (3) Identify the Steps in an Outbreak Investigation and how they guide hypothesis generation.
- (4) Identify the problem using person, place, and time triad to formulate case definitions.
- (5) Interpret epi curves, line listings, cluster maps, subdivided tables, PFGE gels, SNP mapping and the PulseNet concept.
- (6) Understand the agent, host, environment triad and chain of transmission.
- (7) Evaluate data by calculating and comparing simple rates and proportions such as attack rate, relative risk, odds-ratio, and explain their meaning. Determine whether presented data support hypotheses of disease within scenarios, and revise hypotheses as appropriate.
- (8) Apply the Bradford Hill Criteria for Verifying the Cause of presented outbreaks. Compare the accuracy of Bradford Hill criteria, Koch's and Evan's postulates, and newer causality models such as Directed Acyclic graphs, Sufficient/component cause models, and GRADE methods.



- (9) Recognize factors such as study design/biases, errors, and confounding that influence results. Be able to propose appropriate control or comparison groups and data collection methods, and recognize limitations. Be able to interpret and use confidence intervals for measures of association. Competitors need not be able to calculate these confidence intervals. Understand and use methods such as stratification and adjusted rates. Know the experimental and observational methods used to calculate vaccine effectiveness and efficacy and be able to use them.
 - (10) Understand the concept of herd immunity. Be able to calculate and interpret herd immunity threshold, basic and effective reproductive numbers.
 - (11) Nationals Only: Suggest types of control & prevention measures for outbreaks and other public health problems.
- iii. Patterns, Control, and Prevention
- (1) Identify patterns and trends of epidemiologic data in charts, tables and graphs.
 - (2) Using given data, calculate disease risk and frequency ratio, proportion, incidence proportion (attack rate), incidence rate, prevalence death rate and mortality rate.
 - (3) Understand the Strategies of Disease Control as they apply to given disease scenarios.
 - (4) Understand Strategies for Prevention, including the Scope and Levels of Prevention.
 - (5) Propose a reasonable set of prevention strategies for public health problems within the scenarios provided, once the cause has been determined by your analyses.
 - (6) Nationals Only: Identify the strengths and weaknesses of a set of proposed prevention strategies and analyze pre-and post-intervention data, to determine effectiveness of presented strategies.

4. **SCORING:**

- a. High Score wins.
- b. Preselected questions will be used as tiebreakers.
- c. Points will be assigned to the various questions and problems. Both the nature of the questions and scoring will emphasize an understanding that is broad and basic rather than detailed and advanced.
- d. Depending on the problem, scoring may be based on a combination of answers, including graphs/charts, explanations, analysis, calculations, and closed-ended responses to specific questions. Critical reasoning skills and data interpretation with hypothesis generation will be evaluated.
- e. Points will be awarded for both quality and accuracy of answers, the quality of supporting reasoning, and the use of proper scientific methods.

In partnership with the Centers for Disease Control (CDC)



1. **DESCRIPTION:** Teams will complete tasks related to the properties and processes of Earth's fresh waters.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 50 minutes

CALCULATOR: Class II

2. **EVENT PARAMETERS:**

- a. Each team may bring writing utensils.
- b. Each team may bring two Class II calculators.
- c. Each team may bring a binder of any size containing information in any form and from any source. Sheet protectors, lamination, tabs, and labels are permitted. If the event features a rotation through a series of laboratory stations where the participants interact with samples, specimens, or displays, no material may be removed from the binder throughout the event.

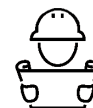
3. **THE COMPETITION:** Participants will be assessed through an exam and/or timed stations on the following topics:

- a. Hydrologic cycle, water budgets, and the critical zone
 - i. Components and associated fluxes of the hydrologic cycle (e.g., precipitation, evaporation, transpiration, evapotranspiration, infiltration, runoff, condensation)
 - ii. Storage reservoirs, residence times, and water budgets
- b. Climatic controls on freshwater distribution
 - i. Precipitation and atmospheric inputs to freshwater systems (e.g., rainfall, snowfall, seasonality, intensity, spatial patterns)
 - ii. Topographic and geographic controls on precipitation and climate (e.g., orographic lift and the rain shadow effect, continentality, lake effect, elevation)
 - iii. The arid-semiarid-humid spectrum and its hydrologic and geomorphic expression (e.g., perennial vs. ephemeral streams, playas, evaporite deposits)
- c. Drainage basins and stream networks
 - i. Watersheds, drainage divides, and drainage basin characteristics
 - ii. Stream order and networks
 - iii. Drainage patterns, density, and texture
- d. Fluvial sediment processes and channel form
 - i. Channel types and planform (e.g., braided, meandering, straight, anastomosing) and calculation of sinuosity
 - ii. Sediment properties (e.g., grain size, shape, sorting, rounding)
 - iii. Sediment transport and deposition (e.g., bed load, suspended load, dissolved load) and the controls on sediment supply, competence, and capacity
 - iv. Fluvial bedforms (e.g., ripples, dunes, antidunes, bars)
- e. Fluvial geomorphology — landforms and valley processes
 - i. Longitudinal profile, gradient, and base level
 - ii. Dynamic equilibrium and graded streams
 - iii. Erosional features (e.g., knickpoints, waterfalls, v-shaped valleys)
 - iv. Erosional processes (e.g. stream capture, downcutting, abrasion, meandering, hydraulic action)
 - v. Channel and floodplain landforms (e.g., point bars, cut banks, oxbow lakes, floodplains, natural levees, deltas, alluvial fans)
- f. Streamflow and floods
 - i. Flow regimes (e.g., perennial, intermittent, ephemeral)
 - ii. Discharge, velocity, and load (water and sediment)
 - iii. Mathematical relationships describing open-channel flow
 - iv. Floods, flood frequency, and recurrence intervals
- g. Groundwater and karst
 - i. Subsurface zones (e.g., zone of aeration, zone of saturation, water table)
 - ii. Aquifers and their properties (e.g., confined, unconfined, and artesian aquifers; porosity, permeability)
 - iii. Hydraulic gradient and groundwater flow
 - iv. Recharge, discharge, and capillarity
 - v. Interactions between surface water and groundwater, including saltwater intrusion
 - vi. Karst processes, hydrology, landforms, and features (e.g., sinkholes, solution valleys, caves, disappearing streams, springs)



- h. Lakes and wetlands
 - i. Lake formation and types (e.g., tectonic/rifting, volcanic, glacial, fluvial damming)
 - ii. Lake water budgets — inflow and outflow
 - iii. Physical and chemical properties and stratification (e.g., seasonal turnover)
 - iv. Shoreline processes (e.g., waves, shoreline features)
 - v. Wetland types, formation, and hydrology (e.g., bogs, marshes, swamps, fens)
 - i. Geologic and paleohydrologic record
 - i. Sedimentary structures as indicators of water and flow conditions (e.g., ripple marks, raindrop imprints, mud cracks, cross-bedding, graded bedding)
 - ii. Paleochannels and preserved fluvial and lacustrine deposits
 - j. Human impacts and water quality
 - i. Effects of land-use change on freshwater systems (e.g., altered runoff, sedimentation, ecological change)
 - ii. Engineering structures, water extraction, and their effects (e.g., dams, levees, channelization, diversion, downcutting; wells and groundwater pumping, with effects such as aquifer depletion and land subsidence)
 - iii. Water pollution (e.g., types, sources, transport)
 - k. Measuring and monitoring freshwater
 - i. Interpreting freshwater features on topographic maps
 - ii. Mapping the water table from well data (e.g., contour maps, flow lines, flow direction)
 - iii. Stream gauging, discharge measurement, and hydrographs (e.g., stage–discharge relationships, baseflow vs. stormflow, peak discharge, lag time)
 - iv. Monitoring networks and techniques (e.g., groundwater observation wells, tracer studies, remote sensing)
4. **SCORING:**
- a. High Score wins.
 - b. Points will be awarded for the quality and accuracy of answers, the quality of supporting reasoning, and the use of proper scientific methods in responses.
 - c. Preselected questions will be used as tiebreakers.

In partnership with the National Oceanic and Atmospheric Administration (NOAA) and the North American Association for Environmental Education (NAAEE)



1. **DESCRIPTION:** Teams design, build, and test one vehicle that uses electrical energy as its sole means of propulsion to travel in a set amount of time, push a bottle past a line, and move backwards to stop close to a Target Point.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 12 minutes

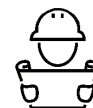
IMPOUND: Yes

2. **EVENT PARAMETERS:**

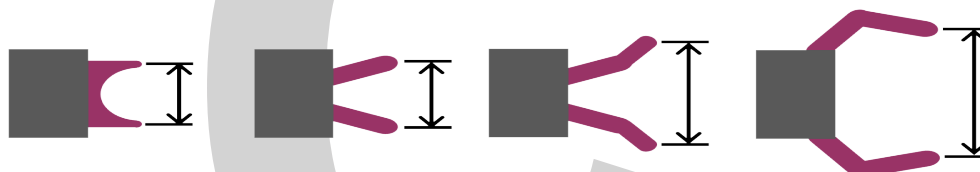
- a. Vehicle will be designed and built by the team prior to the competition.
- b. Impound: Team must impound before the first time slot.
- c. Event Time: Teams will each have 8 minutes to adjust, repair, and run their Vehicle.
- d. Track: The Event Supervisor will provide and set up the Track as listed in the Track section.

3. **CONSTRUCTION PARAMETERS:**

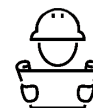
- a. Electrical Energy used by the Vehicle for any purpose must come from a maximum of eight (8) AA batteries.
 - i. No battery containing lithium or lead acid is permitted. Teams using these batteries will not be permitted to run and will receive only participation points.
 - ii. Energy storage methods which allow teams to exceed the instantaneous energy available from the batteries are not allowed. An example would be large capacitors or a spring.
 - iii. Electronic sighting, alignment, or aiming devices may have their own separate power source. Lasers are prohibited.
 - iv. All sources of energy must be in easily accessible locations for inspection by the ES.
- b. Propulsion Energy: must be electric and come from the AA batteries.
- c. Components may be purchased or made by the team members. Electronic components are allowed.
 - i. If a microprocessor is used, students may communicate with their device over WiFi, Bluetooth, or USB.
- d. **Vehicle Length: The Vehicle length including all components must not exceed 80.0 cm.**
- e. Vehicle Width: The Vehicle width must not exceed 35.0 cm at any point.
- f. **Bottle Pusher: A device to catch and push the Water Bottle must be attached to the front of the Vehicle. The Pusher must be the foremost part of the Vehicle. Only one Bottle Pusher can be impounded. The Pusher must be rigid, cannot flex, change locations on the Vehicle, have moving parts, or have the ability to change in size. The Pusher's foremost edge(s) must have rounded edge(s) approximately the diameter of #2 pencil or larger. The Pusher must be able to move bottles not perfectly round. The Bottle does not need to fully fit inside the Bottle Pusher.**
- g. Measurement Point: The Vehicle must have a Measurement Point (MP) for distance measurements at the rear of the Vehicle. **The MP must be the rear most part of the Vehicle.**
- h. Remote Control: The Vehicle must not be remotely controlled or tethered.
- i. Vehicle Parts: All parts of the Vehicle must move as a whole: no anchors, tethers, tie downs, launching ramps, or other separate pieces are allowed. The only parts allowed to contact the floor during the run are wheels/treads, and any parts already in contact with the floor in the ready-to-run configuration. Pieces falling off during the run constitutes a Construction Violation.
- j. Design Knowledge: Answer questions on design, construction, and operation of the device per the Building Policy found on www.soinc.org.
- k. **Purchased Kits or Open Source Kits are allowed for this event with the following requirements:**
 - i. **Teams must make at least two functional modifications to the kit during the build process. These modifications need to be documented on the Kit Modification Form available on soinc.org. The completed form must be impounded.**
 - ii. **A Bottle Pusher from kit sources cannot be used. Teams must design and build their own Bottle Pusher.**
 - iii. **A kit (supplier or open source) can only provide components if these components are common parts available from other sources (e.g. no custom printed circuit boards). This does not include parts that are 3D printed plastic parts.**
 - iv. **Teams violating rules 3.k.i. through 3.k.iii. will receive the Non-modification Penalty to their Run Scores.**



1. The program must be developed by at least 1 member of the 15 member team. Teams may use skeleton or blank shell programs from the Internet or another competitor from the same school as their starting point. Teams may download small code sections or examples from the Internet to develop the larger completed program for their device. Single-source programs are prohibited. Teams will receive a construction penalty for violating this rule.
4. **THE COMPETITION:**
 - a. **Impound before first time slot.**
 - i. Teams must impound one Vehicle, batteries, spare parts, alignment device (if used), **Kit Modification Form** (if used), and any papers (if used).
 - ii. Batteries must be stored separately from Vehicle and presented to ES for inspection if asked. Teams will be allowed to correct if an issue is found.
 - iii. Tools do not need to be impounded.
 - iv. Inspections will NOT take place during impound as any of the 15 member team may impound.
 - b. **After Impound** and before the first time slot, the Target Distance, Target Time, and **Bottle Line Distance** will be announced and posted. All posted values at a competition will be the same for all teams.
 - c. **Check-In**
 - i. Only the participants will enter and may bring tools as defined.
 - ii. Teams will be instructed by the ES when to retrieve their impounded items.
 - iii. Teams will present their Vehicle for inspection and measurement by the ES. Participants will be notified at this point if any construction violations are found.
 - (1) A construction violation will be awarded to all runs if more than one Vehicle was impounded.
 - (2) Teams may use their Event Time to correct any construction violations. A team's Event Time will be paused during re-evaluation by the ES.
 - iv. **ES will measure, to the nearest 0.1 cm, the widest opening of the Bottle Pusher (See diagram below). Teams without a designated Bottle Pusher will use 35.0 cm for the measurement.**



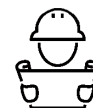
- v. Once participants start the Check-In process, they must not leave or gain any outside assistance or materials until their Event Time is completed.
- vi. Participants may not use an AC outlet power during their time slot.
- vii. **Any type of AI cannot be used by the participants or the device during the team's event time. AI tools may be used to assist with designing or programming the device prior to the day of the competition.**
- d. **Event Time**
 - i. The Event Time ends after 2 runs or the time limit has been reached. Teams may adjust their Vehicle (e.g., repair, wiring, change distance, aiming) within their Event Time.
 - (1) The time limit for the Event Time is 8 minutes.
 - (2) The ES will notify the participants when their Event Time starts.
 - (3) Vehicles in the ready-to-run configuration before the end of the Event Time will be allowed to complete a run.
 - ii. Sighting, aligning, measurement, and guiding devices, if used, must be placed and used within the defined Track area. **Device(s) cannot be secured to the floor of the track area. Alignment devices can be shared by teams from the same school.** If placed on the Vehicle, they may be removed at the team's discretion. Devices remaining on the Vehicle will be considered part of the Vehicle. All items not part of the Vehicle must be moved and placed behind the Start Point prior to starting a run. Sighting, aligning, and guiding devices may use electricity. Lasers are prohibited.
 - iii. Teams may use their own tools to verify the Track dimensions during their Event Time.
 - iv. Substances applied to the Vehicle must be approved by the ES prior to use and must not damage or leave residue on the floor, Track and/or event area. Teams may clean the Track during their Event Time, but it must remain dry. **Competitors are not allowed to use tape on the event floor.**



- v. Teams may access the Internet to program their Device. Any actions not required for programming will cause a team to be awarded a competition violation. Internet access may not be available at all competitions.
- vi. Teams must not roll or test the Vehicle on the floor of the Track on the day of the event **or prior day** without tournament permission. If permitted, only participants may be present. During competition, teams must not roll the vehicle on the floor before or after runs.
- vii. Individual Run Testing
 - (1) Participants must place the Vehicle's MP above the Start Point **within approximately 1 cm. The Vehicle's body must be placed in front of the Start Point (on the side toward the Target Point).**
 - (2) The Vehicle must be able to remain at the starting position without being touched until triggered.
 - (3) Participants will notify the ES when ready to start. At which point the ES will pause the Event Time and review the Vehicle in the Ready-To-Run configuration for any violations.
 - (a) The ES will only notify the participants of competition violations at the end of each run. Teams may ask question(s) before the run if there are specific violations. The ES will not answer generic questions like "Do I have a violation?"
 - (4) Participants will use a #2 unsharpened pencil to start the Vehicle using a vertical motion. Only the pencil may contact the Vehicle when triggered.
 - (a) If the Vehicle fails to actuate or move when triggered, then a run has not occurred. The Vehicle must move to be a measurable run. The participants may continue to work on their device provided they have not reached the end of their Event Time.
 - (5) Once the run starts and until the Vehicle is retrieved, competitors may not use, view, or access their programming tool(s).
 - (6) ES will review the run to determine if the run was Successful or a Failed Run.
 - (7) Run Time, Vehicle Distance, and Bottle Distance will be measured and recorded for Successful runs.
 - (8) **ES will verify the Moving Water Bottle Requirements were followed.**
 - (9) Competitors following their Vehicle down the track before being called to retrieve their vehicle, will receive a Competition Violation.
 - (10) All violations will be recorded. The ES must notify the participants of any violations at the end of each run.
 - (11) Timing resumes once the participants pick up their device or begin working toward the next run.
- viii. The Event Supervisor will review with teams their data recorded at the end of their Event Time.
- ix. Teams filing an appeal must leave their Vehicle and other impounded material in the event area.

5. **SCORING:**

- a. Low Score wins.
- b. Each team's Final Score is the better of the 2 Run Scores + Final Score Penalties.
- c. Run Score = 100 + Distance Score + Time Score + Bonuses + Run Penalties
- d. Distance Score = $(2.0 \text{ pt/cm} \times \text{Vehicle Distance}) + (1.0 \text{ pt/cm} \times \text{Bottle Distance})$.
 - i. For a Failed Run, Distance Score is 2500.
 - ii. **If Water Bottle is not past Target Point or if the Moving Water Bottle Requirements are violated, then the Bottle Distance is equal to 400.0 cm.**
- e. Time Score = Absolute value of $(\text{Target Time} - \text{Run Time}) \times 0.5 \text{ pt/sec}$
 - i. The Run Time will be recorded as 0.00 seconds for Failed Runs.
- f. **Bottle Bonus = -20 points if the Water Bottle is fully beyond the Bottle Line**
- g. **Pusher Bonus = $(\text{Pusher Foremost Opening Width} - 35.0) \times 1.5 \text{ pt/cm}$**
- h. Event Time Bonus (National Only) = $(\text{Event Time Used} - 480) / 30 \text{ points}$
 - i. **This bonus will only be applied to the Final Score.**
- i. Run Penalties:
 - i. Competition Violation: 150 points added to the Run Score with one or more violations.
 - ii. Construction Violation: 300 points added to the Run Score with one or more violations.
 - iii. **Non-modification Penalty: 50 points added to Run Score.**
 - iv. Failed Runs can also be assessed Competition and/or Construction violations.



- j. Final Score Penalties: Vehicle not Impounded: 5000 points added to the team's Final Score.
- k. Two or more teams tied with 2 Failed Run scores, without Competition or Construction Violations, will remain scored as ties. Other ties are possible.
- l. Tiebreakers in order: 1. Better **Distance Score** of the scored run; 2. Lower Time Score of the scored run; 3. **Better Vehicle Distance of the scored run**; 4. **Lower Bottle Pusher Measurement**.

6. SCORING EXAMPLES:

- a. A Vehicle has 2 Successful runs during the team's Event Time. Target time is 14 seconds. The Vehicle's Bottle Pusher had a Foremost Opening of 16.0 cm.
 - i. In the 1st run, the Vehicle stopped 17.6 cm from the Target Point with a Run Time of 12.27 seconds. The Water Bottle was 10.6 cm from the Bottle Line but not past the line.
 - ii. In the 2nd run, the Vehicle stopped 22.5 cm from the Target Point with a Run Time of 16.37 seconds. The Water Bottle was 12.1 cm from the Bottle Line and past the line.

Base Score	= 100	100.00
Distance Score	= 17.6 cm × 2.0 pts/cm + 10.6 cm x 1.0 pts / cm	45.80
Time Score	= abs(14 – 12.27) x 0.5 pts/sec	0.87
Bottle Bonus	= Not Beyond	0.00
Pusher Bonus	= (16.0 cm – 35.0) x 1.5 pts /cm	-28.50
1st Run Score		118.17

Base Score	= 100	100.00
Distance Score	= 22.5 cm × 2.0 pts/cm + 12.1 x 1.0 pts / cm	57.10
Time Score	= abs(14 – 16.37) x 0.5 pts/sec	1.19
Bottle Bonus	= -20.0 (Beyond)	-20.00
Pusher Bonus	= (16.0 cm – 35.0) x 1.5 pts /cm	-28.50
2nd Run Score		109.79

Final Score = 2nd Run Score (Better Score) = 109.79 pts

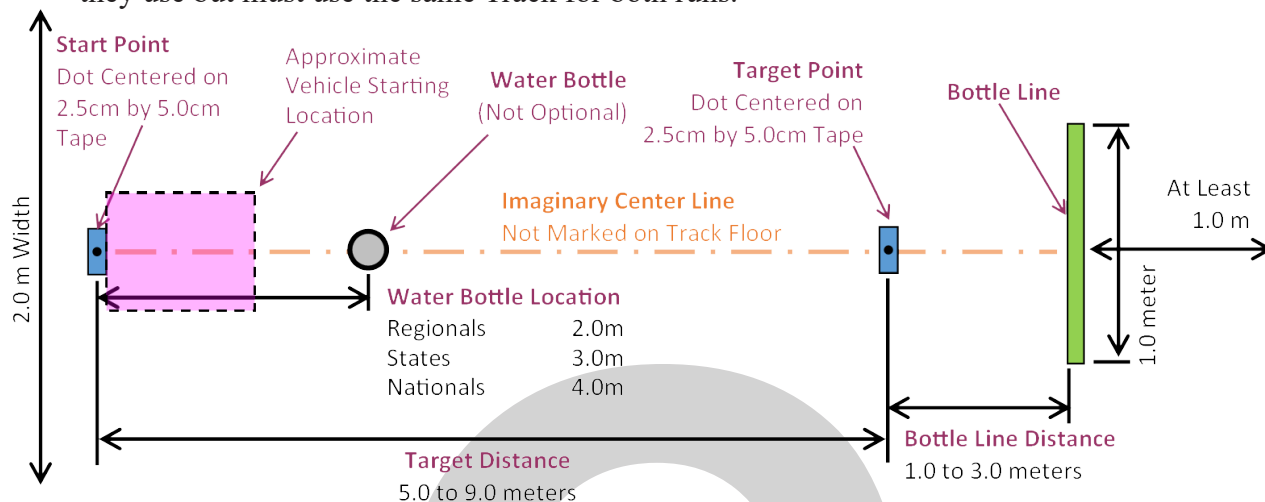
7. THE TRACK:

- a. In-Person tournaments: The ES will supply and set up the Track.
- b. Virtual tournaments: Team will supply the Track that fully complies with the requirements of this section. Any deviation will be scored as a competition violation for all runs.
- c. Track Definition:
 - i. The Track will be on a smooth, level, and hard surface.
 - ii. The Event Supervisors must mark the track as follows:
 - (1) Start Point – an approximately 5 cm × 2.5 cm tape with the Start Point marked at the center of the tape.
 - (2) Target Point – an approximately 5 cm × 2.5 cm tape with the Target Point marked at the center of the tape. The Target Point is placed at the Target Distance from the Start Point.
 - (3) **Imaginary Center Line is a line not placed on the Track that starts at the Start Point and extends through the Target Point.**
 - (4) **Bottle Line – an approximately 2.5 cm width tape placed at the Bottle Line Distance beyond the Target Point. The Bottle Line is perpendicular and centered to the imaginary center line. The Bottle line is at least 1.0 meter wide. The edge of the line closest to the Target Point defines the line.**



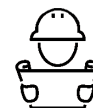
(5) **Water Bottle Location** will be along the imaginary center line at different distances from the Start Point based on the tournament level. For Regionals/Invitationals, the distance is 2.0 m. For States, 3.0 m. For Nationals, 4.0 m. The non-optional Water Bottle is placed centered at this location.

- iii. The Track Area will have a width of 2.0 m and extend at least 1.0 m beyond the Target Point. More space may be available but is not guaranteed.
- iv. At the ES's discretion, more than one Track may be used. If so, the team may choose which Track they use but must use the same Track for both runs.

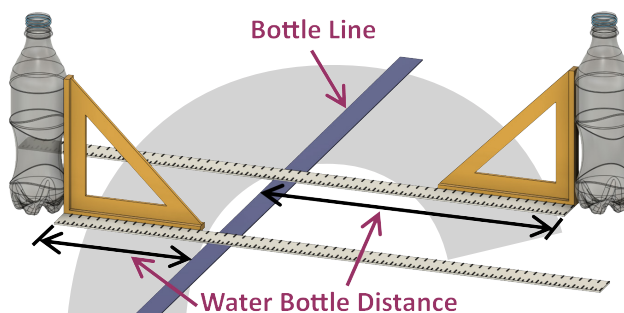
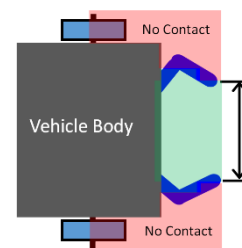


8. DEFINITIONS:

- a. *Event Supervisor (ES)* is responsible for ensuring the event runs correctly. The ES can assign one or more event helpers to perform various competition tasks.
- b. *Batteries* are commercial batteries, including rechargeables, labeled as AA batteries. Individual batteries must have a voltage between 1.2 and 1.5 volts when fully charged. The battery label must be the manufacturer's original label and easily viewable by the ES. The batteries must be individual batteries and removable from a battery pack. Batteries containing lithium or lead are prohibited.
- c. *Components* can be, but not limited to, motors, gearboxes, motor controls, bodies, and chassis.
- d. *Electronic components* can be, but not limited to, solid-state devices such as transistors, integrated circuits, diodes, and microprocessors. Cameras of any type or purpose are prohibited. Sensors with lasers are prohibited.
- e. **Vehicle Front** is defined as the side facing the Target Point when the Vehicle is placed in the Ready-to-Run position.
- f. *Measurement Point (MP)* must meet the following requirements:
 - i. The MP may be made of any material (e.g. wooden skewer, toothpick, pencil, dowel). Examples include but are not limited to: tip of a skewer/toothpick, tip of a pen, tip of a nail/paper clip/wire, corner of a wedge, the edge of a dowel.
 - ii. The MP must be less than or equal to 1.0 cm above the Track.
 - iii. The MP **must be the rearmost** part of the Vehicle. The MP must be easily accessible for measurements while the Vehicle is on the Track.
- g. *Tools* may be electric and electronic tools. To qualify as a Tool, the item must not travel on the Track with the Vehicle and cannot be used to align or aim the Vehicle.
- h. *Target Distance* is selected by the ES and will be between 5.00 and 9.00 meters. At Regionals/Invitationals the interval will be 0.25 m, for States 0.10 m, and for National 0.01 m.
- i. *Target Time* is selected by the ES and will be between 10.0 and 20.0 seconds. The interval will be 0.5 seconds for all tournaments.
- j. **Bottle Line Distance** is selected by the ES and will be between 1.00 and 3.00 meters. At Regionals/Invitationals the interval will be 0.25 m, for States 0.10 m, and for National 0.01 m.



- k. **Water Bottle** is a single-use clear plastic 16.9 oz (500ml) water bottle with a diameter between 6 and 7 cm and a height between 18 and 22 cm. The bottle's weight will be between 80 to 100 grams. Metal (non-lead) BBs or similar will be used to increase the bottle's weight. Liquids cannot be used.
- l. **Moving Water Bottle Requirements** are defined as:
- The Bottle must remain in contact with the event floor at all times. No part of the Vehicle may extend under the Bottle.
 - The Bottle must remain upright at all times.
 - The Bottle may only be pushed.
 - Only the Bottle Pusher may contact the Water Bottle. No part of the Bottle Pusher or Vehicle beyond the measured opening may contact the Water Bottle. See example.
- m. **Vehicle Distance** is a point-to-point measurement from the Vehicle Measurement Point to the Target Point, measured to the nearest 0.1 cm.
- n. **Water Bottle Distance** is a point-to-line measurement from the Bottle Line to the Water Bottle's closest side, measured to the nearest 0.1 cm. Use a square to find the Bottle's side. If the Bottle is resting on top of the Bottle Line, then both the front and rear side of the Bottle is measured to the Bottle Line and the smaller distance is used.



- o. **Run Time** begins when the team actuates the Vehicle and ends when the Vehicle comes to a complete stop.
- Any pause between the actuating the Vehicle and the Vehicle moving will be included in the Run Time. This pause can be longer than 3 seconds.
 - Recoil or oscillations will be included in the Run Time.
 - The Run Time is recorded in seconds to the precision of the timing device used.
 - Three timekeepers should be utilized with the middle time used as the official Run Time.
- p. **Complete Stop** is when the Vehicle is stationary for 3 or more seconds. The 3 seconds is not part of the Run Time.
- q. **Successful Run**: Any run when the Vehicle moves before the end of the Event Time and is not a Failed Run.
- r. **Failed Run**: A run will be scored as a Failed Run when one of the following occurs. Construction and/or Competition Violations must still be recorded for Failed Runs.
- The Vehicle travels the wrong direction from the Start Point.
 - The Vehicle starts before the Event Supervisor is ready.
 - The Vehicle's distance(s) cannot be measured (e.g., the participants pick it up before it is measured).
 - The team pushes the Vehicle down the track.
 - A team having only one successful run during the 8-minute Event Time will be assessed a Failed Run for a 2nd run score. If the Vehicle does not move during the Event Time, the team will be assessed 2 Failed Runs.
 - Teams may request a run to be scored as a Failed Run. If this request is for the 2nd run, then the team's event time will stop.

This event is sponsored by International Motors.



1. **DESCRIPTION:** Teams will read a set of engineering drawings and collaborate to CAD parts in Onshape and then incorporate these parts with provided components to create an assembly.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 50 minutes

2. **EVENT PARAMETERS:**

- a. Teams will use Onshape (onshape.com/edu) on two computers with mice to create the model.
- b. Each participant must bring a laptop capable of running Onshape for the purposes of completing the event unless specified by the Event Supervisor.
- c. Each participant may bring a wired or wireless mouse, a laptop charger, and writing utensils.
- d. No resource materials, except those provided by the Event Supervisor, may be used.
- e. Teams will be provided with a printed set of engineering drawings and a starting Onshape document with an assembly that has parts needed for creating the finished assembly. Teams will either be required to use their own Onshape accounts or use provided Onshape credentials at the discretion of the Event Supervisor.
- f. **Teams will provide the Event Supervisor with two Onshape account email addresses prior to the event. These account addresses will be used to provide testing materials during the event session.**

3. **THE COMPETITION:**

- a. Teams will use Onshape to CAD parts and an assembly based on the engineering drawings which specify the geometry, materials, and units for each part.
 - i. For Regionals, teams will be required to model **3 to 6** parts for use in the assembly.
 - ii. For State and Nationals, teams will be required to model **4 to 8** parts for use in the assembly.
- b. Students on the team will work collaboratively on the parts and assembly.

4. **SCORING:**

- a. The high score wins.
- b. Event Score = Individual Part Scores (IPS) + Individual Assembly Score (IAS)
- c. IPS is determined by comparing the volume of named parts as specified in the engineering drawing to the correct values. A perfect match for the volume is 100 points. Answers cannot earn negative points and will be worth a minimum of 0 points. Points for each part will be calculated as:

$$IPS = 100(1 - |\Delta_{volume}| \text{ where } \Delta_{volume} = \left| \frac{volume_{student} - volume_{correct}}{volume_{correct}} \right|$$

- d. IAS is determined by comparing the X, Y and Z component of the center of mass (CoM) of the named assembly as specified in the engineering drawing to the correct values. A perfect match for the center of mass is 100 points and the minimum score is 0 points. Points for each assembly will be calculated as:

$$IAS = 100 \left(1 - \frac{|\Delta_{X CoM}|}{3} - \frac{|\Delta_{Y CoM}|}{3} - \frac{|\Delta_{Z CoM}|}{3} \right) \text{ where } \Delta_{CoM} = \left| \frac{CoM_{student} - CoM_{correct}}{CoM_{correct}} \right|$$

- e. The tiebreaker is modeling time. The team with the lowest modeling time wins the tiebreaker. Time is measured from the moment the “Start assignment” button is pressed to the moment the “Submit” button is pressed. Work done after the last submission is not graded.

This event is sponsored by Onshape.



1. **DESCRIPTION:** Participants will be assessed on the ability to design, conduct, and report the findings of an experiment entirely on-site.

A TEAM OF UP TO: 3

EYE PROTECTION: C

APPROXIMATE TIME: 50 minutes

CALCULATOR: Class III

2. **EVENT PARAMETERS:**

- a. Participants must bring goggles and writing utensils. Experiments will not require any other safety equipment.
- b. Teams may bring one timepiece, one linear measuring device, and one stand alone Class III calculator. Teams CANNOT use any of these as part of the experiment - they must only be used for their intended function.
- c. The Event Supervisor will provide each team with identical sets of materials either at a distribution center or in an individual container.
- d. The Event Supervisor must provide the 2 part student report form posted on the event page at soinc.org, for teams to record their experimental information and data. This student report form must be printed single-sided to facilitate separation by individual team members.

3. **THE COMPETITION:**

- a. The teams must design, conduct, and report the findings of an experiment conducted on site that addresses the assigned question/topic area provided by the Event Supervisor. The assigned question/topic area should be the same for all teams and allow the participants to conduct experiments involving relationships between independent and dependent variables (i.e., height vs. distance).
- b. During the first 20 minutes of the event, participants will receive the assigned question/topic area, materials, and Part I of the student report form. Participants will focus on designing and conducting their experiment.
- c. After the first 20 minutes, participants will receive Part II of the student report form and will focus on analyzing their experiment and reporting findings. Participants may continue experimenting throughout the entire event.
- d. The identity of the materials will be unknown until the start of the event.
- e. When a team finishes, all materials must be returned to the Event Supervisor including both parts of the student report form.

4. **SCORING:**

- a. High Score wins. Scoring will be done using the Experimental Design Checklist found on the Science Olympiad website (soinc.org).
- b. Points will be awarded depending upon the completeness of the response. Zero points will be given for no responses as well as illegible or inappropriate responses.
- c. Ties will be broken by comparing the point totals in the scoring areas of the checklist in the following order:
 - i. Analysis of Claim/Evidence/Reasoning
 - ii. Procedure and Set-Up Diagrams
 - iii. Variables
 - iv. Data Table
 - v. Graph
- d. Any participant not following proper safety procedures will be asked to leave the room and will be disqualified from the event.
- e. Any team not following clean-up procedures will have their final score multiplied by 0.95.
- f. Any team not addressing the assigned question/topic area will have their final score multiplied by up to 0.75 based on the extent to which the report deviates from the assigned topic.
- g. Any team not collecting data by conducting an experiment on-site or falsifying/making up fake data will have their final score multiplied by 0.25.



2027 Experimental Design Division C Checklist

(Note: The maximum points available for each task are shown. Consider using the electronic checklist on www.soinc.org)

Shaded cells are included only at the State/National levels.

Part I. Design and Construction of the Experiment (60 pts)

A. Statement of the Problem (2 pts)

		2	1	0	Statement addresses the experiment including variables (not a yes/no question)
--	--	---	---	---	--

B. Hypothesis (6 pts)

		2	1	0	Statement predicts a relationship between the IV & DV
--	--	---	---	---	---

		2	1	0	Statement gives specific direction to the prediction
--	--	---	---	---	--

		2	1	0	A rationale is given for the hypothesis
--	--	---	---	---	---

C. Variables (18 pts)

a. Independent (IV) and Dependent (DV) Variables

	3	2	1	0	IV correctly identified and operationally defined
--	---	---	---	---	---

4	3	2	1	0	Levels of IV given
---	---	---	---	---	--------------------

	3	2	1	0	DV correctly identified and operationally defined
--	---	---	---	---	---

b. Controlled Variables (CV)

		2	1	0	1st CV correctly identified and relevant
--	--	---	---	---	--

		2	1	0	2nd CV correctly identified and relevant
--	--	---	---	---	--

		2	1	0	3rd CV correctly identified and relevant
--	--	---	---	---	--

c. Constant

		2	1	0	Constant correctly identified and relevant
--	--	---	---	---	--

D. Experimental Control (Standard of Comparison) (4 pts)

		2	1	0	SOC locally identified for the experiment
--	--	---	---	---	---

		2	1	0	Reason given for selection of SOC
--	--	---	---	---	-----------------------------------

E. Materials (4 pts)

		2	1	0	All materials used are listed and quantified
--	--	---	---	---	--

		2	1	0	No unused or extra materials are listed
--	--	---	---	---	---

F. Procedure and Set-Up Diagrams (13 pts)

		2	1	0	Procedure is presented in list form
--	--	---	---	---	-------------------------------------

		2	1	0	Procedure is in a logical sequence
--	--	---	---	---	------------------------------------

		2	1	0	Steps for repeated trials are included
--	--	---	---	---	--

		2	1	0	Multiple diagrams of setup are provided
--	--	---	---	---	---

		2	1	0	All diagrams are appropriately labeled
--	--	---	---	---	--

	3	2	1	0	Procedure detailed enough to repeat experiment accurately
--	---	---	---	---	---

G. Qualitative Observations (6 pts)

		2	1	0	Observations about set-up provided
--	--	---	---	---	------------------------------------

		2	1	0	Observations about the procedure provided
--	--	---	---	---	---

		2	1	0	Observations about the results provided
--	--	---	---	---	---

H. Quantitative Data - Data Table (7 pts)

	3	2	1	0	All raw data provided with units and labels
--	---	---	---	---	---

		2	1	0	Condensed data table with only the data to be graphed is provided, one sample calculation per derived variable
--	--	---	---	---	--

		2	1	0	Significant figure rules properly applied to table
--	--	---	---	---	--

--	--	--	--	--	--

of 4s # of 3s # of 2s # of 1s

Tiebreakers in Order:
 L. CER
 F. Procedure
 C. Variables
 H. Data Table
 I. Graph

Multipliers
 0.95 Non Clean Up
 Up to 0.75 - Off Topic
 0.25 - Non Lab/Fake/Falsified Data

Part II. Data, Analysis, and Conclusions (57/80 pts)

I. Graph (12 pts)

4	3	2	1	0	Appropriate graph is provided
---	---	---	---	---	-------------------------------

4	3	2	1	0	Graph properly titled and labeled
---	---	---	---	---	-----------------------------------

4	3	2	1	0	Appropriate scale and units included
---	---	---	---	---	--------------------------------------

J. Statistics (11/14 pts)

4	3	2	1	0	Statistics of Central Tendency used
---	---	---	---	---	-------------------------------------

		2	1	0	One example calculation is given for each statistic with units, done on level of IV
--	--	---	---	---	---

4	3	2	1	0	Statistics of variation are included
---	---	---	---	---	--------------------------------------

		2	1	0	One example calculation is given for each statistic with units, done on level of IV
--	--	---	---	---	---

		2	1	0	Significant figure rules properly applied to statistics
--	--	---	---	---	---

K. Possible Experimental Errors (6 pts)

	3	2	1	0	1st specific error is identified & effect on results discussed
--	---	---	---	---	--

	3	2	1	0	2nd specific error is identified & effect on results discussed
--	---	---	---	---	--

L. Analysis of Claim/Evidence/Reason (CER) (12/18 pts)

		2	1	0	Data trend claim completed logically
--	--	---	---	---	--------------------------------------

		2	1	0	Data trend evidence completed logically
--	--	---	---	---	---

		2	1	0	Data trend reasoning completed logically
--	--	---	---	---	--

		2	1	0	Outliers claim completed logically
--	--	---	---	---	------------------------------------

		2	1	0	Outliers evidence completed logically
--	--	---	---	---	---------------------------------------

		2	1	0	Outliers reasoning completed logically
--	--	---	---	---	--

		2	1	0	Variation claim completed logically
--	--	---	---	---	-------------------------------------

		2	1	0	Variation evidence completed logically
--	--	---	---	---	--

		2	1	0	Variation reasoning completed logically
--	--	---	---	---	---

M. Conclusion (8 pts)

		2	1	0	Hypothesis is restated
--	--	---	---	---	------------------------

		2	1	0	Hypothesis claim completed logically
--	--	---	---	---	--------------------------------------

		2	1	0	Hypothesis evidence completed logically
--	--	---	---	---	---

		2	1	0	Hypothesis reasoning completed logically
--	--	---	---	---	--

N. Applications and Recommendations for Further Use (8/12 pts)

4	3	2	1	0	Suggestions to improve experiment with rationale provided
---	---	---	---	---	---

4	3	2	1	0	Suggestions for practical applications of experiment provided
---	---	---	---	---	---

4	3	2	1	0	Suggestions for future experiments are provided
---	---	---	---	---	---

O Abstract (0/10 pts)

	3	2	1	0	Contains the statement of the problem, hypothesis, and practical application
--	---	---	---	---	--

	3	2	1	0	Summarizes the procedure, data trend, is well organized
--	---	---	---	---	---

4	3	2	1	0	Discussion of errors and improvements
---	---	---	---	---	---------------------------------------

--	--	--	--	--	--

of 4s # of 3s # of 2s # of 1s
 Team Name Team Number

--	--	--	--	--	--

Total Score Multiplier Final Score

--	--	--



1. **DESCRIPTION:** Given a scenario and some possible suspects, students will perform a series of tests. These tests, along with other evidence or test results, will be used to solve a crime.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 50 minutes

EYE PROTECTION: C

CALCULATOR: Class III

2. **EVENT PARAMETERS:**

- a. Each participant may bring writing utensils.
- b. Each participant may bring a Class III calculator.
- c. Each participant may bring one 8.5" x 11" sheet of paper, which may be in a sheet protector sealed by tape or laminated, that may contain information on both sides in any form and from any source without any affixed labels.
- d. Each team may bring any or all of the items listed on the Division C Chemistry Events Lab Equipment List, posted on soinc.org. Teams not bringing these items may be at a disadvantage. The Supervisor will not provide them.
- e. Participants must wear goggles, an apron or a lab coat and have skin covered from the neck down to the wrist and toes. Gloves are optional, but if the host requires a specific type, they will notify teams. Pants should be loose fitting; if the host has more specific guidelines they will notify teams in advance of the tournament. Shoulder length or longer hair must be tied back. Participants removing safety clothing/goggles or unsafely handling materials or equipment will be penalized, up to and including potential disqualification.
- f. The Supervisor will provide:
 - i. iodine reagent (**I_2 dissolved in KI solution or 2% tincture of Iodine**)
 - ii. 1M HCl
 - iii. 1M NaOH
 - iv. Benedict's solution
 - v. a hot water bath
 - vi. a Bunsen burner or equivalent BTU heat source to perform flame tests
 - vii. a waste container
 - viii. chromatography materials (e.g., beakers, Petri dishes, etc.)
 - ix. a wash bottle with distilled water
- g. The Supervisor may provide:
 - i. other equipment (e.g., a microscope, probes, etc.)
 - ii. candle & matches if fibers given
 - iii. differential density solutions or other method of determining density of polymers if plastics given
 - iv. reagents to perform other tests

3. **THE COMPETITION:**

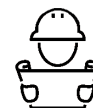
- a. The competition will consist of evidence from Parts 3.b. - e. and analysis of the evidence in Part 3.f. Analysis or questions can only be on the evidence topics included in the competition. The collected evidence and other data given may be used in a mock crime scene. The amount of evidence included will be according to the following table:

Level	Part b. # of samples	Part c. # of samples	Part d. # of chromatograms	Part e. # of topics	Part f.
Regional	3-8	5-9	1 type + Mass Spectra	1-2	Required
State	6-10	6-12	1-2 types + Mass Spectra	1-3	Required
National	10-14	10-18	1-3 types + Mass Spectra	3-5	Required

- b. Qualitative Analysis: Participants may be asked to identify the following substances: sodium acetate, sodium chloride, sodium hydrogen carbonate, sodium carbonate, lithium chloride, potassium chloride, calcium nitrate, calcium sulfate, calcium carbonate, cornstarch, glucose, sucrose, magnesium sulfate, boric acid, and ammonium chloride (there will be no mixtures). All teams will have the same set of solids to identify.
- c. Polymers: Participants may be asked to identify:
 - i. Plastics: PETE, HDPE, non-expanded PS, LDPE, PP, PVC, PMMA, PC – Participants will not perform any burn tests on these plastics, but the Supervisor may provide burn test results on them.



- ii. Fibers: cotton, wool, silk, linen, nylon, spandex, polyester - burn tests will be permitted on the fibers.
 - iii. Hair: human, bat, cow, squirrel, and horse - participants will need to know hair structure including medulla, cortex, cuticle, root and hair scale classification.
 - d. Chromatography/Spectroscopy: Participants will be expected to separate components using paper chromatography, TLC, and/or analyze mass spectra. Participants may be expected to measure R_fs.
 - e. Crime Scene Physical Evidence:
 - i. Fingerprint Analysis: Participants will be expected to know the 8 specific fingerprint patterns (plain arch, tented arch, radial loop, ulnar loop, plain whorl, central pocket whorl, accidental whorl, and double loop whorl). Participants should also be familiar with the common fingerprint development techniques of dusting, iodine fuming, ninhydrin, and cyanoacrylate fuming. Participants should understand terminology such as bifurcation, ridges, island, enclosure, loop, eye, spur, trifurcation, delta, and double bifurcation. Participants should be able to answer questions about skin layers and how fingerprints are formed. Participants may be asked questions on the different methods of detecting fingerprints and the chemistry behind each of these methods.
 - ii. DNA: Participants may be asked to compare DNA chromatograms/electropherograms from materials found at the scene to those of the suspects. Participants will be expected to know how DNA is copied. See <http://educationalgames.nobelprize.org/educational/chemistry/pcr/>
 - iii. Glass analysis: Participants may be asked to use index of refraction to determine the type of a glass found broken at a crime scene. They may be asked to analyze which hole or fractures occurred before others based on a piece of glass available for examination or a picture of a piece of glass.
 - iv. Entomology: Participants may be asked to identify how long an animal has been dead based on the type of insects found on the body at the scene.
 - v. Spatters: Participants may be asked to analyze actual spatters or photographs of spatters to determine the angle and velocity with which the liquid approached the solid object bearing the spatter & the spatter origin direction.
 - vi. Seeds and Pollen: Participants may be asked to compare pictures of seeds/pollen found at the scene with either seeds/pollen found on the suspects or seeds/pollen from different country regions.
 - vii. Tracks and Soil: Participants may be asked to match tire tracks or footprints found at the scene to tires or shoes of the suspects. Participants may be given the composition of soil found at the scene or on the suspects and asked to determine if this implicates any of the suspects.
 - viii. Blood: Participants may be asked to identify the ABO blood type using artificial blood (Event Supervisor required to provide instructions on how the typing system works) or participants may be asked to identify if a blood sample, either prepared microscope slide or pictures of microscope slide, is human, avian, mammalian, or reptilian/amphibian.
 - ix. Bullet striations: Participants may be asked to match the striations on bullets or casings found at the crime scene and fired from a given gun.
 - f. Analysis of the Crime: Participants will be asked to write an analysis of the crime scene explaining not only which pieces of evidence implicate which suspect(s) and why the suspect(s) was (were) chosen as the culprit(s), but also why the other suspects were not chosen. They will also answer any other crime scene analysis questions posed by the Event Supervisor.
 - g. Teams will dispose of waste as directed by the Event Supervisor.
4. **SCORING:**
- a. High Score wins. Time will not be used for scoring.
 - b. The score will be composed of the following elements (percentages given are approximate): Part 3.b. ≈ 20%, Part 3.c. ≈ 20%, Part 3.d. ≈ 15%, Part 3.e. ≈ 15%, and 3.f. ≈ 30%.
 - c. Ties will be broken by the highest score on the analysis of the crime scene, which includes the reasons why certain suspects have been eliminated or others remain in the pool of possible criminals.
 - d. A 10% penalty may be given if the area is not cleaned up as designated by the Event Supervisor.
 - e. A penalty of up to 10% may be given if a team brings prohibited lab equipment to the event.



1. **DESCRIPTION:** Prior to the competition, participants will design, construct, and calibrate a self-propelled, air-levitated vehicle **to carry nickels** down a track.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 10 minutes

EYE PROTECTION: B

IMPOUND: Yes

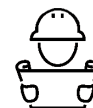
CALCULATOR: Class III

2. **EVENT PARAMETERS:**

- a. Each team must impound only one vehicle, **any spare parts**, and any papers/notes necessary for the competition time. The vehicle must be impounded with the batteries stored separately and presented to the Event Supervisor for inspection. No additional papers/notes may be brought into the competition area after impound.
- b. Teams may bring tools and two Class III calculators. These items need not be impounded. **Tools are items that do not travel with the hovercraft on its competition runs. Tools utilizing lasers are not permitted.**
- c. Participants must wear Category B eye protection during vehicle setup and operation. Teams without proper eye protection must be immediately informed and given a chance to obtain eye protection if time allows.
- d. The Event Supervisor will provide the testing materials listed in the **ES-PROVIDED MASSES and TRACK** sections. Teams should not bring their own **masses**, track, or ramp.

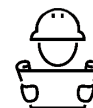
3. **CONSTRUCTION PARAMETERS:**

- a. The vehicle must fit into a 40.0 cm x 40.0 cm x 40.0 cm box when levitated in the ready-to-run configuration. Vehicles must not modify or damage the track.
- b. Electrical components shall be limited to batteries, wires (**including battery holders/connectors**), **mechanical** switches, **relays**, resistors (including potentiometers **and rheostats**), capacitors, and up to two motors (including brushless motors). Integrated circuits (other than those that are an **integrated, manufactured** part of a commercial motor) are not permitted.
 - i. **Mechanical switches are switches with moving parts that connect or disconnect an electrical circuit. They may be operated manually by a person or by contact with another part of the hovercraft.**
 - ii. **Components utilizing lasers are not permitted.**
- c. The vehicle must levitate on a cushion of air as it moves down the track. Participants may be asked to demonstrate levitation by pushing the vehicle slightly down. If it then rises, it is levitating. **Contact of vehicle components with the track surface or inside edge of the track's side rails is permitted. Vehicle components cannot contact the top or outside edge of the side rails.**
- d. **All thrust and lift on the vehicle must be achieved with air pressure. Non-electric sources of energy are permitted to power other vehicle functions such as a timing system.**
- e. For timing and measurement purposes, the vehicle must have a 1/4" or larger wooden dowel vertically attached within 3.0 cm of its front edge such that the top end is at least 20.0 cm above the track's surface when levitated. The dowel must be placed on the hovercraft so that it will be the first part of the vehicle to break a laser timing beam when the vehicle is traveling forward.
- f. **Safety Parameters (devices failing to meet these may not run, scored as Participation points):**
 - i. All propellers/impellers, including those under the vehicle, must have shielding that prevents a 3/8" dowel from touching them.
 - ii. Only commercial batteries, including rechargeables, may be used to energize each of the vehicle's motors. The label on each battery must be the manufacturer's original and easily viewable by the Event Supervisor. Lithium and lead batteries are prohibited. Multiple batteries may be connected together as long as the expected voltage across any two points does not exceed 12.0V, as calculated by their individual labels.
 - iii. All motors must have an **accessible manually-operated mechanical** switch to permit safe starting and stopping. Relying on inserting batteries or twisting wires together to start is not allowed. If more than one motor is used, they may be combined on the same switch or wired on separate switches.
- g. Participants must be able to answer questions regarding the design, construction, and operation of the vehicle per the Building Policy found on www.soinc.org.



4. THE COMPETITION:

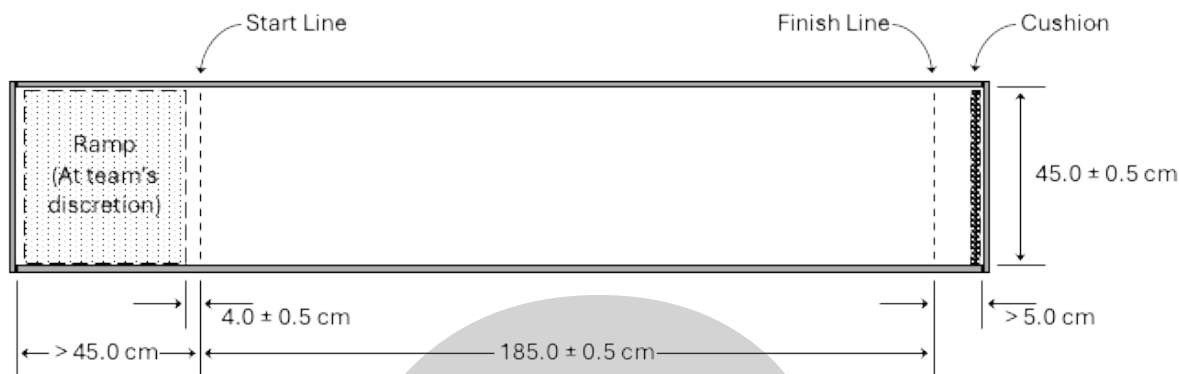
- a. Once participants enter the event area to compete, they must not leave or receive outside assistance, materials, or communication until they are finished competing.
- b. Event Supervisors will check the vehicle specifications before the team's testing period. Teams must be notified as soon as possible if a vehicle is out of spec. Teams may modify the vehicle to bring it into compliance during their testing period.
- c. Vehicles violating any of rule 3f or that are otherwise deemed unsafe in construction or operation by the Event Supervisor will not be allowed to run unless brought into compliance. If the team is unable to bring the vehicle into compliance with 3f, the vehicle will not run and will only receive **Participation points**.
- d. At the start of their **competition time**, each team will be given a Target Time (TT). The Target Time will be between **6.0-26.0** seconds in intervals of **1.0** seconds for Regionals and 0.5 seconds for **States/Nationals**. The TT will be the same for all teams.
- e. Teams have a **competition time** of 8 minutes to adjust and repair their vehicle and make their runs. **The team's competition time ends after two complete or two incomplete runs or the competition time limit has been reached (whichever comes first)**. Practice runs are not allowed.
- f. **During their competition time, teams may dry-sweep the track, but may not wet it or apply any substance. Teams may take measurements of the track and the surface it is on, but may not request changes to the track's position or leveling.**
- g. **The vehicle may carry an ES-provided mass consisting of a combination of both loose nickels and rolls of nickels. Participants will select how much mass, if any, the vehicle carries and this amount may vary by run. Participants will not be allowed to unroll, break, or alter the packaging of the nickels nor will they be allowed to use any adhesive or tape to affix the rolls or nickels to the vehicle; any vehicle that requires such measures to carry the nickels and/or rolls will not be allowed to use masses during their competition runs.**
- h. An incomplete run occurs if a vehicle fails to move for 3 seconds before crossing the finish line or the dowel fails to cross the finish line within (2 x TT) seconds. Teams are not allowed to declare a run as incomplete. A run is complete if it is not incomplete.
- i. If the vehicle's dowel does not cross the starting line within 3 seconds of launching, it does not count as a run (either complete or incomplete). A run **will be allowed to finish and** will be counted in the scoring as long as it is started before the 8-minute period has elapsed.
- j. If any part of the vehicle, **nickel rolls, or individual nickels** fall off during a run, the team incurs **one construction penalty (not one per item)** for the run. **Any nickels or rolls that fall off before the Event Supervisor tells the team to turn off the vehicle at the completion of the run do not count towards the Mass Score.**
- k. If they choose, teams may utilize an Event Supervisor-provided ramp (as described in 6.f) for the launching of their vehicle. In such cases, the Event Supervisor will place the ramp in the same position for all teams. Using the ramp will impact the team's score (7.f.v and 7.d.iii), and the time for placing the ramp will come out of the 8-minute period. The team may change this decision for each run.
- l. To begin a run, the team will place their vehicle, **including the nickel load**, on the track fully behind the start line. **The team may start anywhere behind the start line. The Event Supervisor will provide and hold a wood block against the front of the vehicle to hold it in place.** The team then activates their vehicle's motor(s) **and removes their hands from the vehicle.**
- m. The team will give a countdown of "3, 2, 1, launch"; then the Event Supervisor will remove the wood block. Timing starts when the vehicle's dowel crosses the start line and stops when the dowel crosses the finish line. If photogates are used, the dowel must be the first part of the vehicle to break the laser beams at both the start and finish lines.
- n. **After removing their hands at the vehicle's start**, the team must not touch their vehicle **or interfere with its run in any manner** until the Event Supervisor states they may do so. If touched **or interfered with**, the run is counted as a complete run with a DS, TS, and MS of 0.
- o. If a run is declared incomplete, the Event Supervisor will place the starting block in front of the hovercraft at the point where the run was declared incomplete to prevent the vehicle from moving. The ES will ask the competitors to turn off their vehicle without moving it, and the ES will then record the distance from the finish line to the position of the dowel. The team's testing period time will halt while the Event Supervisor makes measurements and will resume when the measurement time is done.
- p. The Event Supervisor will review with the team the data recorded on their scoresheet.
- q. A team filing an appeal must **alert the Event Supervisor and** leave their vehicle in the **impound area**.



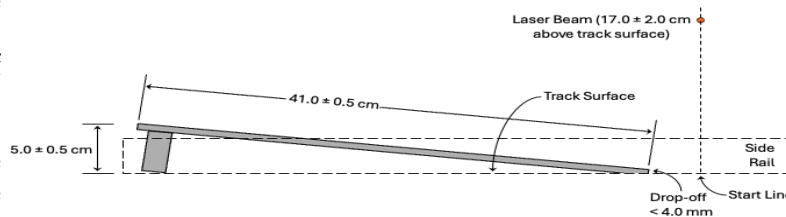
5. THE ES-PROVIDED MASSES:

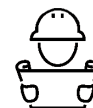
- The ES will provide 4 full rolls of U.S. nickels, 1 half roll of U.S. nickels, and 20 loose U.S. nickels.
- One full roll contains 40 nickels. Each roll should be unopened, wrapped in a bank paper roll wrapper, folded snugly against the nickels.
- One half roll contains 20 nickels. The one half-roll should be wrapped in a bank paper roll wrapper with the ends trimmed to minimize the excess paper, and folded snugly against the nickels.
- One full roll of U.S. nickels ≈ 200 g, each individual nickel ≈ 5 g. In combination, the ES-provided nickels can make a mass approximately between 5 g and 1 kg.

6. THE TRACK:



- The Event Supervisor will supply a 45.0 ± 0.5 cm wide and at least 240 cm long track on a non-carpeted smooth floor or other firm smooth base surface, such as a countertop. The outside boundary of the track is composed of rails each with a height at least 30.0 mm (standard 2x4 framing studs or steel framing studs recommended). The Event Supervisor will also supply a cushioned barrier to stop vehicles and a wood block to hold the vehicle at the start line. Example setups are at www.soinc.org.
- Each rail must be securely affixed to the floor, base, or each other.
- Event Supervisors may provide a track that has either a built-in base or utilize a table or the room's floor as a base. In either case, Event Supervisors should clean the base of the track (or the floor) at the start of the day and should provide as level a surface as possible. Small imperfections in the track's base surface, though, do not disqualify the track for tournament use; the point of a hovercraft is to traverse uneven terrain. Tournaments are encouraged to make available ahead of time information about the type of surface the tracks will have.
- The length of the timed portion of the track is fixed at 185.0 ± 0.5 cm. A start line must be marked that is at least 45.0 cm from the end of the track. The finish line must be marked 185.0 ± 0.5 cm from the start line and a cushioned barrier at least 5.0 cm past it must block the channel.
- A photogate timing system is recommended. If used, the system will be installed at the start and finish lines with the beams at a height of 17.0 ± 2.0 cm. At least one manual timer should be used as a backup. If photogates are not being used, three timekeepers should be utilized with the median time used as the official Run Time; lasers are recommended to be placed at the start and finish lines so the timekeepers only have to watch for the flash of light as the dowel cuts through the laser beam. Time is recorded in seconds to the device precision if photogates are used, or to the tenth of a second if manual timers are used.
- The Event Supervisor must provide a removable ramp that competitors may use for the launching of their vehicle. The ramp should have a smooth, flat surface and must span the width of the track between the rails. The height of the ramp is 5.0 ± 0.5 cm and the length of the ramp's surface is 41.0 ± 0.5 cm. When the ramp is placed on the track, the edge where the vehicle exits the ramp must be 4.0 ± 0.5 cm behind the start line. At the bottom of the ramp, where it meets the track surface, the transition should be reasonably smooth, with a drop-off from the ramp surface to the track surface of no more than 4.0 mm. See www.soinc.org for information on how to build a ramp and track.

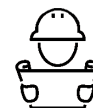




g. Multiple tracks may be used to facilitate teams competing in a timely manner.

7. **SCORING:**

- a. High Score wins.
- b. Final Score (FS) = Best Run Score. A scoring spreadsheet is available at www.soinc.org.
- c. Run Score = DS + TS + MS
- d. Distance Score (DS):
 - i. Complete Run: Distance Score = **40**
 - ii. Incomplete Run: Distance Score = **40** x (185 - (distance from the finish line in cm)) / 185
 - iii. If the ramp is utilized, the Distance Score is multiplied by 0.5 for that run.
- e. Time Score (TS):
 - i. Runtime = Time from the Vehicle dowel crossing the start line until the dowel crosses the finish line
 - ii. Complete Run: Time Score = **40** x (1 - abs(Runtime - TT)/TT)
 - iii. Incomplete Run: Time Score = 0
 - iv. The smallest possible Time Score is 0.
- f. **Mass Score (MS) = (4R + .1N) x (185 - (distance from the finish line in cm)) / 185**
 - i. **R = # nickel rolls. The half-roll of nickels counts as .5 of a nickel roll (i.e. if the vehicle carries 2 complete rolls and the half roll, the # of nickel rolls is 2.5)**
 - ii. **N = # loose nickels**
 - iii. **The Maximum Mass Score for a Complete run is 20. The Mass Score formula prorates the Mass Score for the distance traveled in incomplete runs**
 - iv. **The nickel rolls and loose nickels must stay on the vehicle for the entirety of the run to be counted.**
 - v. **Mass Score for a run using the ramp = 0**
- g. **Teams unable to pass the safety parameters in 3f will receive only Participation Points.**
- h. Teams without any runs will receive a Final Score of 0, with ties broken as indicated in 7.1.. Ties are possible.
 - i. The DS, TS, and MS for each run will be multiplied by 0.7 if the team misses impound.
 - j. The DS, TS, and MS for a run will be multiplied by 0.8 for each CONSTRUCTION violation present during that run.
 - k. The DS, TS, and MS for a run will be multiplied by 0.9 for each COMPETITION violation present during that run.
- l. Tie Breakers: 1st – Best TS over all runs; 2nd – Fewest # construction violations **in the team's run that has the most construction violations, or, in the case of a vehicle that does not have any runs in the competition time, the number of construction violations per 4b**; 3rd – Best MS, 4th – 2nd Best Run Score



1. **DESCRIPTION:** Prior to the competition, competitors will design, build, test, and document a Rube Goldberg[®]-like device that begins with a specific Start Action and ends with a specific Final Action with an optional series of specific actions between the Start and Final Actions.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 40 minutes

EYE PROTECTION: C

IMPOUND: State & National only

2. **EVENT PARAMETERS:**

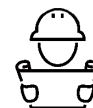
- a. All participants must properly wear eye protection at all times. Participants without proper eye protection must be immediately informed and given a chance to obtain eye protection if time allows. Participants without eye protection will not compete and be ranked in Tier 3 based on device size.-
- b. Each device must pass a safety inspection before operation.
- c. Event supervisors will need a voltmeter, meter sticks, stopwatches, and measuring tape as well as an unaltered US nickel, dime and quarter.
- d. Uncontrolled flames, cell phones, as well as hazardous liquids, gases and materials (e.g., rat traps, lead objects, combustible fuses, dry ice, liquid nitrogen, etc.) and unsafe handling of chemicals will not be permitted. No flammable gases allowed. Devices with potential hazards or safety concerns must not be permitted to run unless safety concerns are resolved to the satisfaction of the event supervisor; otherwise they must receive only participation points.

3. **CONSTRUCTION PARAMETERS:**

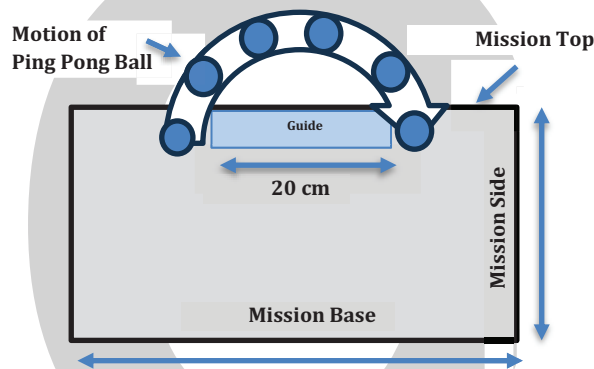
- a. During operation, device dimensions can be no greater than 80.0 cm (D) x 80.0 cm (W) x 80.0 cm (H).
- b. All actions used for scoring must be visible and/or verifiable. The top and at least two vertical walls must be open or transparent for viewing all actions.
- c. Any action in the device not designed to contribute to the completion of the Final Action will not count for points. Parallel and dead-end actions are not allowed and will not count for points.
- d. Each movable/adjustable physical object in the device must be utilized by at most one assigned action. An object at the end of one action may initiate the next action but must not be used in any further action.
- e. Other non-scorable actions may be incorporated into the device but must contribute to the completion of the Final Action, receive no points and be listed on the Action Sequence List (ASL).
- f. Energy devices (i.e., springs/mousetraps) may be set prior to starting the device.
- g. Arduinos, Raspberry Pis, or integrated circuits are not allowed. Timing must not be powered by electricity or springs. Use of electricity is limited to the first 30 seconds of operation. If electricity is used beyond 30 seconds, a penalty will be assessed and the scorable action(s) using electricity will not count.
- h. Only commercial batteries, not exceeding 9 volts as labeled, may be used to energize each of the Device's electrical circuits. Multiple batteries may be connected in series or parallel as long as the expected voltage output across any two points does not exceed 9 volts as calculated using their labeled voltage. Teams must be able to show the event supervisors the labeled voltage. All energy storage devices must be contained in the device. All use of batteries must comply with the Science Olympiad Battery Policy at www.soinc.org. Non-compliant batteries must be removed prior to and during device operation.
- i. Students must be able to answer questions regarding the design, construction, and operation of the device per the Building Policy found on www.soinc.org.

4. **DEVICE OPERATION:**

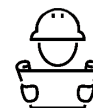
- a. **Start Action** (25, 50, or 200 points) –To get any points for starting the device, the device must have a vertical coin slot on one side that is only large enough to allow one coin to be added at a time. All coins used in the Official Run will be supplied by the ES and returned to the ES after the Official Run. The scoring for the start sequence is as follows:
 - i. When signaled by the team, the ES will put a US nickel or dime (per ES selected Coin Drop Order) in the coin slot. If the coin initiates the first action, then award 25 points and let it continue. The first action must be one that is clearly visible to the ES because timing starts along with the first action, not the insertion of the coin.
 - ii. If the device does not start by adding the first coin, the ES will add the second coin (a dime or nickel). If the first action is initiated, timing will begin and 50 points will be awarded instead of 25.
 - iii. If the device still does not start, the ES will add a US Quarter through the slot. If the Quarter initiates the first action, then award 200 points and begin timing when the first action is initiated.



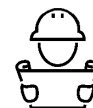
- iv. If the device does not start with the quarter, the ES will implement a 20 second countdown for a member of the team to drop another un-altered US Quarter into the device from a point entirely above the device to initiate the first action. Timing will begin with the first action and no starting points will be awarded.
 - v. If the team must touch the device to start it, then no points will be awarded for the start and a penalty for the first touch will be assessed. If the task that is touched is scorable, then it will not count either.
 - vi. All coins must be returned to the ES at the end of the operation of the Device or a Competition Violation will be assessed.
- b. **Scorable Actions** (50 points each) – Participants may have up to 12 scorable unique actions to count for points.
- i. Use vinegar and baking soda to inflate a balloon so that it expands 20 cm, then the balloon strikes an object that originally was at least 20 cm above the balloon, causing the object to initiate the next action.
 - ii. Use an endothermic action that initiates the next action because of the reduction in temperature.
 - iii. Drop two Alka Seltzer tablets into water so that the reaction triggers the next action.
 - iv. Use a 1st class lever to launch an unaltered ping pong ball completely out of the top of the Device, flies at least 20 cm horizontally before re-entering, initiating the next action, and staying in the Device. A guide of 20 cm must be included at the top level of the Device to ensure compliance. The 1st class lever is considered part of the Device for Device dimension measurement.
 - v. Use an infrared beam where the transmitter and receiver are at least 20 cm apart to initiate the next action. The transmitter must be an IR LED and the receiver must be a photodiode-based receiver.



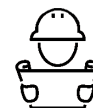
- vi. Push or pull an object with a mass ≥ 400 g up an inclined plane with an IMA ≥ 2 at least 10 vertical cm before the object initiates the next action. The mass must be removeable for the ES to verify.
 - vii. Use a pulley system with an ideal mechanical advantage (IMA) of at least 3 to lift an object with a mass ≥ 400 g at least 10 vertical cm before the object initiates the next action. The mass must be removeable for the ES to verify.
 - viii. Use the mechanical advantage of all 3 classes of levers in any sequence to initiate the next action.
 - ix. Use gravity to clearly rotate a screw so that it operates as a screw converting rotational force into linear force. The screw must complete at least two full rotations and move an object at least 2 cm before that object initiates the next action. The screw must be marked to make its rotational movement clearly visible.
 - x. Use an Archimedes Screw to raise a marble at least 20 cm before the marble initiates the next action. The entire action must be visible.
 - xi. Use electricity to directly or indirectly break a string or fishing line so that the breaking of the line drops an object that initiates the next action.
 - xii. Remove a magnet from a surface so that a magnetic object falls due to the removal of the magnetic force. That falling object must initiate the next action.
- c. Participants may designate one water timer, an action taking over 10 seconds, to be eligible for bonus points. This timer must not be one of the scorable actions.
- i. A 1-point bonus will be awarded for every full second the water timer runs before the Target Time.
 - ii. The timer may run past the Target Time but will not receive points for seconds after the Target Time.
 - iii. The timer must initiate the next action for any bonus points to count.
 - iv. For State/National tournaments, the team must demonstrate how this timer is adjusted to account for the increased length of Target Time for bonus points to count.



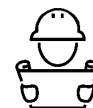
- d. **Final Action** (250 points) After attempting all scorable actions, the device will dispense (drop) a commercially purchased, unaltered 5-stick pack of chewing gum (team supplied) at least 50 cm vertically onto a platform ("tray") made of flat, rigid, solid, unpadded material that is no more than 12 cm x 12 cm x 0.5 cm with no sides. The platform must have no barriers/walls within 4 cm of the edge. The top of the platform is at least 2 cm above the top of the base. Teams will not be allowed to touch the action that leads to releasing the pack of gum to complete the Final Action. If this action is touched by a team member, then no points will be awarded for the Final Action and zero time points will be scored.
 - e. To receive any points for the Final Action, the only thing allowed on the platform is the dropped pack of gum. The pack of gum must be supported only by the platform, clearly visible on the platform and there must be an opening on the same side of the device as the coin slot that is large enough for a team member to reach in and remove the pack of gum. The Final Action points will not be awarded if the gum is not removed.
 - f. An Action Sequence List (ASL) must be submitted to the event supervisor at impound. The ASL must be a:
 - i. Legible, neat, and accurate documentation of each intended scorable and non-scorable action of the device's operation.
 - ii. Scorable and non-scorable actions must be numbered and documented in the ASL and correspondingly labeled in the device.
 - iii. Scoring will be based only on the Scorable Actions listed (by numeral) in the ASL. See www.soinc.org for an example of the format required.
5. **COMPETITION:**
- a. Impound before first time slot (Only for State and National tournaments).
 - i. Teams must impound their Device, Action Sequence List (ASL), and any tools or parts that they will use during their set-up time or run.
 - ii. Inspections will NOT take place during impound as any of the 15 member team may impound.
 - b. Before the first time slot begins:
 - i. The Target Operation Time will be selected by the ES. The Target Operation Time at a competition will be the same for all teams. Teams will be told the Target Operation Time at the start of their Setup Time.
 - (1) Regionals/Invitationals will be 60 seconds.
 - (2) State tournaments will be 61 to 90 seconds.
 - (3) National tournament will be 91 to 120 seconds.
 - ii. The Coin Drop Order (Nickel, Dime, then Quarter or Dime, Nickel, then Quarter) will be selected by the ES. Teams will be informed of the Coin Drop Order at the start to their set up time. The order at a competition will be the same for all teams.
 - c. Check-In
 - i. Only the participants will enter the event area. Once participants start the Check-In process, they must not leave or gain any outside assistance or materials until their Event Time is completed.
 - ii. Teams will be instructed by the ES when to retrieve their impounded items (State & National).
 - iii. Each Device must pass a safety inspection before operation. Devices with potential hazards and/or safety concerns must not be permitted to run unless safety concerns are resolved to the satisfaction of the Event Supervisor, otherwise they must receive only participation points.
 - iv. ES will take all measurements including dimensions and masses before or after the Device's Official Run at the ES discretion. ES may re-measure after the Official Run if the dimensions change.
 - v. Participants must be able to answer questions regarding the design, construction, and operation of the Device per the Building Policy found on www.soinc.org.
 - d. Setup Time
 - i. The ES will notify the participants when their time starts. The ES will record the team's start time for scoring.
 - ii. Participants will start preparing their Device for an Official Run. Teams are permitted to adjust, repair, and test their Device during their Setup Time.
 - iii. Participants must notify the ES when ready to start their Device for scoring. ES will record the number of minutes of Setup Time used.



- e. Official Run
 - i. Timing and scoring begin when the first action starts after the coin is deposited into the device. Timing stops when the pack of gum comes to rest on the platform, or after 2x the Target Time in seconds has elapsed, whichever comes first. No points will be earned after timing stops.
 - ii. Teams that have a run time of twice the Target Time will receive no (zero) points for running time.
 - iii. If the Device stops, jams, or fails, the participants will be allowed to adjust it to continue operation up to three times. An adjustment may consist of multiple physical touches and is only completed once the Device runs again on its own. Adjusting only to stall or impact operation time will result in disqualification.
 - iv. If a participant completes a scorable action or makes an adjustment that leads directly to the completion of that action, then that action will not count for points, even if it is part of the Final Action.
 - f. If an action starts out of order on the ASL, all actions skipped in the listed sequence, even if completed, earn zero (0) points.
 - g. The Event Supervisor will review with teams their data recorded at the end of their Event Time.
 - h. Teams filing an appeal must leave their Device and ASL in the event area.
6. **SCORING:**
- a. High Score wins.
 - b. Award 50 points if not more than 30 minutes is used to set up their Device.
 - i. At State & Nationals only: Award 75 points if participants use not more than 15 minutes to set up their Device or 50 points if not more than 30 minutes.
 - c. Award 25 points if the ASL is submitted on time at device impound or at check-in for Regionals.
 - d. Award 25 points if ASL is in proper format.
 - e. Award 25 points if all scorable and non-scorable actions are included and accurately described.
 - f. Award 25 points if the original number (column 1) in the ASL are properly labelled in the Device and the action number (column 3) is the roman numeral of the 12 optional intermediate tasks (i to xii).
 - g. Award 50 points the first time each unique action in Section 4 is successfully completed as described in the rules.
 - h. Award points for completing the Start Action as outlined below:
 - i. Award 25 points if the device starts with insertion of the first coin by the Event Supervisor.
 - ii. Award 50 points if the device starts with insertion of the second coin in the slot after trying to start with the first coin by the Event Supervisor.
 - iii. Award 200 points if only the insertion of a quarter starts the first action after both other coins have been tried by the Event Supervisor.
 - i. Award 250 points for completing the Final Action if the device dispenses an unaltered pack of gum in the tray as described after all scorable actions had been attempted.
 - j. Award 2 points for each full second (rounded down) of operation up to the Target Operation Time. Devices running twice the Target Time will receive zero time points.
 - k. Award 1 point per full second that a Water Timer runs before the Target Operation Time if all conditions are met, and the next action is initiated by the timer.
 - l. Award 0.1 point for each 0.1cm that the Device dimensions are under 80 cm x 80 cm x 80 cm. The maximum score awarded for each dimension is 30 points, for a total of 90 points (Only at in-person tournaments.)
 - m. Award 75 points for a Device that has no adjustments during operation.
 - n. Teams receive only participation points for impounding a Device but not competing, unsafe Devices, or Devices that are remotely timed/controlled.
7. **PENALTIES:**
- a. Deduct 2 points for each full second (rounded down) that the Device operates past the Target Operation Time up to 2x the Target Time seconds.
 - b. Deduct 25 points for each dimension of the Device that exceeds its limit of 80 cm.
 - c. Deduct 25 points if the top and 2 vertical walls are not open or transparent.
 - d. Deduct 25 points for each touch/adjust up to 3 times. If the device stops after the third touch, it will not be allowed to be touched/adjusted and the time will be scored as 0 (zero) seconds. No other scoring is allowed after the third touch.



- e. Deduct 50 points for the first solid or liquid leaves the measured dimensions of the Device with the exception of the ping pong ball during the operation of action 4.b.iv.
 - f. Deduct 250 points for use of electricity after 30 seconds, and allow no scoring for any action that is initiated by electricity after 30 seconds.
 - g. Devices with any dimension greater than 82 cm will be allowed to run but will be scored in Tier 2.
 - h. Teams failing to impound their Device on-time will be scored in Tier 3.
 - i. Teams failing to answer build questions will be scored in Tier 3.
8. **TIERS:**
- a. Tier 1: Devices without any violations
 - b. Tier 2: Devices with construction or run violations that were not already penalized.
 - c. Tier 3: Devices impounded after the deadline, participants without eye protection, teams failing to answer build questions.
9. **TIEBREAKERS:**
- a. Ties are broken as follows: a) Fewest penalty points and b) Smallest overall dimension (L+D+H) of the device.
10. **DEFINITIONS:**
- a. *Event Supervisor (ES)* is responsible for ensuring the event runs correctly. The ES can assign one or more event helpers to perform various competition tasks.
 - b. *Electricity:* A form of energy that results from the movement/flow of charged particles, electrons to create electric current.
 - c. *Endothermic:* process is a physical or chemical change that absorbs heat energy from its surroundings, causing them to cool down.
 - d. *Ping Pong Ball* must be a commercially available product and sold as a ping pong ball.
 - e. *Ideal Mechanical Advantage (IMA)* is a theoretical calculation of a machine's force-magnifying ability, assuming no energy is lost to friction or other inefficiencies.
 - f. *Action Sequence List (ASL)* The list must include all sequential actions in the machine including but not inclusive of this list:
 - i. Indicate the Start, the action initiated by the coin(s), the Water Timer (if one is included), the sequence of unique scorable actions, all non-scorable actions, and the action that drops the pack of gum, etc.
 - ii. ASL column 1 should be an ordinal number that matches the number in your Mission Possible Device for easy tracking.
 - iii. ASL column 2 should specifically call out the object by name in 4.b.i. to 4.b.xii. that is to be moved.
 - iv. ASL column 3 should be the specific task that the points are to be requested (e.g. 4.b.iv.).
 - g. *Touches:* A single touch can consist of multiple physical contacts with the Device; but is only considered over when the Device resumes operation on its own. If a participant completes a scorable action or makes an adjustment that leads directly to the completion of the action, then that action will not count for points, even if it is part of the Final Action.
 - h. *Parallel and Dead End Paths:* Parallel actions have no direct relationship to one another and if one of the two actions fails, the overall sequence of events can still continue or lead to a "dead-end" path. Parallel actions are not measured in a chronologic manner but in a causality manner. That is to say, if one action causes the next action, then they are not parallel.



1. **DESCRIPTION:** Prior to the tournament, teams will design, build, and bring up to two bottle rockets to the tournament to launch a ping pong ball attached to a parachute to stay aloft for the greatest amount of time.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 5 minutes

EYE PROTECTION: B

2. **EVENT PARAMETERS:**

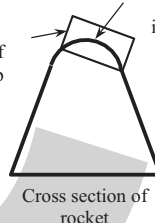
- a. Teams must provide up to two rockets, two unaltered standard ping pong balls, and two parachutes.
- b. Parachutes must be attached to ping pong balls with tape only. The ping pong ball attached to the parachute assembly makes up the parachute payload system.
- c. Event Supervisors must provide a launcher (that uses a Schrader valve), an air pump, a pressure gauge, and timing devices. Teams may bring their own manual bicycle pump with a pressure gauge to use, but it must attach to the launcher provided by the Event Supervisor. **Event supervisors are encouraged to make multiple launchers available with slightly differing connection widths (21.5 mm - 22.0 mm diameters). Connection of the air pump to the launcher must be completed before timing begins for the students. Connection to the launcher must be completed before timing begins for the students. If a rocket is not maintaining pressure with one launcher, students must be allowed to change launchers (if available) while the timer is stopped.**
- d. This event should be held inside with a high ceiling (greater than 20 feet recommended). Tournament directors must provide the ceiling height (in feet) to teams **with as much advance notice as possible**. Extreme care must be taken to protect the floor and ceiling of any inside facilities used for practice and competition. **Tournament officials are encouraged to reduce air flow (HVAC, doors opening, etc.) as much as possible.**

3. **CONSTRUCTION PARAMETERS:**

- a. Rocket pressure vessels must be made from a single 1-liter or less plastic carbonated beverage bottle with a nozzle opening internal diameter of approximately 2.2 cm and a standard neck height from flange to bottle's opening of under 1.6 cm. The bottle label must be presented.
- b. The structural integrity of the pressure vessel must not be altered. This includes, but is not limited to: physical, thermal or chemical damage (e.g., cutting, sanding, using hot or super glues, spray painting).
- c. The nose of the rocket must be rounded or blunt at the tip and designed such that when a standard bottle cap (~3.1 cm diameter x 1.25 cm tall) is placed on top of the nose, no portion of the nose touches the inside top of the bottle cap (see Figure 1).
- d. Only tape must be used to attach fins and other components to the pressure vessel. No glues of any type may be used on the pressure vessel. Metal of any type is prohibited anywhere on the rocket or parachute payload system.
- e. Fins and other parts added to the bottle must be 5 cm or higher above the level of the bottle's opening, to ensure rockets fit on the launcher (see Figure 2).
- f. All energy imparted to the rocket/parachute payload system must originate from air pressure provided by the **air pump through the launcher; water is not allowed to be used**. Gases other than air, explosives, liquids, chemical reactions, pyrotechnics, electrical devices, elastic powered flight assists, throwing devices, remote controls, and tethers are prohibited at **all times**.
- g. **Mylar/ultrafilm materials must not be used in any part of the parachute or rocket.**
- h. **Participants must be able to answer questions on design, construction, and operation of the device per the Building Policy found on www.soinc.org**

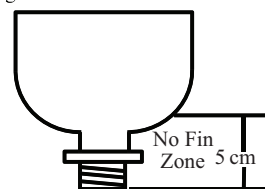
Figure 1

Cross section of bottle cap



Nose not touching inside top of cap

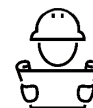
Figure 2



No Fin Zone 5 cm

4. **PRACTICE LOG:**

- a. During inspection, each team must present a flight log of recorded data. Data must include five or more parameters (three required and at least two additional) for 10 or more test flights prior to the competition. The required parameters are: 1) pressure (psi), 2) estimated peak flight height (feet), 3) time aloft (seconds). The additional parameters are chosen by the team (examples include: number of fins, parachute diameter, etc.)



- b. An incomplete flight log is a flight log that is missing at least one of the required or additional parameters or does not include data for at least 10 total flights.
- c. **Teams must use their data to justify their pressure choice. Teams without a flight log or an incomplete flight log will not be allowed to be pressurized beyond an event-supervisor determined safe pressure that will be the same for all teams without complete practice logs and will ensure rockets will not reach the ceiling. Additionally, teams without a flight log will have their final score multiplied by 0.7 and teams with an incomplete flight log will have their final score multiplied by 0.85.**

5. THE COMPETITION:

- a. Teams must arrive at the competition site ready to launch with proper eye protection to have their rocket(s) inspected for safety.
- b. Teams will have 5 minutes to make a total of two launches using the same rocket or two different rockets.
- c. When called to launch, teams will load their rocket onto the launcher. Once the rocket is loaded, but NOT pressurized, teams will place the parachute payload system on or in the rocket. After the payload parachute system is loaded it cannot be manipulated. Teams will then pressurize the rocket to the pressure (psi) of choice based on their practice log data. The Event Supervisor will check the gauge on the pump to ensure the rocket is pressurized to the psi chosen and justified by the team's data.
- d. The Event Supervisor will make sure 3 timers are ready and then signal a team member to make a loud announcement of, "3, 2, 1, LAUNCH!" Then a team member will proceed to launch the rocket. After launching, the team will prepare for the next launch.
- e. Timing begins when the rocket separates from the launcher and stops when the parachute payload system lands. The parachute payload system must separate from the rocket.
- f. If the parachute payload system does not separate from a rocket, timing is from when the rocket separates from the launcher to when any part of the rocket touches the ground. **This launch is penalized with a multiplier of 0.90.**
- g. If any part of a rocket or parachute payload system hits the ceiling or any part connected to the ceiling (e.g., a rafter, light, basketball hoop) **during the ascending portion of the launch**, then timing is stopped at the instant of contact. That launch is **penalized with a multiplier of 0.25.**
- h. If a rocket fails to separate from the launcher because of a problem with the supplied launcher then the launch never occurs and the launch can be restarted.
- i. All times for each launch must be recorded for breaking ties. Time aloft is recorded in hundredths of a second. The middle value is the officially recorded time.
- j. Teams filing an appeal must leave their rocket(s), parachute payload system(s), and Practice Log(s) in the event area.

6. SCORING:

- a. High Score wins.
- b. Ranking is determined by the greatest time aloft of a parachute payload system from a single launch.
- c. Rockets and/or parachute payload systems violating 2.b. and/or 3.a.-g. will not be launched. Teams with a rocket unable to make any launches will receive participation points only.
- d. Ties will be broken by the greatest time aloft of the parachute payload system from each tied team's other launch.
- e. **Penalties will be assigned to a launch as follows:**
 - i. **A 0.90 multiplier will be applied to launches where the parachute payload system does not separate from the rocket.**
 - ii. **A 0.85 multiplier will be applied to launches where the team presents an incomplete practice log.**
 - iii. **A 0.70 multiplier will be applied to launches where the team does not present a practice log.**
 - iv. **A 0.25 multiplier will be applied to launches where any part of the rocket or parachute payload system strikes the ceiling or any part connected to the ceiling on the ascending portion of the launch.**



1. **DESCRIPTION:** Participants will be assessed on a written exam and prior to the competition will use computer visualization and online resources to construct a physical model of the sialic acid binding site of the 2009 H1N1 influenza hemagglutinin protein.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 50 minutes

IMPOUND: Yes

2. **EVENT PARAMETERS:**

- a. Each team may bring writing utensils.
- b. Each participant may bring one 8.5" x 11" sheet of paper, which may be in a sheet protector sealed by tape or laminated, that may contain information on both sides in any form and from any source without any annotations or labels affixed, along with writing utensils for each participant.
- c. **Each team will impound a pre-built model of the sialic acid binding site of the Hemagglutinin protein. This year's prebuild will focus on residues 153–252 of chain A from the crystal structure with PDB ID 3UBE, as described in the article "Structural Characterization of the Hemagglutinin Receptor Specificity from the 2009 H1N1 Influenza Pandemic" (DOI: 10.1128/JVI.06322-11), freely available at <https://journals.asm.org/doi/10.1128/JVI.06322-11>.**
- d. Each team will bring one device (laptop, Chromebook, tablet, or phone) that can connect to WIFI with a browser that can run the JUDE Jmol program. A wired/wireless mouse may also be used.
- e. Event Supervisors will provide written exams, WiFi information, and the website address for the molecular exploration portion of the event.

3. **THE COMPETITION:**

Part I: The Pre-Built Model

- a. Participants will use the program JUDE (Jmol User Design Environment) to visualize **residues 153–252 of chain A from the crystal structure with 3UBE.pdb. A modified file for easier modeling by participants can be found on the 3D Molecular Designs Protein Modeling Event website (<https://learn.3dmoleculardesigns.com/science-olympiad-protein-modeling-event>).** In addition, **The atomic coordinate data file can be accessed for free from the RCSB Protein Data Bank (www.rcsb.org).** A constructed model of this protein must be brought to all competitions; as the competition level increases, the scoring rubrics for the pre-built model will reflect higher expectations for model accuracy, detail, and enhancements. (See SCORING for more details.) The pre-built model must be based on the protein's alpha carbon backbone display, using a scale of 2 cm per amino acid. Students will manually fold their pre-built model using Modeling Ties or comparable bendable material (Mini-Toobers, Kwik Twists, 12-gauge dimensional house wire, etc.). The backbone should primarily be gray with a diameter of 0.75 – 1.25 cm.
- b. Three-dimensional (3D) printed materials may NOT be used to build the protein backbone but may be used for functionally relevant features.
- c. Participants will use materials of their choosing to add functionally relevant features to their model (e.g., selected amino acid sidechains, DNA, or associated molecules). These additions should highlight the significance of structure to the protein's function.
- d. When an amino acid sidechain is displayed, its chemical property should be annotated on the alpha-carbon of the backbone. For example, hydrophobic sidechains are yellow, hydrophilic white, acidic red, basic blue, and cysteine green.
- e. Participants must explain their functionally relevant features in a Model Description Legend using clear and concise descriptions on a printed 8.5" x 5.5" paper, in the form of a table with three columns, headed:
 - i. What is displayed?
 - ii. How is it displayed?
 - iii. Why is it displayed?
- f. All models, including all functionally relevant features, must fit within a 50.0 cm x 50.0 cm x 50.0 cm space.
- g. The model must be sufficiently sturdy so that judges can remove it from any scaffolding or support, pick it up, and rotate it so that they can view it unobstructed from any angle and properly evaluate it.
- h. Teams must impound their pre-built model and Model Description Legend. They must pick up pre-built



models after the competition.

Part II: Written Exam

- a. Teams will complete a written exam to assess their protein structure and function knowledge and their ability to explore protein structures in the molecular visualization environment, JUDE. JUDE can be accessed at learn.3dmoleculardesigns.com/digital-modeling-hub/jude_online_jmol for free.
 - b. The exam will consist of multiple-choice and short-answer questions.
 - c. Protein Structure topics addressed include:
 - i. The principles of chemistry drive protein folding and stability.
 - ii. The chemical properties of amino acid side chains and how their interactions stabilize protein structures.
 - iii. **The virus infection process and immune response, specifically influenza.**
 - iv. **At State and National Tournaments, students will be asked questions about the role of hemagglutinin binding of sialic acid receptors in the influenza virus infection mechanism.**
 - d. Molecular Visualization topics addressed include:
 - i. Ability to identify different secondary structural elements of a protein and specific amino acids found in these structures.
 - ii. Chemical environment around specific amino acid sidechains that stabilize structure.
 - iii. **Interactions of the protein with ligands and other molecules.**
 - e. Participants may not visit websites outside of the JUDE environment.
4. **SCORING:**
- a. High Score wins. The final score will be derived from both parts of the competition:
 - b. The pre-built model (Part I) accounts for approximately 40% of the final score.
 - i. The pre-built score is based on the accuracy and scale of the secondary structures, the tertiary arrangement of these structures, and the relevant functional features added.
 - ii. Features that are irrelevant or do not explain the protein's structure/function relationship will not receive credit.
 - iii. The pre-built model's scoring rubric will change with each competition's level.
 - (1) For Regional competitions, scoring will be based primarily on the accuracy of the 3D folded structure of the protein's alpha-carbon backbone (secondary and tertiary structures).
 - (2) For State competitions, approximately 20% of the score awarded to the pre-built model will be based on functionally relevant features, such as the inclusion of key sidechains that have been added to the alpha-carbon backbone model to explain the protein's structure.-
 - (3) For the National competition, approximately 30% of the score awarded to the pre-built model will be based on the inclusion of key sidechains that explain the interactions between specific amino acids that stabilize the protein's structure.
 - c. The written exam (Part II) accounts for approximately 60% of the score.
 - d. Preselected questions, on the written exam (Part II) will be used as tiebreakers.

This event is sponsored by 3D Molecular Designs and supported by the National Institute of General Medical Sciences of the National Institutes of Health under Award Number R44GM154526.



1. **DESCRIPTION:** Participants will use remote sensing imagery, data, and computational process skills to complete tasks related to climate change processes in the Earth system.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 50 minutes

CALCULATOR: Class II

2. **EVENT PARAMETERS:**

- a. Each team may bring writing utensils.
- b. Each team may bring two Class II calculators
- c. Each team can bring two protractors and two rulers.
- d. Each team may bring one three-ring binder of any size containing information in any form and from any source attached using the available rings. Sheet protectors, lamination, tabs and labels are permitted. If the event features a rotation through a series of laboratory stations where the participants interact with samples, specimens, or displays; no material may be removed from the binder throughout the event.

3. **THE COMPETITION:**

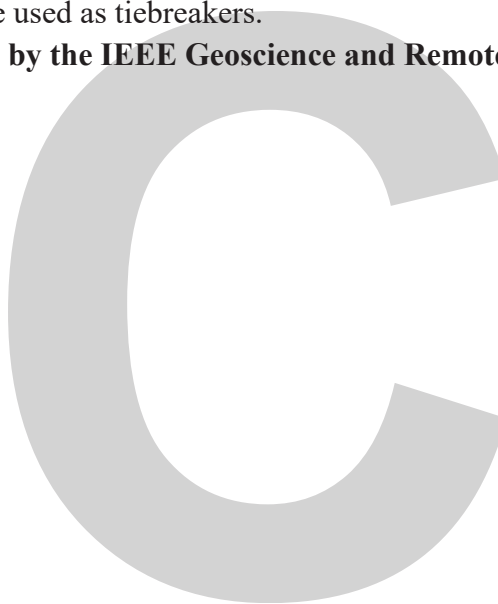
- a. The event will consist of questions and activities testing **the principles of remote sensing data collection, and the usage and interpretation of data from the specified satellites and instruments** to observe and study climate change processes and their effects on Earth systems.
- b. The test should be divided so that approximately 25% focuses on topics under 3.c., 35% focuses on topics under 3.d., 20% focuses on topics under 3.e., and 20% focuses on topics under 3.f. Students should not be expected to know or use calculus to answer any questions.
- c. Remote sensing instrumentation and physics.
 - i. Active vs. Passive sensors
 - ii. Optical and infrared imagers, radiometers, LIDAR, space-based gravimetry
 - iii. Radar Altimetry, precipitation radar, Doppler radar (ground, airborne, and satellite-based)
 - iv. Reflection, refraction, attenuation, transmission, and scattering of light by the atmosphere, aerosols, water, vegetation
 - v. Spectroscopy, including regions of the electromagnetic spectrum, basic types of spectra (continuous, emission, and absorption), band ratios, usage in identifying spectral composition (atmospheric gases, **atmospheric interactions with the terrestrial biosphere**, etc.)
 - vi. Characteristics of blackbody radiation (Wien's Law, Stefan-Boltzmann Law)
 - vii. Orbital properties, including Kepler's Laws, types of orbits (LEO, MEO, HEO, geosynchronous, geostationary, polar, sun-synchronous)
 - viii. Imaging fidelity/quality, including satellite altitude or semi-major axis and angular resolution, instantaneous field of view, ground sample distance, field of view, revisit period vs. orbital period vs. frame rate to cover a swath, types of resolution
- d. Interpretation of images and datasets from **only the listed satellites and instruments**, as relevant to the topics of this event. Appropriate context (e.g. sensors, bands, etc.) should be provided as necessary for satellite imagery.
 - i. Optical, infrared, and Doppler radar imagery of clouds, precipitation, and extreme weather events (MODIS on Aqua, CALIPSO, CloudSat, GOES, Suomi NPP, GPM)
 - ii. Sea level rise and surface waves (PACE, JASON, GRACE)
 - iii. Distributions and radiative effects of greenhouse gases, trace atmospheric gases, and aerosols (OCO-2, Aura, MODIS on Aqua, GOES, Suomi NPP, CALIPSO, PACE, **Tropomi on Sentinel-5P, ASTER on Terra**)
 - iv. Atmospheric and sea surface temperatures (MODIS on Aqua, ECOSTRESS, **ASTER on Terra**, GOES, CrIS on Suomi NPP)
 - v. Radiative energy flux (CERES on **Aqua**/Suomi NPP, PACE)
- e. Climate processes.
 - i. Atmospheric concentrations and distributions of greenhouse gases, trace gases, and aerosols (CO₂, H₂O, CH₄, NO₂, SO₂, Ozone, CFCs, HFCs, NH₄, Halogenated gas), and relevant influencing factors (e.g. solar activity, volcanic activity, Earth's orbit and seasons, atmospheric and ocean circulation)
 - ii. El Niño - Southern Oscillation and its effects on regional climate
 - iii. Carbon cycle and Nitrogen cycle, the impacts of natural phenomena and anthropogenic activity on



these cycles, and relevance of these cycles to climate processes

- iv. The effects of climate change on existing weather processes (e.g. clouds and rainfall, flooding, sea level rise, heat waves, atmospheric effects of wildfires)
 - v. Ecological impacts, such as on vegetation and biome distributions
 - f. Earth's energy balance.
 - i. The effects of atmospheric and surface conditions (e.g. clouds, aerosols, surface albedo) on climate change, including radiative forcing and climate feedback loops
 - ii. Blackbody radiation (using the Stefan-Boltzmann Law) absorbed by the Earth from the Sun, and emitted from the Earth.
 - iii. Albedo-based reflection by surface, ice, and clouds
 - iv. Modeling planetary equilibrium temperature as a result of the interaction between incoming and outgoing radiation, and changes to equilibrium temperature due to atmospheric and surface conditions
 - v. **Interpreting changes in planetary and atmospheric heating and cooling rates over time (linear, piecewise-linear, and exponential)**
4. **SCORING:**
- a. High Score wins.
 - b. Points will be awarded for the quality and accuracy of responses.
 - c. Preselected questions will be used as tiebreakers.

This event is sponsored by the IEEE Geoscience and Remote Sensing Society (GRSS).





1. **DESCRIPTION:** Teams will identify and classify rocks and minerals and demonstrate knowledge of how rocks and minerals help to understand geologic processes, interpretation of earth's history, the development of natural resources, and use by society.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 50 minutes

CALCULATOR: Class II

2. **EVENT PARAMETERS:**

- a. Each team may bring writing utensils.
- b. Each team may bring one Class II calculator.
- c. Each team may bring one magnifying glass
- d. Each team may bring one three-ring binder of any size containing information in any form and from any source attached using the available rings. Sheet protectors, lamination, tabs and labels are permitted.
- e. Each team may also have one commercially produced field guide which may be tabbed and annotated.
- f. In addition to a binder and a field guide, each team may bring one copy of the **2027 National Rocks and Minerals List** which does not have to be secured in the binder and may be annotated.
- g. Teams are not permitted to bring samples or specimens to the event.
- h. If the event features a rotation through a series of stations where the participants interact with samples, specimens or displays; no material may be removed from the binder, except for the **2027 National Rocks and Minerals List**.

3. **THE COMPETITION:**

- a. Emphasis will be placed upon task-oriented activities such as identification of rocks and minerals based on observations of properties and characteristics, interpretation of graphs and charts, and analyzing data.
- b. Where possible, participants will move from station to station, with the length of time at each station predetermined and announced by the Event Supervisor. Participants may not return to stations but may change or add information to their original responses while at other stations.
- c. Identification will be limited to specimens appearing on the Official Science Olympiad **2027 National Rocks and Minerals List**, but other rocks or minerals may be used to illustrate key concepts.
- d. If identification of a specimen is not possible through observation, key characteristics/properties of the specimen will be provided.
- e. Identification of rocks and minerals should be at least 30% but not more than 50% of available points.
- f. Questions/tasks should be balanced between rock and mineral topics.
- g. Written descriptions of how a specimen might react if tested with hydrochloric acid (HCl) may be given. However, HCl will not be used or provided. Participants will not be allowed to do a taste test.
- h. **Mineral Topics**
 - i. **Definition of a mineral: naturally occurring inorganic solid with definite composition and crystalline structure.**
 - ii. Identification - specimens or images used should show observable properties. **Where observable properties are insufficient to identify a specimen, other diagnostic characteristics will be provided**
 - iii. Physical Properties - color, hardness, luster, streak, cleavage/fracture, density/specific gravity/ heft, diaphaneity, and tenacity
 - iv. Other properties - limited to reaction with acid, fluorescence, magnetism, smell, taste, double refraction, piezoelectricity, and radioactivity.
 - v. Mineral habit - limited to acicular (needlelike), bladed, botryoidal, cubic, dendritic, dodecahedral, doubly terminated, druzy, geodic, hexagonal, hopper, massive, micaceous, octahedral, pisolitic, prismatic, radiating, rosette, stalactitic, twinning, and tabular
 - vi. Chemical composition – chemical formulas, relationships between chemistry and properties (e.g., effect of trace elements on mineral color)
 - (1) Solid solution series: **limited to: plagioclase and alkali feldspars** (e.g., feldspar ternary diagrams); **olivine group (magnesium-rich forsterite and iron-rich fayalite); carbonate group (calcite, siderite and rhodochrosite); amphibole group (actinolite and tremolite); and sulfate group (barite and celestite)**
 - (2) Phase diagram interpretation: temperature/pressure, temperature/composition (limited to two component systems)



- vii. Polymorphs (e.g., aragonite/calcite; diamond/graphite, orthoclase/microcline; sillimanite/andalusite/kyanite)
- viii. Classification - mineral families based on composition (see 2027 National Rocks and Minerals List)
 - (1) Mineral groups (e.g., feldspar, garnet, tourmaline) - similarities of chemical composition and shared properties
 - (2) Silicate tetrahedra and their structures: isolated tetrahedra (nesosilicates), island (sorosilicates), chain (inosilicates), ring (cyclosilicates), sheet (phyllosilicates), and framework (tectosilicates)
 - (3) Crystal Systems – cubic, tetragonal, orthorhombic, monoclinic, triclinic, trigonal, and hexagonal; emphasis on how crystalline structures result in certain physical properties (e.g., cleavage planes, crystal shape)
- ix. Methods of formation and environments (e.g., hydrothermal, chemical weathering, crystallization from magma, evaporites, chemical precipitation, alteration under heat & pressure)
- x. Minerals associated with rock-forming environments (e.g., evaporite minerals in sedimentary settings; mafic minerals in oceanic crust; minerals that form under metamorphic conditions)
- xi. Bowen's Reaction Series – relationship between mineral crystallization and temperature in magma
- xii. Uses of minerals
 - (1) Ores, industry, jewelry, geochronology, medicine, manufacturing, construction, electronics, etc.
 - (2) Precious and semi-precious gemstone minerals on the **2027 Rocks and Minerals List** as well as the following **varieties**, limited to: emerald, aquamarine, and morganite (**beryl**); kunzite and hiddenite (spodumene), peridot (olivine); ruby and sapphire (corundum); pearl; and amber

i. Rock Topics

- i. **Definition of a rock: naturally occurring solid mass composed of one or more minerals or mineraloid matter; rocks can be an aggregate (mixture) of one (monomineralic) or more minerals (polymineralic); unlike minerals, rocks can be made of organic matter.**
- ii. Identification - specimens or images used should show observable characteristics. **Where observable characteristics are insufficient to identify a specimen, other diagnostic characteristics will be provided (e.g., mineral composition of fine-grained igneous rocks, reaction to acid of chalk or diatomite)**
- iii. Classification - igneous, sedimentary, and metamorphic including observable diagnostic characteristics that facilitate classification (e.g., glassy or vesicular texture in igneous; rounded grains, fossils, or layers in sedimentary; and foliation or banding in metamorphic)
- iv. Igneous Rocks:
 - (1) Textures - including aphanitic (fine-grained), glassy, vesicular, porphyritic, pyroclastic, phaneritic (coarse-grained), and pegmatitic
 - (2) Composition and essential minerals - felsic, intermediate, mafic, and ultramafic
 - (3) Intrusive (plutonic) and extrusive (volcanic) environments; and igneous formations limited to batholith, lava flow, and pyroclastic deposits
 - (4) Relationship between textures and environments of formation (e.g., intrusive/plutonic, extrusive/volcanic and relative rates of solidification.)
 - (5) Types of igneous rocks and constituent minerals at various types of plate boundaries (e.g., basalt at mid-ocean ridge; andesite/diorite at subduction zones; granite at convergent boundaries)
 - (6) **Evolution of Igneous Melts**
 - (a) **Magmatic differentiation: limited to fractional crystallization governed by Bowen's Reaction Series; assimilation (rising magma melts and incorporates country rock); and magma mixing.**
 - (b) **State and Nationals only: Basic understanding of incompatible elements (trace elements with atomic sizes or charges that make them unable to fit into crystal structures of common minerals and are concentrated in the liquid phase of magma) and their significance in pegmatites (large crystals, exotic minerals, gemstones, and economically valuable elements).**
- v. Sedimentary Rocks:
 - (1) Textures - limited to clastic (detrital or terrigenous), chemical (crystalline), and biochemical (organic/bioclastic)



- (2) Composition and essential minerals
- (3) Grain sizes (e.g., clay, silt, sand, pebble, cobble, boulder) **and their relationship to water velocity in rivers and streams**; sorting, and shape (round vs angular) and their implications for energy and conditions of environments of deposition (e.g., fine grained-low energy; coarse-grained-high energy)
- (4) Relationship between textures and composition to environments of deposition
- (5) Environments of deposition - including, but not limited to alluvial fan, delta, river/stream (fluvial), lake (lacustrine), swamp, wind (aeolian), floodplain, beach, shallow marine/shelf, and deep marine (**abyssal**)
- (6) Primary sedimentary structures and their implications about depositional processes and environments (e.g., plane bedding, crossbedding, ripple marks, mud cracks, graded bedding, and fossil tracks & trails)
- vi. Metamorphic Rocks:
 - (1) Textures - foliated (e.g., slaty cleavage, schistose, banding) and non-foliated
 - (2) Mineral composition
 - (3) Metamorphic minerals that form in existing rocks due to heat and pressure (e.g., garnet, corundum, kyanite, staurolite, epidote, andalusite, sillimanite, chlorite)
 - (4) Protoliths (parent rocks); e.g. shale for slate, limestone for marble, sandstone for quartzite, **peridotite for serpentinite, basalt or gabbro for amphibolite.**
 - (5) Regional and contact metamorphism
 - (6) Grade (intensity) of metamorphism (low, medium, high) and index minerals (**limited to** chlorite, **muscovite, biotite**, garnet, staurolite, kyanite, and sillimanite) that indicate the degree of metamorphism
 - (7) Relationship of temperature, pressure, depth to types of metamorphism and metamorphic facies (hornfels, zeolite, greenschist, amphibolite, granulite, blueschist, and eclogite) based on interpretation of graphs and charts
 - (8) Environments of metamorphism in the context of plate tectonics - regional metamorphism and mountain building at convergent continental-continental boundary; blueschist and eclogite formation in subduction zones; greenstone/greenschist formation from basalt or gabbro at ocean crust divergent boundaries
- vii. Rock Cycle – emphasis on the geologic processes that form rocks (e.g., melting & solidification; uplift, erosion & deposition; burial, compaction & cementation; heat & pressure resulting in recrystallization & deformation)
- viii. Economic importance and uses of rocks (e.g., building stone, ores, ornamental, agriculture, fossil fuels)
- ix. **States and National Only** - Thin Sections of Rocks; using photographs taken through a microscope (photomicrographs)
 - (1) Identify minerals using their optical properties and features in polarized light (twinning, extinction, cleavage planes, birefringence); limited to microcline, plagioclase, calcite, augite, and garnet.
 - (2) Distinguish rock types and characteristics of igneous, sedimentary, metamorphic rocks by their microscopic textures limited to:
 - (a) Igneous - fine grained crystalline (holocrystalline), vesicular, glassy, porphyritic (e.g., basalt vs. pumice)
 - (b) Sedimentary – rounded, angular, well sorted vs. poorly sorted, skeletal fragments (e.g., oolites, sandstone vs. arkose)

4. **SCORING:**

- a. High Score wins.
- b. Point values of each question or section may be specified.
- c. Preselected questions/sections will be used as tiebreakers.



MINERALS

BORATES

Ulexite

CARBONATES

Aragonite
Azurite
Calcite
Dolomite
Malachite
Magnesite*¹
Rhodochrosite*
Siderite*

NATIVE ELEMENTS

Copper
Diamond
Gold
Graphite
Silver
Sulfur

HALIDES

Fluorite
Halite⁵

OXIDE/HYDROXIDES

Corundum
Goethite/Limonite
Hematite
Magnetite
Pyrolusite*
Rutile*
Zincite*

PHOSPHATES

Turquoise*
Apatite Group
Fluorapatite
Pyromorphite*
Vanadinite*

SULFATES

Barite
Celestite/**Celestine**
*Gypsum*⁵ *varieties:*
Alabaster (massive)
Satin Spar (fibrous)
Selenite (crystalline)

SULFIDES

Bornite*¹
Chalcopyrite
Galena
Pyrite
Sphalerite
Stibnite*

SILICATES

Amphibole Group
Actinolite*
Hornblende
Tremolite*
Apophyllite*
Beryl
Chrysocolla*
Diopside*
Epidote
Feldspar Group
Plagioclase feldspars
Albite
Labradorite
Potassium feldspars
Amazonite
Orthoclase/Microcline²
(tan/pink variety)
Garnet Group
Almandine
Grossular (green variety)

SILICATES (cont.)

Kaolinite
Kyanite
Mica Group
Biotite
Lepidolite*
Muscovite
Olivine
Prehnite*
Pyroxene Group
Augite
Rhodonite*
Spodumene*
Quartz varieties:
Aventurine
Agate/Chalcedony
Amethyst
Citrine
Jasper*
Milky Quartz
Opal
Rock Crystal
Rose Quartz
Smoky Quartz*
Tiger's Eye*
Sodalite
Staurolite
Stilbite*
Talc
Topaz
Tourmaline³
Willemite*¹
Wollastonite*
Zircon*



ROCKS

IGNEOUS

Andesite
Basalt
Diorite
Gabbro
Granite
Obsidian
Pegmatite
Peridotite
Pumice
Rhyolite
Scoria
Syenite*¹
Tuff*¹
Tuff Breccia^{1*}

SEDIMENTARY

Banded Iron Formation
Bauxite⁴
Breccia
Chert/Flint
Conglomerate
Diatomite*¹
Dolostone
Rock Salt (Halite)⁵
Rock Gypsum⁵
Shale
Coal varieties:
Anthracite
Bituminous
Lignite
Limestone varieties:
Chalk
Coquina
Fossil Limestone
Oolitic Limestone
Travertine
Sandstone varieties:
Arkose
Greywacke*
Quartz Sandstone

METAMORPHIC

Amphibolite*
Gneiss
Marble
Phyllite
Quartzite
Schist varieties:
Garnet Schist
Mica Schist
Talc Schist (Soapstone)*
Serpentinite*
Slate

Specimens marked with an asterisk () are for State and National Tournaments

1. For identification purposes, information will be provided such as diagnostic properties, chemical formulas, grain size, or composition.
2. The pink/tan variety of feldspar should be identified as **Orthoclase/Microcline** or Potassium feldspar. **White or pink/tan Orthoclase/Microcline can be found in granite.**
3. Although Tourmaline is the generic name for a group of related mineral species, for identification purposes, tourmaline will be accepted.
4. Bauxite has been reclassified as a sedimentary rock.
5. Rock Salt and Rock Gypsum for identification purposes are considered the same, respectively, as the minerals Halite and Gypsum and do not need to be distinguished



1. **DESCRIPTION:** Teams will construct a device prior to the tournament designed to collect heat and complete a written test on thermodynamic concepts.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 50 minutes

IMPOUND: Yes

CALCULATOR: Class III

2. **EVENT PARAMETERS:**

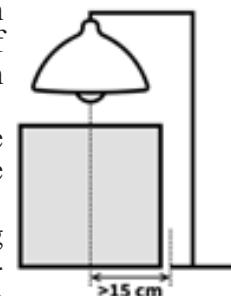
- a. Each team may bring writing utensils.
- b. Each team may bring two Class III calculators for use during any part of the event.
- c. Each team may bring a collection of notes and resources, written/printed on paper, of any size containing information in any form and from any source. Binders, books, notebooks, folders, sheet protectors, lamination, tabs, and labels are permitted. Participants are responsible for organizing and containing their notes efficiently. They may separate or remove the pages from containers for use during any part of the event.
- d. Each team may bring one pre-constructed device, spare parts, tools (defined as items aiding setup, not part of the build), and any prepared graphs or tables for use in their calculations of the device's results. Only the device and spare parts must be impounded, and all items must be at room temperature when impounded.
- e. The Event Supervisors will provide the testing materials listed in the COMPETITION AREA section. Teams should not bring these materials.

3. **CONSTRUCTION PARAMETERS:**

- a. The team's device will be designed to collect heat from an Event Supervisor-provided lamp setup to raise the temperature of room-temperature water.
- b. Devices may be constructed of and contain anything except the following materials and components:
 - i. Hazardous materials, including but not limited to: asbestos, mineral wool, and fiberglass
 - ii. Thermometers, probes, commercially available thermoses, coolers, or vacuum sealed devices.
 - iii. A team's use of any of the hazardous materials listed in 3.b.i. means that their device will not be allowed to be set up, modified or tested. Such teams will only be allowed to participate in the written test.
- c. Within the device, participants must be able to insert and remove a 9 oz disposable plastic party cup provided by the Event Supervisor filled with room-temperature water.
- d. The device, including the cup of water, must fit within a 30.0 cm x 30.0 cm x 30.0 cm cube when set up for testing.
- e. Parts of the device may be inside the cup, but no part of the device may contact the water. Nothing may be added to the water.
- f. Devices must have no energy sources (e.g., no electrical components, battery powered heaters, chemical reactions, etc.) to help warm the water. At the Event Supervisor's discretion, teams must disassemble their devices at the end of the testing period to verify the materials used in construction. A team's use of any of the materials listed in this rule will result in TS and PS scores of zero for the team.
- g. No part of the device may be significantly different from room temperature at the start of the event.
- h. Competitors must be able to answer questions regarding the design, construction, and operation of the device per the Building Policy found on www.soinc.org.

4. **COMPETITION AREA:**

- a. For each testing setup, the Event Supervisor will provide a clamp lamp with metal reflector and with a 100-125 watt heat lamp bulb. The bulb must not be tinted and may not be LED, halogen, or ceramic.
- b. The lamp will be mounted, facing down, above the Testing Surface. The Testing Surface is the surface that the team's device is placed on, like a table or countertop. Multiple identical setups may be used.
- c. The heights of the bulb and reflector above the Testing Surface must be the same for all teams. The height above the Testing Surface of the bottommost point of the bulb must be:
 - i. Regionals: 40 - 50 cm height, in 5 cm increments
 - ii. States: 40 - 55 cm height, in 5 cm increments
 - iii. Nationals: 40 - 65 cm, in 1 cm increments



- d. The Event Supervisor must mount the lamp in such a way so that the team's device can sit centered under the lamp. Any protruding parts of a lamp stand (such as the feet of the stand) must be no closer than 15 cm to the center of the bulb. See the diagram for an example.
- e. The Event Supervisor will provide a source water bath, which is expected to be approximately room temperature. Event Supervisors will do their best to keep this the same for all teams.
- f. The Event Supervisor will provide the thermometer for the measurement of the starting and final temperatures. The Event Supervisor thermometer may be a traditional, non-mercury thermometer or an electronic probe capable of measuring the temperature to a tenth of a degree Celsius. Multiple thermometers may be used, but, for each team, the same thermometer must be used to measure their starting and final temperatures. Non-contact thermometers are allowed. Competitors will not be allowed to use the Event Supervisor thermometers.
- g. The Event Supervisor will provide one cup per team, containing 100 mL of water from the source water bath. The cup will be a disposable, clear plastic, 9 ounce party cup (approximately 3.9 - 4.0 inches tall, with a mouth of approximately 2.9-3.0 inches).

5. THE COMPETITION:

Part I: Written Test

- a. The written test will assess the team's knowledge of thermodynamics.
- b. Unless otherwise requested, answers must be in metric units with appropriate significant figures.
- c. The test will consist of at least three questions from each of the following areas:
 - i. Thermodynamic systems, intensive and extensive properties of a system, definition of temperature, the zeroth law of thermodynamics, temperature scales and conversions, definitions of heat-related units
 - ii. Phases of matter, phase transitions, phase diagrams, latent and sensible heat, ideal gas law
 - iii. Kinds of heat transfer, thermal conductivity, heat capacity, specific heat
 - iv. Processes (e.g., adiabatic, isothermal, isochoric, isobaric), thermodynamic cycles, engines, efficiency, and the first and second laws of thermodynamics
 - v. Historical perspective of the contributions of Kelvin, Joseph Black, Joule, Carnot, Planck, Clausius, Boltzmann, and Maxwell to the field of thermodynamics
 - vi. **State/National only:** Radiant exitance, blackbody radiation, the Stefan-Boltzmann law, and the third law of thermodynamics
 - vii. **State/National only:** Entropy and enthalpy
- d. Questions on the test will use the following mathematical content:
 - i. Standard arithmetic operations, including ratios, exponents, and absolute value.
 - ii. Algebraic manipulations, including derivations involving multiple steps and variables.
 - iii. Geometric calculations, including computing the volume or area of common shapes.
 - iv. No questions may require the use of trigonometry or calculus.

Part II: Device Testing

- a. At the start of each competition block, the Event Supervisor will announce the height of the bulb above the Testing Surface, the Heating Time, room temperature, and the temperature of the source water bath. The Event Supervisor will also announce the bulb's specifications, which should include the wattage and bulb measurements. These parameters and bulb specifications must be the same for all teams, except for slight incidental variations in temperature. It is recommended that the Event Supervisor provide this information in written form. The Heating Time is:
 - i. 10 minutes at Regionals,
 - ii. 10 - 15 minutes at States/Nationals in 1 minute increments
- b. The teams will be given 3 minutes of Device Setup Time to set up or modify their device and begin making their prediction of the water temperature at the end of the Heating Time. Teams may attempt to fix any construction violations during this time. This time is not paused for any construction re-checks. No modifications may be made to the device after the Device Setup Time finishes except for inserting/removing the cup of water.



- c. The Event Supervisor, using their own measuring device and thermometer, will dispense 100 mL of water into a cup for each team and measure and record the starting temperature to the nearest tenth of a degree Celsius. The Event Supervisor will inform the team of the starting temperature of the team's water. The water may be distributed to teams at the same time or in a staggered manner. Event Supervisors will not leave a thermometer in teams' devices or measure the water temperature while the cups are in teams' devices.
 - d. Once a team receives their water and its Event Supervisor-measured temperature, they will have up to 1 minute of Water Setup Time to insert the cup of water into their device. Teams may not alter or manipulate the cup or the water in it beyond inserting the cup into their device.
 - e. At the end of the Water Setup Time, the Event Supervisor will turn on the lamp and start a timer for the Heating Time. The team may not touch their device or the lamp testing setup during the Heating Time. Teams that do so will receive a score of zero (0) for their TS and PS scores.
 - f. The team must finalize their prediction of the final temperature of their cup of water before the end of the Heating Time, or they will receive no Prediction Score (PS). Teams must make their prediction in degrees Celsius. Teams may also be allowed to work on the written test during the Heating Time at the discretion of the Event Supervisor.
 - g. At the end of the Heating Time, the Event Supervisor will turn off the lamp. The team must then completely remove the cup of water from their device and place it on the Testing Surface next to their device. The cup must be completely separated from all device components before the Event Supervisor measures the final temperature. The Supervisor will then record the final temperature to the nearest tenth of a degree Celsius. Both the team and the Supervisor should attempt to accomplish these tasks within one minute to minimize water cooling.
 - h. The Event Supervisor will review with the team the Part II data recorded on their scoresheet.
 - i. Teams filing an appeal regarding Part II must leave their device in the competition area.
6. **SCORING:**
- a. High Score wins.
 - b. All scoring calculations are to be done in degrees Celsius.
 - c. Final score (FS) = ES + TS + PS; The maximum possible FS is 100 points. A scoring spreadsheet is available at www.soinc.org.
 - d. Exam Score (ES) = (Part I score / Highest Part I score for all teams) x 50 points
 - e. Temperature Score (TS) = (TG / Highest TG of all teams) x 30 points
 - i. Temperature Gain (TG) = final cup water temp - starting cup water temp
 - ii. The lowest possible TG is 0 points.
 - iii. A team utilizing any of the materials in rule 3.f will receive a TS of 0.
 - iv. A team touching their device or the lamp testing setup during the Heating Time will receive a TS of 0.
 - f. Prediction Score (PS) = (PE / Highest PE of all teams) x 20 points
 - i. Prediction Estimate (PE) = 1 - abs(final cup water temp - predicted final cup water temp) / final cup water temp.
 - ii. The lowest possible PE is 0 points.
 - iii. A team utilizing any of the materials in rule 3.f will receive a PS of 0.
 - iv. A team that does not give a final temperature prediction before the end of the Heating Time or that touches their device or the lamp testing setup during the Heating Time will receive a PS of 0.
 - g. If a team violates any COMPETITION rules, their TS and PS values will be multiplied by 0.9 per violation when calculating the scores
 - h. If a team has any CONSTRUCTION violations at the end of their Setup Time, their TS and PS values will be multiplied by 0.7 per violation when calculating the scores.
 - i. Teams that are not allowed to compete in Part II due to unsafe operations or do not bring a device receive zero points for their TS and PS scores. Teams will be allowed to compete in Part I.
 - j. Tie Breakers: 1st - Best TS; 2nd - Best ES; 3rd - preselected test questions



1. **DESCRIPTION:** Participants will be assessed on their understanding and evaluation of marine and estuary aquatic environments.

A TEAM OF UP TO: 2

EYE PROTECTION: C

APPROXIMATE TIME: 50 minutes

CALCULATOR: Class II

2. **EVENT PARAMETERS:**

- a. Each team may bring writing utensils.
 - b. Each team may bring two Class II calculators.
 - c. Each team may bring one (1) 8.5" x 11" sheet of paper that may contain information on both sides in any form and from any source. The sheet of paper may be laminated or placed in a sheet protector to increase durability. Affixed labels, as well as multiple sheets of paper (whether in a single sheet protector or not) are prohibited.
 - d. Each team may bring one student built salinometer/hydrometer for testing.
 - e. Participants must wear Category C eye protection during Salinometer Testing (3.Part IV). Teams without proper eye protection must be immediately informed and given a chance to obtain eye protection if time allows.
3. **THE COMPETITION:** This may include analysis, interpretation or use of charts, graphs and sample data. Supervisors are expected to utilize **marine and estuary** scenarios and have students analyze and evaluate comparative macroinvertebrates and water quality data. Process skills may include equipment use, collecting and interpreting data, measuring, analyzing data, and making inferences.

Part I: Marine and Estuary Ecology – 30% of the total score

- a. This part will use multiple choice, matching, fill-in-the-blank and/or short answers to assess participants' knowledge.
- b. Core knowledge areas may include aquatic ecology, water cycle, nutrient cycling, aquatic chemistry and its implications for life, potable water treatment, waste water treatment, aquatic food chains/webs, community interactions, population dynamics, and the importance of watersheds.
- c. Additional applied topics may include: importance of estuaries, watershed resource management issues, sedimentation pollution, harmful species, recently killed coral, harmful algal blooms (HABs), emerging contaminants (e.g. medicines and chemicals like PFAS), and the five ocean garbage patches.
- d. **Understand and apply the concept of indigenous knowledge or traditional ecological knowledge (TEK) to given scenarios.**
- e. **State and Nationals Only: life history strategies (e.g., age structure, survival curves, life tables, succession, R and K strategies).**

Part II: Coral Reef Macroflora and Fauna Identification – 30% of the total score

- a. This part will assess participants' knowledge of coral reefs and the ecological factors that have harmful effects on reef ecosystems. It will also include the identification (common name only) of Coral Reef organisms and their importance as indicators of reef health.
- b. Participants are also expected to know the general ecology, life cycles, and feeding habits of the following organisms (note: spp. is an abbreviation for multiple species):
 - i. Organisms found around the globe:
 - (1) Banded coral shrimp (*Stenopus hispidus*),
 - (2) Butterfly fish (*Chaetodon* spp.),
 - (3) Crown of thorns starfish (*Acanthaster planci*),
 - (4) Fleshy algae (**Rhodophyta, Chlorophyta, Phaeophyta**),
 - (5) Grouper >30 cm (Serranidae, Epinephelinae),
 - (6) Hard/Stony corals (**Scleractinia**),
 - (7) Lobsters (**Nephropidae**),
 - (8) Long-spined black sea urchins (Diadematidae/Diadema spp.),
 - (9) Moray eel (*Muraenidae*),
 - (10) Parrotfish (>20 cm) (Scaridae),
 - (11) Pencil urchins (**Eucidairis spp.**),
 - (12) Sea Cucumber (Holothuroidea),



- (13) Snapper (*Lutjanidae*),
- (14) **Demosponges** (*Demospongiae*)
- (15) Sweetlips (*Haemulidae*)
- (16) Triton (*Charoniidae*/Charonia spp.).
- ii. Organisms found in the Indo-Pacific region only:
 - (1) Barramundi cod (*Cromileptes altivelis*),
 - (2) Bumphead parrotfish (*Bolbometopon muricatum*),
 - (3) Giant clams (*Tridacna* spp.),
 - (4) Humphead wrasse (*Cheilinus undulatus*).
- iii. Organisms found in the Atlantic region only:
 - (1) Flamingo Tongue Snail (*Cyphoma gibbosum*),
 - (2) Gorgonia (Octocorallia),
 - (3) Nassau grouper (*Epinephelus striatus*).

Part III: Water Monitoring and Analysis – 30% of the total score

- a. Participants are expected to understand and interpret data related to testing procedures and purposes for collecting data related to salinity, pH, phosphates, turbidity, dissolved oxygen, temperature, nitrates, fecal coliform, total solids, biochemical oxygen demand and aragonite saturation and their relationships to one another.
- b. No physical, laboratory tests will be performed on these topics by participants.

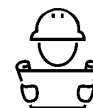
Part IV: Salinometer Testing – 10% of the total score

- a. Teams must build, calibrate, bring and demonstrate a salinometer/hydrometer capable of measuring saltwater (most likely NaCl) concentrations between 1-10% (mass/volume). Students may build a simple electronic salinometer if they wish. No programmable devices or those that can communicate over wifi are allowed. No laptops, tablets, or other similar devices may be used in the competition.
- b. There are no restrictions on size except that the team must build the device to operate within a standard 400 – 600 mL beaker filled with **at least 400 mL of** the saltwater solution.
- c. Teams will be expected to estimate the percent salinity measured by their device to the nearest tenth of a percent. Full credit will be given $\pm 1\%$ at Regionals and $\pm 0.5\%$ at State/Nationals. Calibration solutions may or may not be provided by the Event Supervisor.

4. **SCORING:**

- a. High Score wins.
 - i. Points will be assigned to the various questions and problems for Parts I, II, and III.
 - ii. Points for bringing a salinometer for testing will be 5% of the total score.
 - iii. Points for making an accurate salinity measurement per 3.Part IV.c will be 5% of the total score.
- b. Preselected questions will be used as tiebreakers.

This event is sponsored in partnership with Current and the NSF Great Lakes RENEW Engine.



1. **DESCRIPTION:** Prior to the tournament, teams will construct, collect data on test flights, analyze and optimize free flight rubber-powered Airplanes to achieve maximum time aloft.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 15 minutes

2. **EVENT PARAMETERS:**

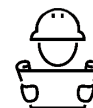
- a. Teams may bring up to 2 Airplanes, Flight Log, transportation boxes, tools, and equipment.
- b. Teams must bring one or more Measurement Boxes; transportation and measurement boxes may be the same box.
- c. The Event Supervisor will provide all other measurement tools and timing devices for scoring purposes.

3. **CONSTRUCTION PARAMETERS:**

- a. Airplanes may be constructed from published plans, commercial kits, competitor's designs, and/or other sources of design. Kits, if used, must not contain any pre-glued joints or pre-covered surfaces.
- b. Any materials except Boron filaments may be used in construction of Airplanes and boxes.
- c. The Airplane in its flight configuration, must fit fully into a team-provided Measurement Box without a lid/cover.
 - i. The external dimensions of the Measurement Box must fit within a right, rectangular prism of 32.0 cm x 24.0 cm x 47.0 cm, including any external protuberances on the lidless box.
- d. "Flight Configuration" means the Airplane is fully assembled and ready to fly. For example, no change in chord, span, length, or total lifting area can occur after removing the Airplane from its box and throughout the flight itself. Components that rotate during flight, such as propellers, may be rotated to allow the Airplane to fit into the box. The rubber motor(s) does not have to be on the Airplane or wound.
- e. Total mass of the Airplane, excluding the rubber motor(s), must be 8.0 g or more.
- f. The Propeller assemblies may be built by the participants or purchased pre-assembled. This may include a Propeller, a shaft, a hanger, and/or a thrust bearing. Variable-pitch Propellers that include mechanisms to actively change the propeller diameter or blade angle must not be used.
- g. All Airplane lifting forces must be generated by wing(s).
- h. The sole power for the Airplane must come from rubber motor(s) that remain attached to the Airplane throughout the flight.
 - i. The total mass of the rubber motor or rubber motor set, including any attachments such as O-rings, must not exceed 1.5 g.
 - ii. For a multi-motor Airplane, rubber motors must be massed together as a set, and individual rubber motors may not be interchanged between sets.
 - iii. Motors will be massed separately from the Airplane. Motors may be lubricated before and/or after check-in.
 - iv. Up to 6 motors, or sets of motors for a multi-motor Airplane, may be checked in.
- i. Participants must be able to answer questions on design, construction, and operation of the device per the Building Policy found on www.soinc.org.

4. **THE COMPETITION:**

- a. Prior to day of Tournament
 - i. An indoor location will be selected by the Tournament officials. The following will be announced:
 - (1) The room dimensions (approximate length, width, and ceiling height).
 - (2) Competition factors such as air currents.
- b. Day of Tournament and prior to the first time slot:
 - i. The Event Supervisor will determine if:
 - (1) Multiple official flights may occur simultaneously.
 - (2) Practice flights may occur throughout the Competition, at the Event Supervisor's discretion, but must yield to any official flights.
 - (3) No practice flights will occur in the final half-hour of the event's last period. (Teams that declare a trim flight during their 10-minute Flight Period may still fly).
 - ii. Once participants enter the cordoned off competition area to trim, practice, or compete, they must not receive outside materials (except as permitted by the Event Supervisor), assistance, or communication. Only participants may handle the Airplane until the event ends. Teams violating this rule will be ranked below all other teams. Spectators will be in a separate area.



c. Self-Check Inspection Station (Optional)

- i. Prior to check-in with the Event Supervisor, a self-check inspection station may be made available to participants for checking their Measurement Box(es), and Airplane. Modifications may be made prior to check-in.

d. Check-In

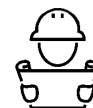
- i. At check-in, participants will present their Airplane in Measurement Box(es) and Flight Logs for inspection immediately prior to their Flight Period. Teams from the same school can share the same Measurement Box.
- ii. The Event Supervisor will verify the external dimensions of the Measurement Box(es) and that the Airplane fits fully inside the Measurement Box while in its flight configuration. Teams may not manipulate the configuration of the Airplane in order to fit into the box, except to rotate the rotor. The Airplane's overall dimensions must not change after being removed from the box. This may be verified by showing that the Airplane slides into and out of the box without changing shape at the discretion of the Event Supervisor.
- iii. The participants will remove the Airplane from the box to allow for the mass to be measured. The mass may be adjusted with ballast provided by the team to account for scale differences.
- iv. All motor(s) will be collected and returned to the team at the start of their 10-minute Flight Period.
- v. Only participants should handle the Airplane or Measurement Box(es).

e. Flight Period

- i. The 10-minute Flight Period begins when the Event Supervisor returns the motors to the team.
 - ii. Any flight beginning within the 10-minute Flight Period will be permitted to fly to completion.
 - iii. Participants may make adjustments/repairs/trim flights during their official Flight Period.
 - iv. Before each launch, participants must indicate to the Timers whether a flight is an official flight or a trim flight. A flight is considered official if a team fails to notify the Timer(s) of the flight's status.
 - v. Teams must not be given extra time to recover or repair their Airplane.
 - vi. Teams may make up to a total of 2 official flights using 1 or 2 airplanes.
 - vii. Time aloft for each flight starts when the Airplane leaves the participant's hand and stops when any part of the Airplane touches the floor, the lifting surfaces no longer support the weight of the Airplane (such as the Airplane landing on a girder or basketball hoop) or the Event Supervisors otherwise determine the flight is over.
 - viii. Event Supervisors are strongly encouraged to utilize three (3) timers on all flights. The median flight time in seconds to the precision of the device used is the official Time aloft.
 - ix. Participants must not steer the Airplane during flights.
 - x. Students must be on the floor to launch from their hands. Artificial aids cannot be used to increase launch height.
 - xi. In the unlikely event of a collision with another Airplane, a team may elect a re-flight. The decision to re-fly may be made after the Airplane lands. Timers are allowed to delay a launch to avoid a possible collision. The 10-minute Flight Period does not apply to such a flight.
- f. If requested by the Event Supervisor, the participants must demonstrate that each Airplane still fits fully inside the Measurement Box(es) in the flight configuration. Motor(s) may be removed from the Airplane or left in place during the demonstration. ES may recheck the mass of the Airplane.
- g. The Event Supervisor will review with teams their data recorded at the end of their Event Time.
- h. Teams filing an appeal must leave their Airplane(s), Motors, Measurement Box(es), and Flight Log in the event area.

5. FLIGHT LOGS:

- a. Teams may present a Flight Log of recorded data for a BONUS. This data must include 6 or more parameters (3 required and at least 3 additional) with data for 10 or more test flights prior to the competition.
 - i. The required parameters are:
 - (1) Motor Size before wind-up
 - (2) Number of Turns on Launch or Torque at Launch
 - (3) Flight Time
 - ii. The team must choose 3 additional data parameters beyond those required (e.g. turns remaining at landing, estimated/recorded peak flight altitude, the motor torque at landing, propeller pitch, etc.)



- b. All numeric data must include units which may be listed with the column headers of the log. Torque units may be “arbitrary” if an uncalibrated torque meter with arbitrary numbers is utilized.
 - c. All logs will be returned to teams after inspection.
6. **SCORING:**
- a. High Score wins.
 - b. A team’s Final Score is the larger of the team’s Flight Scores.
 - c. Flight Score for each official flight = Flight Time x Bonuses (6.d.).
 - d. **Bonuses:**
 - i. Teams with a complete Flight Log will receive a 20% bonus multiplier (x 1.2)
 - ii. Teams with a partial Flight Log will receive a 10% bonus multiplier (x 1.1).
 - iii. Teams without a Flight Log will receive no Flight Log bonus multiplier (x 1.0)
 - e. Example Score: For a flight time of 68.00 seconds with a completed Flight Log the score would be: 68.00 x 1.2 = 81.6 points.
 - f. Teams that violate rule(s) under “CONSTRUCTION PARAMETERS” or “COMPETITION” that do not have a specific penalty will be ranked after all teams that do not violate those rules.
 - g. Ties will be broken by the longest non-scored official Flight Score.
7. **DEFINITIONS:**
- a. *Airplane*: A heavier-than-air device that is held aloft by wing(s) that provide lift as the device is propelled through the air by a propeller.
 - b. *Wing*: A surface that creates lift by the flow of air across the surface.
 - c. *Event Supervisor (ES)* is responsible for ensuring the event runs correctly. The ES can assign one or more event helpers to perform various competition tasks.
 - d. *Flight Configuration* means the Airplane is fully assembled and ready to fly. For example, no change in chord, span, length, or total lifting area can occur after removing the Airplane from its box and throughout the flight itself. Components that rotate during flight, such as propellers or rotors, may be rotated to allow the Airplane to fit into the box. The rubber motor(s) does not have to be on the Airplane or wound.
 - e. *Propeller*: A rotating assembly consisting of a hub and at least one, typically two, angled and/or twisted blades designed to convert rotational energy (from rubber motor) into thrust.
 - f. *Trim Flight*: A test or practice flight that is not used for official scoring.
 - g. *Steering*: Participants using something to control the direction of motion once the device is released. (ex. Balloons, poles, rods, air, hands, etc.)
 - h. *Ballast*: A substance used to provide stability, balance, or control. (e.g. clay, tape, etc.)

This event is sponsored by the National Free Flight Society (NFFS).



TRIAL EVENT RULES EXPLANATION

See General Rules, Eye Protection & other Policies on www.soinc.org as they apply to every event.

Science Olympiad is continually in the process of researching, developing and evaluating new events. We are looking for events, activities and projects that engage students in all aspects of the scientific endeavor while presenting them with exciting and challenging problems to solve and content to master. In an effort to ensure our events meet those standards, we have established a process that moves an event from a creative concept through a series of pilots and trials, with the ultimate goal of making it into rotation as a current event.

For the 2026-2027 season, we are publishing a selection of Trial Events in the 2027 Rules Manual. The events presented here are not a comprehensive list of all the events under development. For a full list please visit: <https://www.soinc.org/learn/trial-events>. These particular events are being showcased here because of the topics they address, their approach to challenging Science Olympiad participants and their potential to become part of the competition in the next few seasons. Right now, they still need additional testing and trial. Besides being incorporated into this manual the rules for these events and additional resources are posted at <https://www.soinc.org/learn/trial-events>.

We have incorporated the rules for these Trial Events into the 2027 Rules Manual so that all teams, event supervisors, and tournaments have easy access to them. If conditions allow, we encourage State Chapters and Tournament hosts to run some of these Trial Events as they offer participants looking for an extra challenge the ability to compete against like-minded peers while contributing important information to prepare these events to become part of the competition in 2028 and beyond.

If a Tournament does choose to run one of the Trial Events published here, a Trial Event from the Trial Event page, or one of their own creation we would ask that you have both event participants and Event Supervisors complete the appropriate post-event evaluation. These evaluations can be found online at soinc.org on the Trial Event page. These brief surveys provide important information to help us fine tune events as well as make decisions about which events are worthy of being part of the Science Olympiad National Competition.



1. **DESCRIPTION:** Participants will participate in quiz and coding activities designed to assess their knowledge of computer science practice module material in many areas, such as, programming concepts, AI, cryptography and programming languages (i.e., Python, Java). The annual practice module will represent the scope of material available for testing.

A TEAM OF UP TO: 2

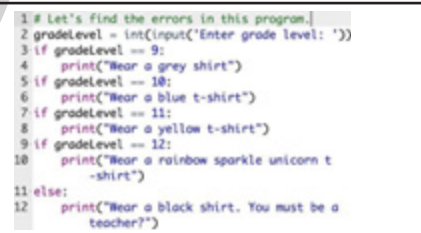
APPROXIMATE TIME: 50 minutes

2. EVENT PARAMETERS:

- a. Participants will use CodeHS.com on a computer to take quizzes, debug and write code.
- b. Each team must bring one device capable of running Google Chrome for the purposes of competing in the event unless specified by the Event Supervisor.
- c. Each team may bring one wired or wireless mouse and keyboard.
- d. No resource materials, except those provided by the Event Supervisor, may be used.
- e. Unauthorized resources or copying code from sources outside of ones provided by the Event Supervisor or those from CodeHS.com will result in a disqualification.
- f. Participants will be provided a URL to access the competition site at the start of the competition.

3. THE COMPETITION:

- a. Teams will use CodeHS.com and the created competition module to complete multiple activity types. Some of the possible activity types follow.

Activity Types & Suggested Times	Activity Description	Image
Gamified Quizzes	Participants will be provided questions in concert with other participants, and both the accuracy and time to result against competitors will be measured.	
Individual Quizzes	Participants will be provided multiple choice questions and coding snippets. Only the accuracy of the response will be measured.	
Individual Debug and Coding Assessments	Participants will be provided coding challenges to debug and author. The accuracy of the result will be measured by an auto grader or event volunteer.	

4. SCORING:

- a. The high score wins as determined by total points.
- b. Ties will be resolved using the time rankings determined in the Gamified Quiz section.

This event is sponsored by CodeHS.



1. **DESCRIPTION:** Teams will complete a written test on density and fluid mechanics concepts and construct a hydraulic device to push objects across a surface.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 50 minutes

IMPOUND: Yes

CALCULATOR: Class III

2. **EVENT PARAMETERS:**

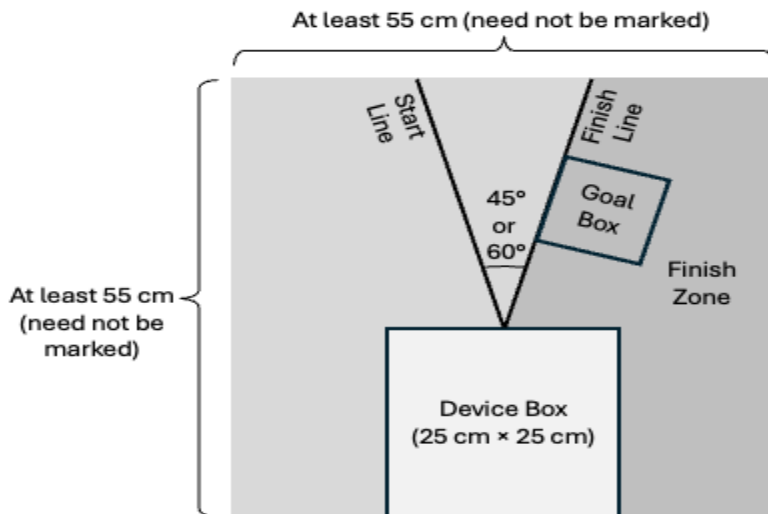
- a. Each team may bring writing utensils.
- b. Each team may bring two Class III calculators for use during any part of the event.
- c. Each team must impound up to one pre-constructed hydraulic device and any tools or spare parts. Teams must have all water that they need already sealed in their hydraulics in order to impound and may not impound additional water or request water from the Event Supervisor.
- d. Each team may bring a collection of notes and resources, written/printed on paper, of any size containing information in any form and from any source. Binders, books, notebooks, folders, sheet protectors, lamination, tabs, and labels are permitted. Participants are responsible for organizing and containing their notes efficiently. They may separate or remove the pages from containers for use during any part of the event.
- e. The Event Supervisors will provide the testing materials listed in the COMPETITION AREA section. Teams should not bring these materials.

3. **CONSTRUCTION PARAMETERS:**

- a. The device must not include any energy storage devices such as batteries, springs, or rubber bands. Devices that include these items will receive a BS and TS of zero (0).
- b. The device must be operated entirely by hand-powered syringes.
 - i. Hydraulic joints are operated in Control/Actuator pairs connected by tubing. Each pair has a Control Syringe, which competitors touch to push or pull on its plunger, and an Actuator Element (usually a syringe, but can be a different hydraulic element such as a balloon), which is mounted within the device to change the joint's position.
 - ii. There is no limit on the number of hydraulic joints or the volume of Control Syringes/Actuator Elements.
 - iii. The syringes may only contain water and/or air. The water may contain coloring.
 - iv. The tubing between each Control Syringe and Actuator Element must be flexible. The Control Syringes must not be attached to the device (including to the base) except via the flexible tubing.
- c. The device must be self-supporting on a flat surface. In the ready-to-run position, the parts of the device touching the Test Surface, excluding the Control Syringes and tubing, must fit within a 25.0 cm x 25.0 cm box with no height restriction. Parts not touching the Test Surface may extend beyond this box.
- d. The device must not damage, mark, or leave residue on the Test Surface.
- e. At the Event Supervisor's discretion, the team must disassemble their device at the end of the testing period to verify the materials used in construction.
- f. Competitors must be able to answer questions regarding the design, construction, and operation of the device per the Building Policy found on www.soinc.org.

4. **COMPETITION AREA:**

- a. The Event Supervisor will provide a flat, smooth Test Surface with several boundary lines marked using 1-inch masking tape. The Test Surface must be a minimum of 55 x 55 cm and have:
 - i. Device Box: A 25.0 x 25.0 cm square taped on the Test Surface. The midpoint of the front of the square must be marked.
 - ii. Travel Area: A wedge-shaped area centered on the front midpoint of the Device Box. At Regionals or States, the angle of the wedge will be 45°. At Nationals, the angle of the wedge will be 60°.
 - iii. Start Line: The left boundary of the Travel Area, indicated by a strip of tape at least 30.0 cm long.
 - iv. Finish Line: The right boundary of the Travel Area, indicated by a strip of tape at least 30.0 cm long.
 - v. Goal Box: A 10.0 x 10.0 cm square taped on the Test Surface. One side of the square is a segment of the Finish Line, extending 10.0 and 20.0 cm from the front midpoint of the Device Box. The other sides of the square are to the right of the Finish Line.
 - vi. Finish Zone: The area extending to the right of the Finish Line, including the Goal Box. The Finish Zone does not include the Device Box.



- b. Four 9-volt batteries will be placed upright on the Start Line with their terminals on top. They may be rotated about their vertical axis and do not have to be parallel to the Start Line. Three batteries will be designated “Movable” and one will be “Unmovable”. During the Competition Time, the team will attempt to move the Movable Batteries to the Finish Zone, while not disturbing the Unmovable Battery. The Unmovable Battery must be clearly marked with tape.
 - i. No part of the nearest battery may be within 3.0 cm of the Device Box, as measured along the Start Line.
 - ii. No part of the furthest battery may be farther than 30.0 cm from the Device Box, as measured along the Start Line.
 - iii. There must be at least 3.0 cm of open space along the Start Line between batteries.
- c. Multiple identical test setups may be used to facilitate teams competing in a timely manner.

5. THE COMPETITION:

Part I: Written Test

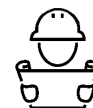
- a. The written test will assess the team’s knowledge of density and fluid mechanics.
- b. Unless otherwise requested, answers must be in metric units with appropriate significant figures.
- c. The test will consist of questions from each of the following areas:
 - i. Density of solids, liquids, and gases, as well as relative density, number density, and area density.
 - ii. Pressure in fluids and gases, hydrostatic pressure, pressure-depth relationship, Pascal’s principle.
 - iii. Behavior of gases according to the gas laws: ideal gas, Boyle’s, Charles’, Gay-Lussac’s, and Avogadro’s.
 - iv. Archimedes’ principle, buoyancy, and stability.
 - v. Hydraulic design principles and materials, mechanical advantage, common hydraulic system components, source of losses.
 - vi. Viscosity: laminar and turbulent flow; Reynolds number; flow rate and the continuity equation; Hagen-Poiseuille equation; Stokes flow and law; Bernoulli’s principle and equation
- d. Questions on the test may use only the following mathematical content:
 - i. Standard arithmetic operations, including ratios, exponents, and absolute value.
 - ii. Simple algebraic manipulations, including solving one equation for one variable.
 - iii. Simple geometric calculations, including computing the volume or area of common shapes.

Part II: Device Testing

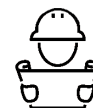
- a. At the start of each competition block, the Event Supervisor will reveal the Competition Area and indicate which battery is Unmovable.
 - i. Teams may not swap out the part(s) of their device expected to contact the batteries after they have seen the layout of the Competition Area.
- b. The team will be given 1 minute of Setup Time to set up their device in the Competition Area and fix any Construction Violations. The batteries will not be placed on the Test Surface until after the Setup Time ends. At the end of the Setup Time, the device must be in the ready-to-run position:



- i. All parts of the device touching the Test Surface, excluding the Control Syringes and tubing, are within the Device Box. (The device may touch the Test Surface outside the Device Box after the Run Time begins.)
 - ii. The part of the device expected to contact the batteries is approximately aligned with or extended over the Finish Line.
 - iii. The competitors are not touching any part of the device, including the Control Syringes.
 - c. The team will have up to 3 minutes of Run Time to attempt to move the Movable Batteries into the Finish Zone, while attempting not to disturb the Unmovable Battery. A Movable Battery is considered to be in the Finish Zone as long as it has crossed the Finish Line and is not touching either the Finish Line or the Device Box.
 - i. Additional points will be earned if the Movable Batteries in the Finish Zone are upright.
 - ii. Additional points will be earned if the Movable Battery that started the farthest from the Device Box ends in the Finish Zone.
 - iii. Points will be deducted if the Unmovable Battery is no longer upright or is no longer touching the Start Line.
 - d. During the Run Time, competitors may touch only the Control Syringes. Competitors must not touch any other part of the device, the batteries, or the Test Surface.
 - e. If the device stops, jams, or fails, the participants will be allowed to adjust it as necessary to continue operation up to two times. An adjustment may consist of multiple physical touches and is only completed once the device runs again on its own. An adjustment may not result in the movement of any batteries. Each adjustment will incur a penalty. The Run Time does not stop during adjustments.
 - f. If the device leaks, testing will be halted immediately and the team will have no additional opportunities to score. If a team's device leaks before the Run Time, the device will not be tested.
 - g. The Run Time ends when either three minutes have passed, the team declares that they are finished, or the device leaks, whichever comes first.
 - i. The Run Time will be recorded as 4 minutes if the device leaks or if all three Movable Batteries are not in the Finish Zone when the Run Time ends. Otherwise, the actual Run Time will be recorded.
 - h. The Event Supervisor will review with the team the Part II data recorded on their scoresheet.
 - i. Teams filing an appeal regarding Part II must leave their device in the competition area.
6. **SCORING:**
- a. High Score wins.
 - b. Final score (FS) = ES + BS + TS. The maximum possible FS is 70 points. A scoring spreadsheet is available at www.soinc.org.
 - c. Exam Score (ES) = (Part I score / Highest Part I score of all teams) x 40 points
 - d. Battery Score (BS) = (Each Movable Battery fully in the Finish Zone x 3 points) + Goal Bonus + Upright Bonus - Unmovable Penalty - Adjustment Penalty
 - i. Goal Bonus = 2 points x (Each Movable Battery entirely within the Goal Box)
 - ii. Upright Bonus = 2 x (Each Movable Battery in the Finish Zone that is upright)
 - iii. Unmovable Penalty = 7 points if the Unmovable Battery is displaced
 - iv. Adjustment Penalty = 4 points each time the device is adjusted
 - v. The lowest BS score = 0
 - e. Time Score (TS) = 9 x (Lowest Run Time of all teams / Team's Run Time)
 - i. Teams with a leaking device or without all three Movable Batteries in the Finish Zone receive a Time Score of 0.
 - f. Teams that are not allowed to compete in Part II due to unsafe operations, do not bring a device, or violate rule 3.a. will receive zero points for their BS and TS scores. These teams will be allowed to compete in Part I.
 - g. If a team has any CONSTRUCTION violations (other than rule 3.a.) at the end of their Setup Time, their BS and TS values will be multiplied by 0.7 per violation when calculating the scores.
 - h. If a team violates any COMPETITION rules, their BS and TS values will be multiplied by 0.9 per violation when calculating the scores.
 - i. If a team does not impound their device and all spare parts and tools, their BS and TS values will be multiplied by 0.5.
 - j. Tie Breakers: 1st - Best TS; 2nd - Best ES; 3rd - preselected test questions.



1. **DESCRIPTION:** Teams must construct a device on-site to solve an engineering challenge
A TEAM OF UP TO: 2
EYE PROTECTION: B
APPROXIMATE TIME: 50 minutes
IMPOUND: Yes
2. **EVENT PARAMETERS:**
 - a. Participants must wear eye protection during device setup and operation. Teams without proper eye protection must be immediately informed and given a chance to obtain eye protection if time allows.
 - b. All materials must be brought in a rectangular material box with a lid. The external dimensions of the material box including the lid must be no larger than 32.0 cm x 26.0 cm x 47.0 cm. The lid must close completely with all materials inside the box.
 - c. The material box and all materials must be impounded.
 - d. Any materials not listed as prohibited items within a challenge are allowed to be in the material box with the exception of:
 - i. Electronic components. Electricity must not be used to complete any challenge.
 - ii. Liquid substances
 - e. The only allowed adhesive is tape, which is limited to the following types:
 - i. Scotch tape
 - ii. Masking tape
 - iii. Duct tape
 - f. Each device must pass a safety inspection before operation.
 - g. Any tools teams want to use during their construction period must be impounded and must be in a separate container. These tools must only be used for their intended function and cannot be used as parts in their device. Electronic tools are not allowed.
 - h. The event supervisor must provide a measuring device, timer, and any equipment required by the specific task.
3. **COMPETITION PARAMETERS:**
 - a. The supervisor will select and announce the single challenge after impound has been completed, depending on the competition level:
 - i. Regionals: Category 1 Challenge Only
 - ii. State: Category 2 Challenge Only
 - iii. Nationals: Category 1 OR Category 2 Challenge
 - b. The device must be built on-site by the teams using only the materials brought in their material box.
 - c. Unless specifically stated in the instructions, devices must be free-standing.
 - d. Teams have a maximum of 30 minutes to build. Teams must not modify their device after the construction period has ended unless allowed in the specific challenge rules. Teams that complete construction early may be judged early.
 - e. The event supervisor will review with teams their data recorded at the end of their event time.
 - f. Teams filing an appeal must leave their materials, device, and material box in the event area.
4. **GENERAL SCORING:**
 - a. The Engineering Challenge instructions will describe the scoring system and will include tie-breaker details for each scoring tier.
 - b. Scoring will be divided into four tiers:
 - i. Tier 1: Devices that meet all requirements and accomplish the challenge
 - ii. Tier 2: Devices that meet all requirements but do not accomplish the challenge
 - iii. Tier 3: Devices that do not meet all requirements but accomplish the challenge
 - iv. Tier 4: Devices that do not meet all requirements and do not accomplish the challenge
 - c. Teams found modifying their device after the construction time has ended will be ranked one tier lower than their score would otherwise warrant.
5. **ENGINEERING CHALLENGES:**
 - a. Category 1 Challenge
 - i. The Challenge: Build a device to accurately measure the mass of an unknown object (Test Mass).
 - ii. Variable Parameters
 - (1) Test Mass will have dimensions no larger than 25 cm x 25 cm x 100. cm
 - (2) Test Mass will have a mass between 10.0 g and 2000.0 g inclusive.



- (3) The Test Mass will be revealed to teams immediately before their construction begins.
- (4) The event supervisor must provide the Test Mass.
- (5) Teams cannot touch the Test Mass until they have completed their construction period.
- (6) Event supervisors must inform team of the level of precision expected on their measurement at the beginning of the construction time. Measurements provided with additional significant figures will be rounded.

iii. Prohibited Materials

- (1) No commercial scales or components thereof.
- (2) Commercial balance weights including pre-measured physics masses.

iv. Demonstration

- (1) When ready, teams will place the Test Mass into their device.
- (2) The teams will have two minutes to measure the mass of the Test Mass and record their findings onto the paper provided by the event supervisor.
- (3) Timing will stop when the teams provide the event supervisor with the written mass as well as the Test Mass.
- (4) To qualify for Tiers 1 and 3, teams must demonstrate that their device actually measures mass and explain how the reading on their device corresponds to their measured mass. Teams may not simply predict the mass by visual inspection or hefting.

v. Scoring

- (1) High Score wins
- (2) Predicted Mass: The mass of the Test Mass in grams as cited by the students.
- (3) Official Mass: The mass of the Test Mass in grams as cited by the event supervisor.
- (4) Tiers 1 and 3: Ranked by the result of the following formula.

$$\text{Final Score} = 5000 - |\text{Official Mass} - \text{Measured Mass}| \text{ (with mass measured in grams)}$$
- (5) Tiers 2 and 4: Ranked by distance in centimeters from the highest point of the device to the ground in centimeters. High Score wins.
- (6) Ties are broken by the number of whole seconds remaining on the two-minute timer truncated down to the nearest whole second.

b. Category 2 Challenge

- i. The Challenge: Build a device capable of sorting five dollars of mixed coins into separate bins.

ii. Variable Parameters

- (1) The number of each type of coin will be determined by the event supervisor such that the total value of the coins equals five dollars.
- (2) The event supervisor must provide the coins.
- (3) Only quarters, dimes, and nickels will be used.
- (4) The number of each coin used will be the same for all teams.
- (5) All coins must enter the device through the same point in the device, which is called the Coin Hopper. Coins may be hand-fed individually or poured collectively into the hopper during demonstration.
- (6) Teams will have one minute to complete the sorting task. Timing begins when the first coin enters the device and ends when team members say "Stop" or one minute has elapsed.
- (7) Once the coins begin to enter the device teams may not touch the device until sorting is complete.
- (8) Each individual coin bin must be clearly marked with the denomination of the coins it will hold.
- (9) Coin bins must be removable after sorting.

iii. Prohibited Materials

- (1) No commercial coin sorting machines or parts thereof.

iv. Demonstration

- (1) When ready the team will have one minute to feed their coins into the device to be sorted.
- (2) Coins may not be sorted by hand prior to feeding into the device.
- (3) To qualify for Tiers 1 and 3 the device must successfully sort at least one coin of each denomination.
- (4) If sorting completed before one minute has elapsed teams must say "Stop" to stop the clock to redeem bonus points.



- (5) At the end of one minute all sorting must stop and a score is tallied. Any coins not sorted at the end of one minute into one of the bins must not be scored.
- (6) Teams will take each labeled bin out of their device after timing is completed and give them to the event supervisor for scoring.

v. Scoring

- (1) High Score wins.
- (2) Total Value: The total value in cents of all change accurately sorted.
- (3) Slot Height: The distance between the bottom of the entry point of the coin hopper and the floor in centimeters.
- (4) Bonus Seconds: The number of seconds remaining on the clock rounded to whole seconds.
- (5) Tiers 1 and 3: Ranked by the result of the following formula.
- (6) Tiers 2 and 4: Ranked by the distance in centimeters from the highest point of the device and the ground. High Score wins.
- (7) Ties are broken by the highest slot height, followed by the highest total value, followed by the highest bonus seconds.





DIV. C CHEMISTRY LAB EQUIPMENT LIST

See General Rules, Eye Protection & other Policies on www.soinc.org as they apply to every event.

Each team may bring any or all of the items listed below for use in Division C Chemistry Events. Teams not bringing these items will be at a disadvantage as Event Supervisors will not provide the listed lab equipment. A penalty of up to 10% may be given if a team brings prohibited lab equipment to the event.

Item	Expected Use
Box	Containing all of the kit materials
Graduated Cylinders (10 - 100 mL)	Measuring volumes
Beakers (50 - 500 mL)	Doing reactions, developing chromatograms
Erlenmeyer Flasks (10 - 250 mL)	Doing reactions
Test Tubes	Mix Chemicals, heat chemicals
Test Tube Brush	Clean Test Tubes
Test Tube Holder	Holds test tubes for heating
Test Tube Rack	Hold Test Tubes
Spot Plates	For semi-micro scale reactions, testing solubility, pH
Petri Dishes	Doing reactions, developing chromatograms
Slides	To put hairs, crystals, or fibers on for use with a microscope
Cover Slips	To cover & prevent items from coming off slides
Droppers	Add small amounts of liquids to reactions
Spatulas or spoons	Getting small amounts of solids out of containers
Metal Tongs, Forceps, or Tweezers	Holding & retrieving objects
Stirring Rods	Stirring mixtures
Thermometer	Determining the temperature of a solution
pH paper/meter	Test acidity or alkalinity of solution
Hand Lens	Magnification of small items for identification
Flame Loop	For identification of ions in a compound
Filter Paper	Filter solids from liquids
Funnel	Hold Filter Paper
9V battery	Electrolysis
Alligator Clip Wires	Connecting meters to metals
Nail	Electrolysis
Piece of Cu metal	Electrolysis
Piece of Zn metal	Electrolysis
Multimeter	Measuring current, voltage, and resistivity
9V or less Battery Conductivity Tester	Determining ionic strength of solution
Calipers - mechanical, not digital	Measuring lengths very precisely
Paper Towels	Cleaning
Pencil	Writing, Marking Chromatogram
Ruler	Measuring lengths
Magnets	For extraction and identification of iron filings
Cobalt Blue Glass	To filter out any sodium that might contaminate flame test from hands

The following document was prepared to offer some guidance to teams as they select calculators for use in different Science Olympiad events. **The calculator class listed in the event rules is the most complex calculator level allowed for the event. It is acceptable to use a lower calculator class in the event (e.g., if Class III calculators are allowed for the event, then students are therefore allowed to use Class I, Class II, or Class III calculators).**

By no means are the calculators listed here inclusive of all possible calculators; instead they are offered as common examples. The decisions of the event supervisors will be final.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators are the most basic type of calculators and often look like the one shown to the right. These calculators are limited to the four basic mathematics functions and sometimes square roots. These calculators can often be found at dollar stores.



Class II - Stand-alone non-programmable, non-graphing calculators look like the calculator to the right or simpler. There are hundreds of calculators in this category but some common examples include: CASIO FX-260, Sharp EL-501, and TI-30X.



Class III - Stand-alone, programmable, graphing calculators and stand-alone non-graphing, programmable calculators, often look like the calculator shown on the right. Some examples are: Casio 975 0/9850/9860, HP 40/50/PRIME, and TI 83/84/89/NSPIRE/VOYAGE.

To identify a stand-alone non-graphing, programmable calculator look for the presence of the 'EXE' button, the 'Prog' button, or a 'file' button. Examples include but are not limited to: Casio Super FXs, numerous older Casio models, and HP 35S. A calculator of this type with the buttons labeled is shown to the right.

PROG Button



EXE Button



Class IV - Calculator applications on multipurpose devices (e.g., laptop, phone, tablet, watch) are not allowed unless expressly permitted in the event rule.



This resource was created to help teams comply with the Science Olympiad Policy on Eye Protection adopted on July 29, 2015 and posted on the Science Olympiad Website (soinc.org).

Participant/Coach Responsibilities: Participants are responsible for providing their own protective eyewear. Science Olympiad is unable to determine the degree of hazard presented by equipment, materials and devices brought by the teams. Coaches must ensure the eye protection participants bring is adequate for the hazard. All protective eyewear must bear the manufacturer's mark Z87. At a tournament, teams without adequate eye protection will be given a chance to obtain eye protection if their assigned time permits. If required by the event, participants will not be allowed to compete without adequate eye protection. This is **non-negotiable**.

Corresponding Standards: Protective eyewear used in Science Olympiad must be manufactured to meet the American National Standards Institute (ANSI) standard applicable at its time of manufacture. The current standard is ANSI/ISEA Z87.1-2015. Competitors, coaches and event supervisors are not required to acquire a copy of the standard. The information in this document is sufficient to comply with current standards. Water is not a hazardous liquid and its use does not require protective eyewear unless it is under pressure or substances that create a hazard are added.

Compliant Eyewear Categories: If an event requires eye protection, the rules will identify one of these three categories. Compliance is simple as ABC:

CATEGORY A

- Description: Non-impact protection. They provide basic particle protection only
- Corresponding ANSI designation/required marking: Z87
- Examples: Safety glasses; Safety spectacles with side shields; and Particle protection goggles (these seal tightly to the face completely around the eyes and have direct vents around the sides, consisting of several small holes or a screen that can be seen through in a straight line)

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+
- Example: High impact safety goggles

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you are able to see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)
- Example: Indirect vent chemical/splash protection goggles

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved in lieu of the primary eye protection devices below regardless of the type of vents they have.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87 mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for all three Categories A, B & C
2. VisorGogs do not seal completely to the face, but are acceptable as indirect vent chemical/splash protection goggles



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NATIONAL TOURNAMENT SCHEDULE

See General Rules, Eye Protection & other Policies on www.soinc.org as they apply to every event.

2027 National Tournament Division C Schedule Ohio State University, Columbus, Ohio May 14-15, 2027

Event	7:00 – 8:00 AM	8:15 – 9:15 AM	9:30 – 10:30 AM	10:45 – 11:45 AM	11:45 -12:30 BREAK	12:30 – 1:30 PM	1:45 – 2:45 PM	3:00 – 4:00 PM	7:30– 9:30 PM	
Anatomy & Physiology C		1-10	11-20	21-30		31-40	41-50	51-60	Closing Ceremony	
Astronomy C		41-50	51-60	1-10		11-20	21-30	31-40		
BooMilever C	Self-Schedule									
Botany C		41-50	51-60	1-10		11-20	21-30	31-40		
Chemistry Lab C		31-40	41-50	51-60		1-10	11-20	21-30		
Circuit Lab C		21-30	31-40	41-50		51-60	1-10	11-20		
Codebusters C		51-60	1-10	11-20		21-30	31-40	41-50		
Designer Genes C		11-20	21-30	31-40		41-50	51-60	1-10		
Disease Detectives C		51-60	1-10	11-20		21-30	31-40	41-50		
Dynamic Planet C		11-20	21-30	31-40		41-50	51-60	1-10		
Electric Vehicle C		Impound	Self-Schedule							
Engineering CAD C		1-10	11-20	21-30		31-40	41-50	51-60		
Experimental Design C		41-50	51-60	1-10		11-20	21-30	31-40		
Forensics C		1-10	11-20	21-30		31-40	41-50	51-60		
Hovercraft C	Impound	Self-Schedule								
Mission Possible C	Impound	Self-Schedule								
Ping Pong Parachute C		Self-Schedule								
Protein Modeling C	Impound	21-30	31-40	41-50		51-60	1-10	11-20		
Remote Sensing C		51-60	1-10	11-20		21-30	31-40	41-50		
Rocks and Minerals C		31-40	41-50	51-60		1-10	11-20	21-30		
Thermodynamics C	Impound	11-20	21-30	31-40		41-50	51-60	1-10		
Water Quality C		21-30	31-40	41-50		51-60	1-10	11-20		
Wright Stuff C		Self-Schedule								



Exploring the World of Science

Science Olympiad wishes to acknowledge the following business, government and education leaders for partnering with our organization. Working together, we can increase global competitiveness, improve science and technology literacy and prepare the STEM workforce of the future. Thanks to: The Ohio State University (2027 National Tournament Host), University of Southern California (2026 National Tournament Host), Science Olympiad USA Foundation, Atwell Gives Foundation, Avantor Foundation, Ward's Science, Air Force, Cleveland-Cliffs Foundation, Combined Federal Campaign, Illinois Quantum and Microelectronics Park (IQMP), P33, International Motors, Aerospace Corporation, Amcor Cares Foundation, Cambridge Centre for International Research (CCIR), Centers for Disease Control and Prevention (CDC), CodeHS, Current and the NSF Great Lakes RENEW Engine, Discovery Education 3M Young Scientists Challenge, Double Good Foundation, IEEE Geoscience and Remote Sensing Society (GRSS), InGenius Prep, Kinder Morgan Foundation, National Free Flight Society, National Oceanic and Atmospheric Administration (NOAA) and North American Association for Environmental Education (NAAEE), OnShape, Prequel, SkyCiv, TKS World, Empowerly, Investing in Communities, MDRT Foundation and Yale Young Global Scholars. Strategic Partners: American Red Cross, 3D Molecular Designs, Google, Japan Science and Technology Agency, mHUB, Midnight Science Club, MxD Digital Manufacturing Institute, NBCUniversal Foundation, STEMConnector.

See the Science Olympiad website: www.soinc.org for current information regarding Policies, Standards, Summer Workshops, Official Kits from Ward's Science and digital items in the Science Olympiad Store.

Science Olympiad

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